S5K3B6SA

Automotive 1/2.4" 3MP CMOS Image Sensor with On-Chip HDR & LFM

Revision 0.03

Jul. 2025

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Chip Handling Guide

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1. Wear antistatic clothes and use earth band.
2. All objects that are in direct contact with devices must be made up of materials that do not produce static electricity.
3. Ensure that the equipment and work table are earthed.
4. Use ionizer to remove electron charge.

Contamination

Do not use semiconductor products in an environment exposed to dust or dirt adhesion.

Temperature/Humidity

Semiconductor devices are sensitive to:

- Environment
- Temperature
- Humidity

High temperature or humidity deteriorates the characteristics of semiconductor devices. Therefore, do not store or use semiconductor devices in such conditions.

Mechanical Shock

Do not to apply excessive mechanical shock or force on semiconductor devices.

Chemical

Do not expose semiconductor devices to chemicals because exposure to chemicals leads to reactions that deteriorate the characteristics of the devices.

Light Protection

In non- Epoxy Molding Compound (EMC) package, do not expose semiconductor IC to bright light. Exposure to bright light causes malfunctioning of the devices. However, a few special products that utilize light or with security functions are exempted from this guide.

Radioactive, Cosmic and X-ray

Radioactive substances, cosmic ray, or X-ray may influence semiconductor devices. These substances or rays may cause a soft error during a device operation. Therefore, ensure to shield the semiconductor devices under environment that may be exposed to radioactive substances, cosmic ray, or X-ray.

EMS (Electromagnetic Susceptibility)

Strong electromagnetic wave or magnetic field may affect the characteristic of semiconductor devices during the operation under insufficient PCB circuit design for Electromagnetic Susceptibility (EMS).

Revision History

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List of Conventions

Register RW Access Type Conventions

Type	Definition	Description
R	Read Only	The application has permission to read the Register field. Writes to read-only fields have no effect.
W	Write Only	The application has permission to write in the Register field.
RW	Read & Write	The application has permission to read and writes in the Register field. The application sets this field by writing 1'b1 and clears it by writing 1'b0.

Register Value Conventions

Expression	Description
x	Undefined bit
X	Undefined multiple bits
?	Undefined, but depends on the device or pin status
Device dependent	The value depends on the device
Pin value	The value depends on the pin status

Reset Value Conventions

Expression	Description
0	Clears the register field
1	Sets the register field
x	Don't care condition

Warning: Some bits of control registers are driven by hardware or write operation only. As a result the indicated reset value and the read value after reset might be different.

List of Acronyms

Acronyms	Descriptions
ADLC	Auto Dark Level Compensation
APS	Active Pixel Sensor
CDS	Correlated Double Sampling
CIS	CMOS Image Sensor
FPN	Fixed Pattern Noise
HDR	High Dynamic Range
HSYNC	Horizontal Sync Signal
OTP	One Time Programmable
PLL	Phase-Locked Loop
TG	Timing Generator
VSYNC	Vertical Sync Signal
LOF	Lateral Over Flow
SAD	Sum of Absolute Differences

1 Product Overview

1.1 Introduction

S5K3B6SA is a highly integrated 3Mp camera chip that includes a CMOS Image Sensor (CIS) with image correction functionality and serial transmission capability using 4-lane MIPI. It is designed for delivering the full resolution pixels of 60 frames per second (fps) with 12-bit ADC, HDR3 or HDR4, intra-scene dual conversion gain, and compressed outputs for High Dynamic Range (HDR) image with LED Flicker Mitigation.

S5K3B6SA is fabricated by the SAMSUNG 65 nm Back Deep Trench Isolation (BDTI) CMOS image sensor process, which is developed for imaging applications to realize a high-efficiency and low-power photo sensor. And it is also fabricated by the SAMSUNG 28nm logic process for low-power. The sensor consists of 1936(H)×1552(V) active pixels and eight pixels on each side for color interpolation. This sensor meets 1/2.4" optical format with 3Mp resolution.

S5K3B6SA has on-chip 12-bit ADC arrays to digitize the pixel output and on-chip Correlated Double Sampling (CDS) to drastically reduce the Fixed Pattern Noise (FPN). It incorporates on-chip camera functions such as Bad Pixel Correction (BPC), exposure control, HDR reconstruction, and so on.

S5K3B6SA is programmable through an I2C serial interface and includes on-chip One-Time Programmable (OTP) Non-Volatile Memory (NVM).

S5K3B6SA is suitable for an automotive camera module with 2.8 V/1.8 V/1.05 V power supply, and the target applications are E-Mirror, Front/Rear-View camera and Surround-View-Monitoring (SVM) applications.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

1.2 Features

S5K3B6SA supports the following features:

- 3 MP sensor with 1/2.4" optics
- Unit pixel size : 3.0 μm Split PD
- Effective resolution of 1920(H) x 1536 (V) with 3MP
- Active resolution of 1936 (H) x 1552 (V) with 3MP
  - Including eight pixels in each side for color processing
- High dynamic range and high sensitivity
- Electronic rolling shutter
- Maximum frame rate
  - Compressed 12-bit 3MP 60 fps (Intra-scene*1) Dual Conversion gain and Lateral overflow with LFM)
- Interfaces
  - Fine interface frequency control using additional dedicated PLL for EMI avoidance and integration flexibility
  - 2~4-lane selectable MIPI CSI-2 interface (Data rate : 4.0Gbps/lane)*2)
- Output formats
  - 24bit Combined RAW, 20/16/14/12-bit compressed combined RAW.
  - 20bit Combined RAW, 16/14/12-bit compressed combined RAW
- Control interface
  - CCI (Camera Control Interface) High speed I2C-compatible-Two-wire serial communication circuit up to 1MHz (FastMode+)
- 128 kbit (16 kbyte) on-chip OTP memories to support defect corrections and chip ID (38 kbit available for user)
- Analog gain up to x16
- Digital gain up to x256
- Vertical flip and horizontal mirror mode
- Static bad pixel correction with OTP
- Statistics data output for Combined image
- Output Defect and Detection map during image data stream. (RAW8)
- Built-in test pattern generation
- Embedded temperature sensor
- Embedded voltage monitoring
- Supply voltage of 2.8 V for pixel and analog, 1.8 V for I/O, and 1.05 V for digital core supply
- Satisfied ASIL B for compliance with funtional safety
- AEC-Q100 Grade2 Compatible (Tj = -40 ℃ to 125 ℃)

* 1) Dual conversion gain readout within single frame. * 2) Only 4 lanes are available to support maximum frame rate operation.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

2 Power Sequence

2.1 System State

Sensor module has five operating modes, such as power-off, hardware standby, software standby, streaming, and safety state as described in *Table 1*. Transition from one state to another is achieved by issuing the appropriate state command through the CCI serial control interface, the RSTN (XSHUTDOWN) signal changing state, and the power supplies. By default, S5K3B6SA is powered up by enabling the CSI-2 serial data interface. *Figure 1* illustrates the valid state changes for the sensor module.

S5K3B6SA supports the safety state to ensure the prohibition data stream when there is an error occurred from the error detection logic. Sensor transitions its state from safety state to software standby if an error is cleared through registers. Then, sensor can stream data.

If an error occurs during the data stream, S5K3B6SA outputs the last data normally and then moves to the safety state.

Table 1 System state summary

Power State	Description
Power-off	Power supplies are turned off.
Hardware standby	RSTN is low. Power supplies are turned on, but no communication with the sensor is possible. Internal registers are set to default values.
Software standby	RSTN is high. Register setting through I2C is possible.
Streaming	The sensor is fully functional and video outputs are available.
Safety state	Sensor enters 'Safety state' when there could be fault in sensor operation. State which not able to stream data until resetting 'Error' flag. Goes to 'Software standby' statue after resetting 'Error' flag.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

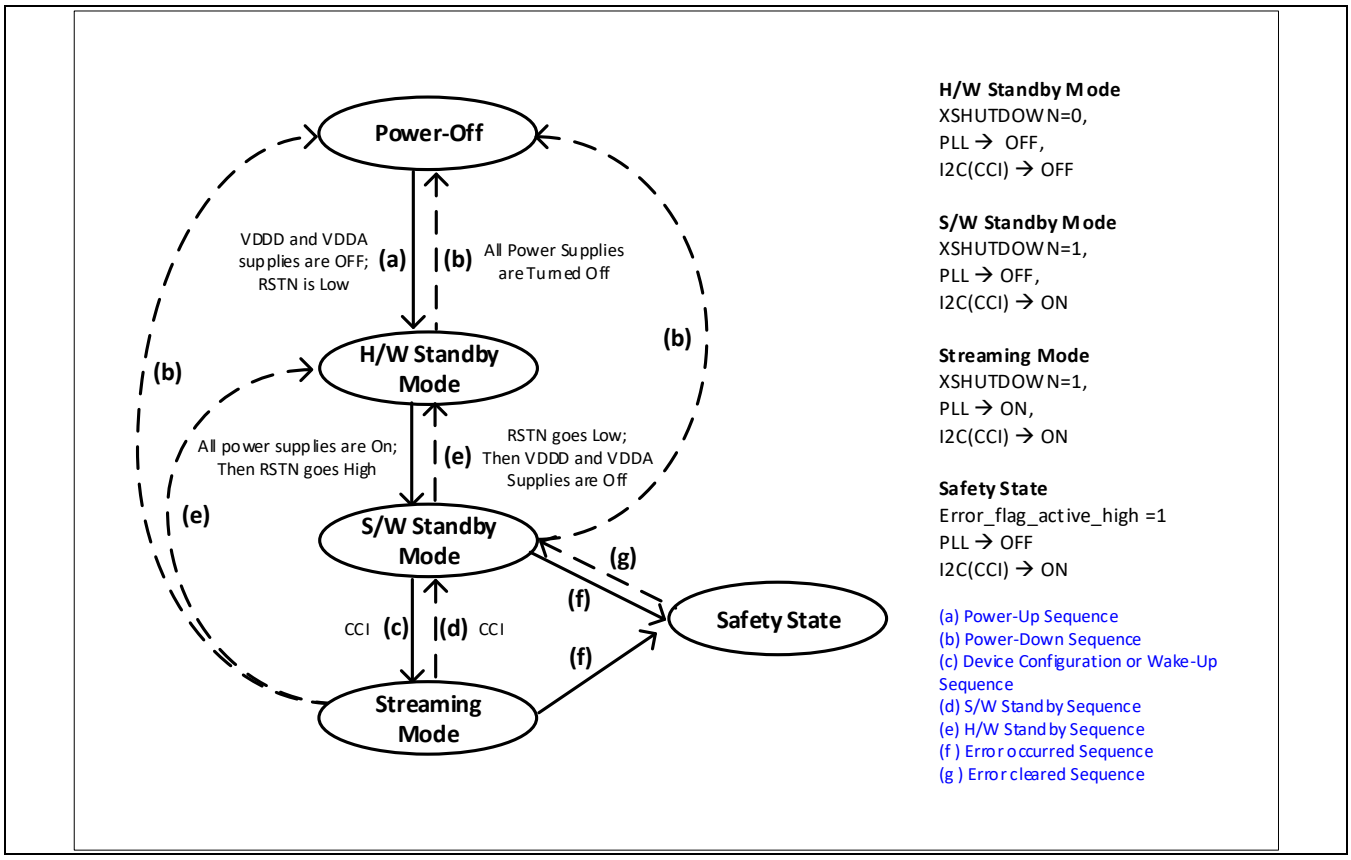


Figure 1 State Diagram of System

The following table describes the registers that control the operating mode of the camera module.

Table 2 Operating Mode Registers

Start	Reset	Name	Type	Width	Description
0x0100	0x0000	tgr_streaming_en	RW	[0:0]	0 = Software Standby 1 = Streaming (Active Video)
0x0102	0x0000	tgr_streaming_en2	RW	[0:0]	stream enable for double lock mechanism

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

2.2 Power-Up Sequence

Figure 2 and Figure 3 illustrate the diagrams of power-up sequence without and with security IP. The digital and analog supply voltages can be powered up in any order, such as first VDDD and then VDDA or first VDDA and then VDDD. After the digital and analog voltage supplies are powered up, RSTN (XSHUTDOWN) should be low, and the sensor module transitions to hardware standby mode. Until RSTN is low and VDDD is down, the sensor module stays in hardware standby mode.

The assertion of RSTN ensures that the CCI register values are initialized correctly to their default values. When RSTN will go 'high', all digital related PADs will exit from FAIL-SAFE mode, and transitions to normal operating mode. The MCLK can be free running or gated before it is enabled during software standby mode.

Table 3 and Table 5 describe the timing constraints of power-up sequence without and with security IP.

NOTE: 1. In hardware standby mode, external VDDD and VDDA should be turned off.
2. To reduce the power consumption, it is recommended to turn off the external clock (MCLK) in software standby state after POST operation done.
3. Maximum time of "POST done to OTP" includes whole data transfer for Security, Bad Pixel Replacement feature.

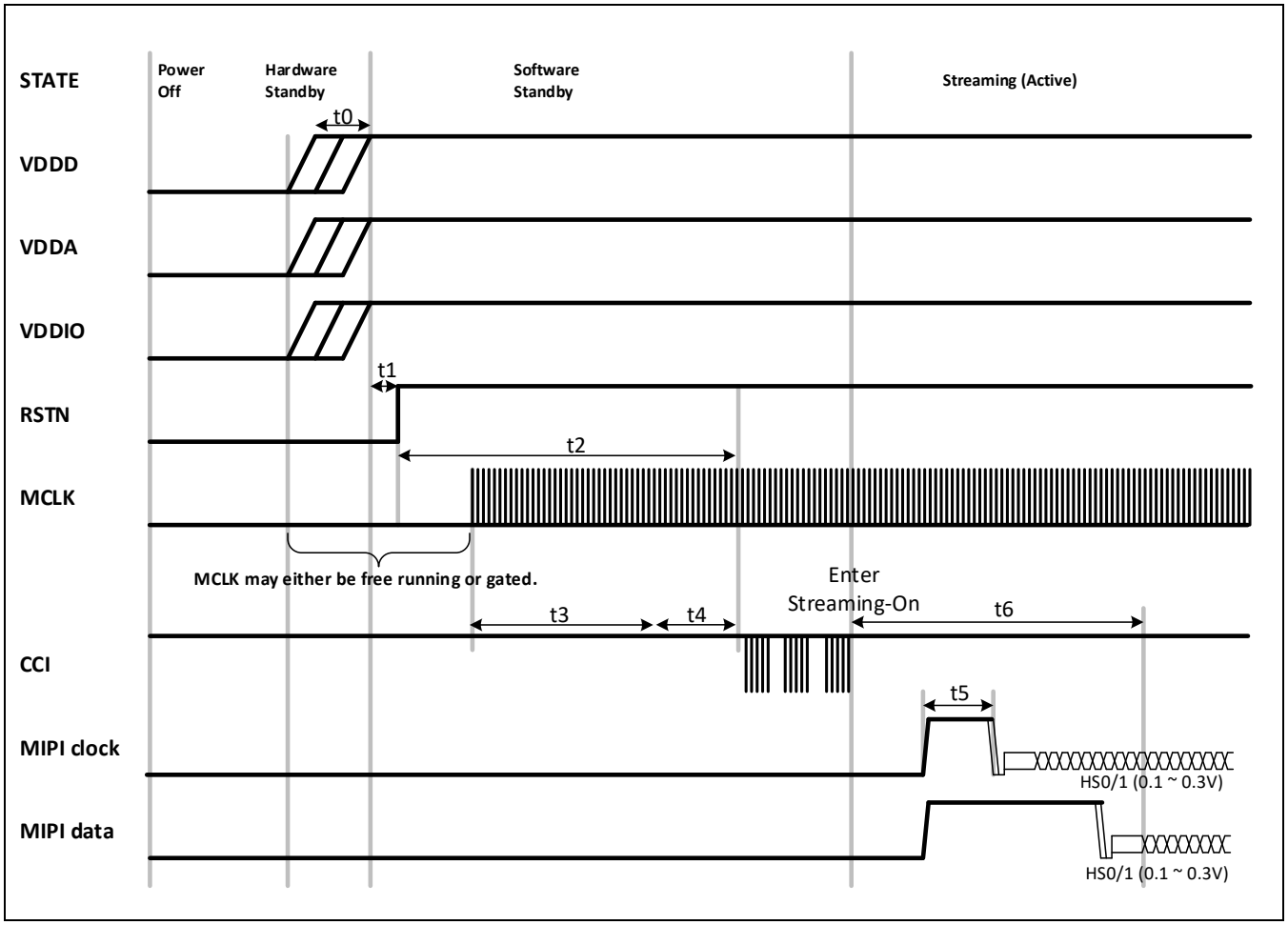


Figure 2 Power-Up Sequence Without Security IP

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Table 3 Power-up Sequence Timing Constraints

Constant	Label	Min.	Max.	Unit
VDDA//VDDD/VDDIO rising time	t0	VDDA/VDDD/VDDIO may rise in any order. The rising separation can vary from 0ns to indefinite.		ns
VDDD rising to RSTN (XSHUTDOWN) rising	t1	0.0	–	ns
RSTN (XSHUTDOWN) rising to first CCI transaction	t2	0.95	27.34	ms
RSTN rising to POST operation. * *POST - Power On Self Test for Memory (External Clock 27Mhz)	t3	0.63 (Disable LBIST,MBIST)	17.66	ms
POST done to OTP operation	t4	0.32	9.68	ms
DPHY initialization period (TINIT)	t5	0.1	–	ms
Entering streaming mode-The first frame start sequence	t6	Internal readout delay + coarse integration time value		–

NOTE: 1. Timing Costraints for General Operation(t1~t6) are same as Power-Up Sequence without Security IP.
2. t7~t9 are required additionally for security function.
3. Image streaming can start before t7, since the additional sequence of RSA decryption(Session Key sharing) can be done also after image streaming. In this case, data authentication, which is only available after session key sharing, starts from the next frame after RSA decryption is finished.
4. *Table 4* I2C timing values area estimated assuming to use I2C Fast Mode.
5. It is recommended that VDDIO rises within 1 second after raising VDDD, because there is no problem with the operation but leakage current may be able to flow.

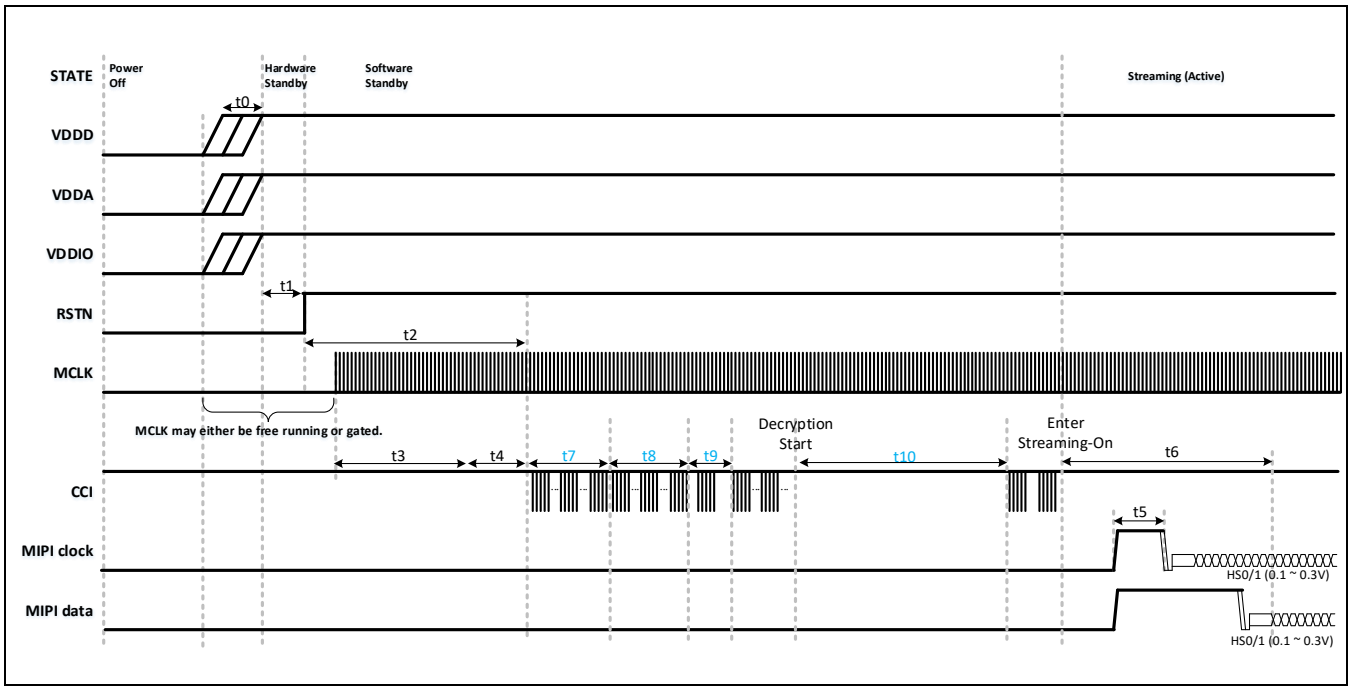


Figure 3 Power-Up Sequence with Security IP

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Table 4 Timing Constraints of Power-Up Sequence with Security Function Enabled

Period	Operation	Time	Description
t7	OTP read by ECU via I2C	RSA 3k: 59.5ms RSA 2k: 40ms	PubK, Certdevice, UID read by ECU to verify RSA Signature
t8	ESsk write by ECU via I2C	RSA 3k: 23.75ms RSA 2k: 16ms	I2C-write of ESsk(ciphertext) by ECU
t9	OTP re-load to RSA memory	RSA 3k: 938us RSA 2k: 625us	PubK & PrivK-load to RSA memory
t10	RSA decryption	RSA 3k: 233.9㎳(128.57㎒) / 1.113s(27㎒;MCLK) RSA 2k: 71.5㎳(128.57㎒) / 340㎳(27㎒;MCLK)	RSA Decryption time including RSA memory-read

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

2.3 Power-Down Sequence

The digital and analog supply voltages can be powered down in any order, such as first VDDD and then VDDA or first VDDA and then VDDD. The sequence diagram and timing constraints of power-down describe *Figure 4* and *Table 5* respectively.

Similar to the power-up sequence, the MCLK (EXTCLK): input clock may be either gated or continuous.

If the CCI command to exit streaming is received while a frame of valid active data is being output, then the sensor module must wait for the frame end code before entering software standby mode. Frame end code may come either after all frame pixels were transmitted or during the frame when next line transmission is completed - based on a configuration register (Refer to *3.5* for details)

If the CCI command to exit streaming mode is received during the inter frame time, then the sensor module must enter software standby mode immediately.

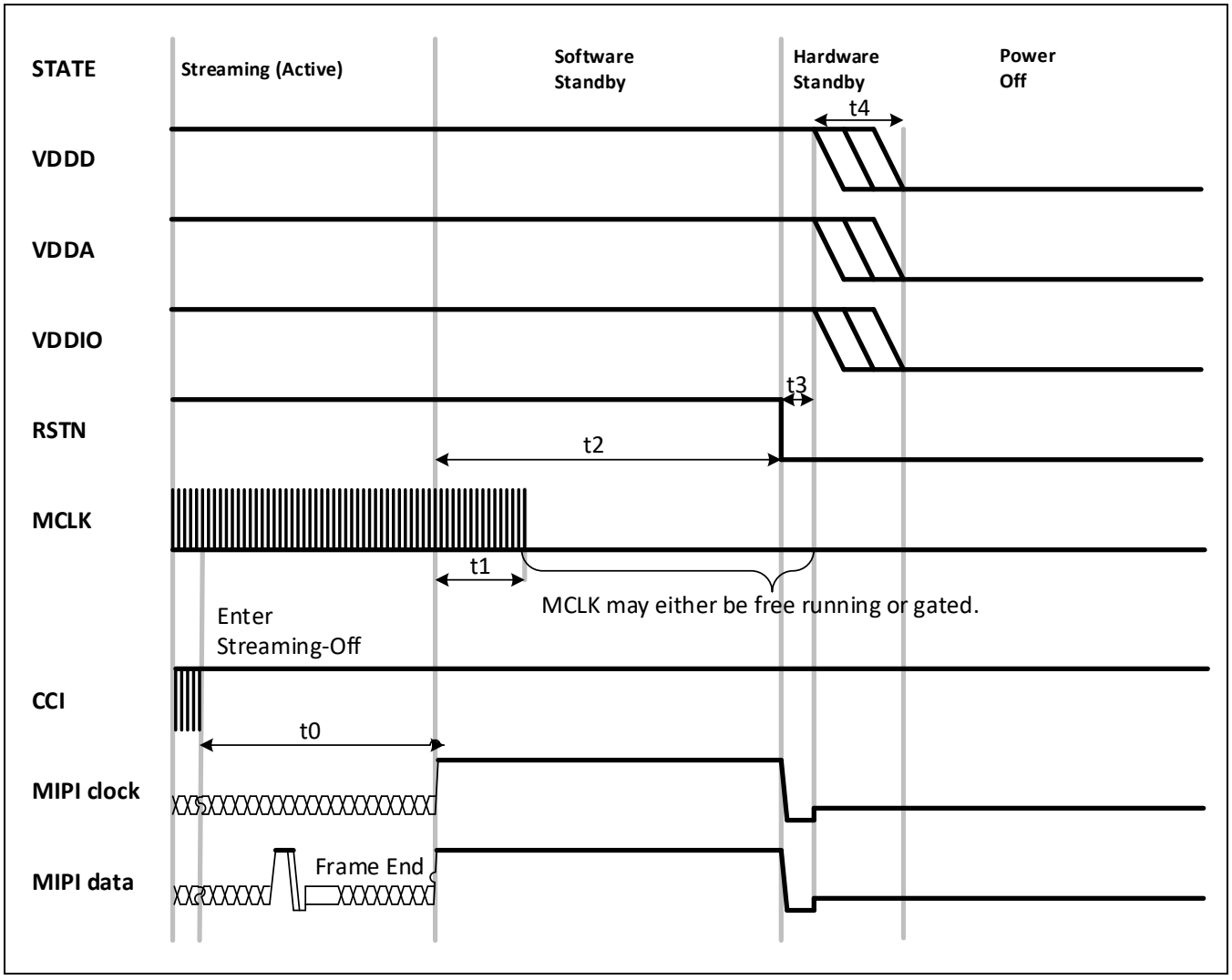


Figure 4 Power-Down Sequence

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Table 5 Power-Down Sequence Timing Constraints

Constant	Label	Min.	Max.	Unit
Enter Software Standby CCI command- Device in Software Standby mode	t0	If outputting a frame of MIPI, the data waits for the MIPI frame end code before entering software standby; otherwise enter software standby mode immediately.		–
Minimum number of MCLK (EXTCLK) cycles after the last CCI transaction or MIPI frame end code.	t1	512	–	MCLK cycles
Last CCI Transaction or MIPI frame end code - RSTN (XSHUTDOWN) falling	t2	512	–	
RSTN (XSHUTDOWN) falling to VDDD or VDDA falling	t3	0.0	–	ns
VDDA/VDDA/VDDD/VDDIO falling time	t4	VDDA/VDDD/VDDIO may fall in any order. The rising separation can vary from 0 ns to indefinite		

NOTE: For minimal power at hardware standby mode external VDDD and VDDA should be OFF.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

3 Control Interface (CCI)

3.1 Camera Control Interface (CCI)

S5K3B6SA supports the Camera Control Interface (CCI), which is an I2C Fast Mode+ compatible interface, to control the transmitter. S5K3B6SA always acts as a slave in CCI bus. CCI is capable of handling several slaves in the bus, but multi-master mode is not supported. Typically, only the receiver and transmitter are connected to CCI bus. This ensures a proper S/W implementation.

Typically, CCI is separated from system I2C bus, but I2C-compatibility ensures that it can connect the transmitter to system I2C bus. CCI is a subset of the I2C protocol including the minimum combination of obligatory features for the I2C slave device specified in the I2C specification. Therefore, transmitters complying with CCI specification are connected to system I2C bus. However, it is important to ensure that the I2C masters do not try to utilize these I2C features, which are not supported in transmitters that are complying with CCI specification. Each transmitter conformed to CCI specification has additional features to support I2C.

3.1.1 Data Transfer Protocol

The data protocol is according to I2C standard specified in I2C specification.

3.1.2 Message Format

The S5K3B6SA CCI supports 16-bit index with 8-bit data with basic I2C standard protocol; START condition, slave address with read/write bit, acknowledge from slave, and STOP condition. In read operation, data byte comes from slave till negative ACK is asserted from master. The device address for the sensor is 0010000b.

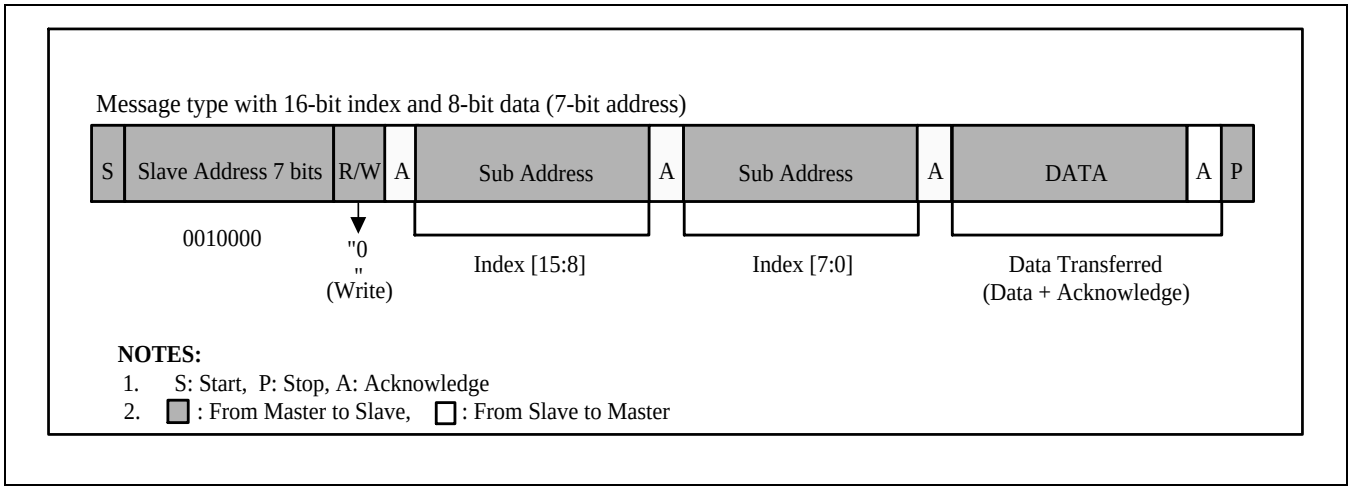


Figure 5 CCI Message Type

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

3.1.3 Read/Write Operation

The S5K3B6SA CCI interface must be able to support four different read operations and two different write operations; single read from random location, sequential read from random location, single read from current location, sequential read from current location, single write to random location and sequential write starting from random location. The read/write operations are presented in the followings. The 16bit index in the slave device has to be auto incremented after each read/write operation.

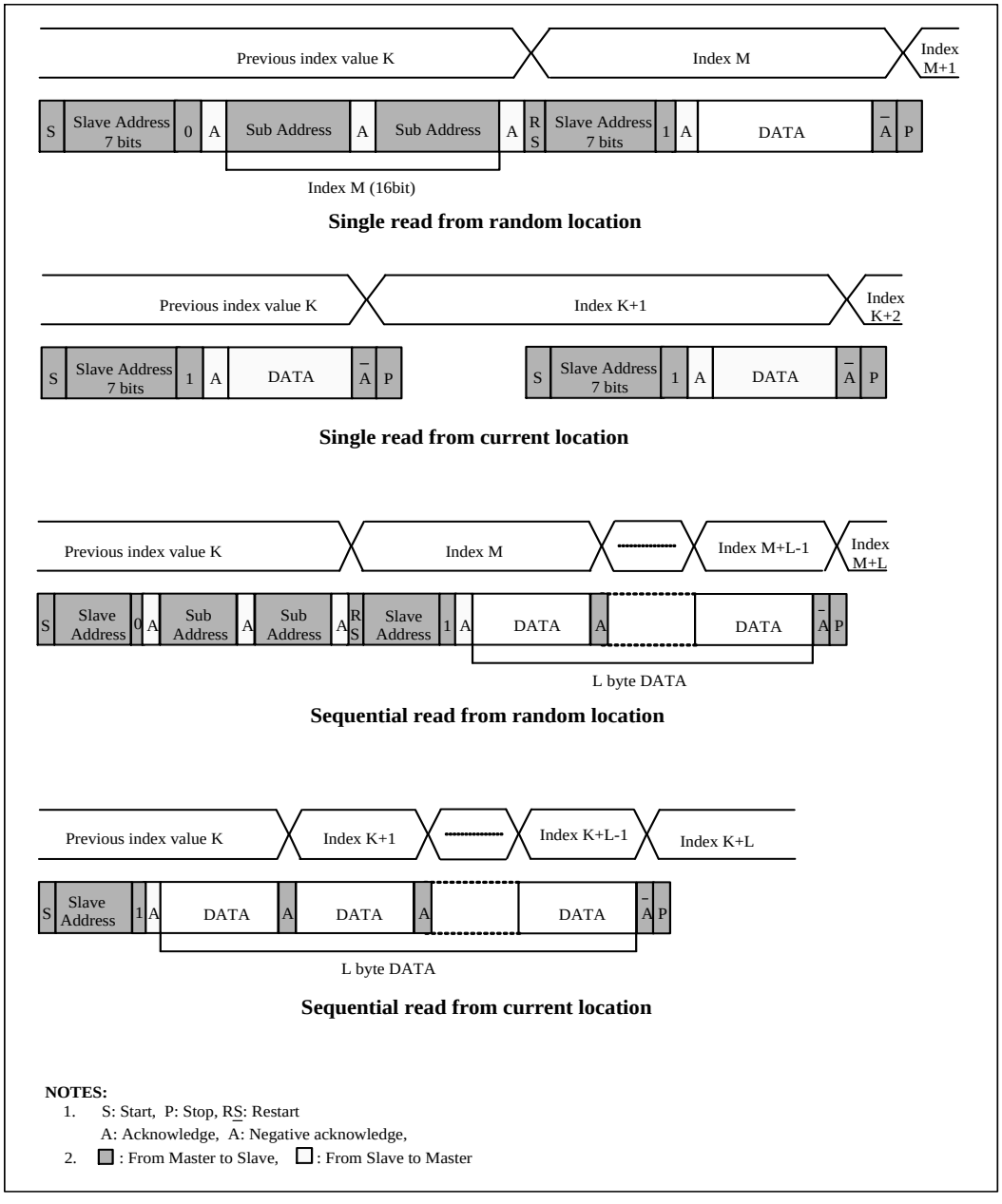


Figure 6 CCI Read Operation

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

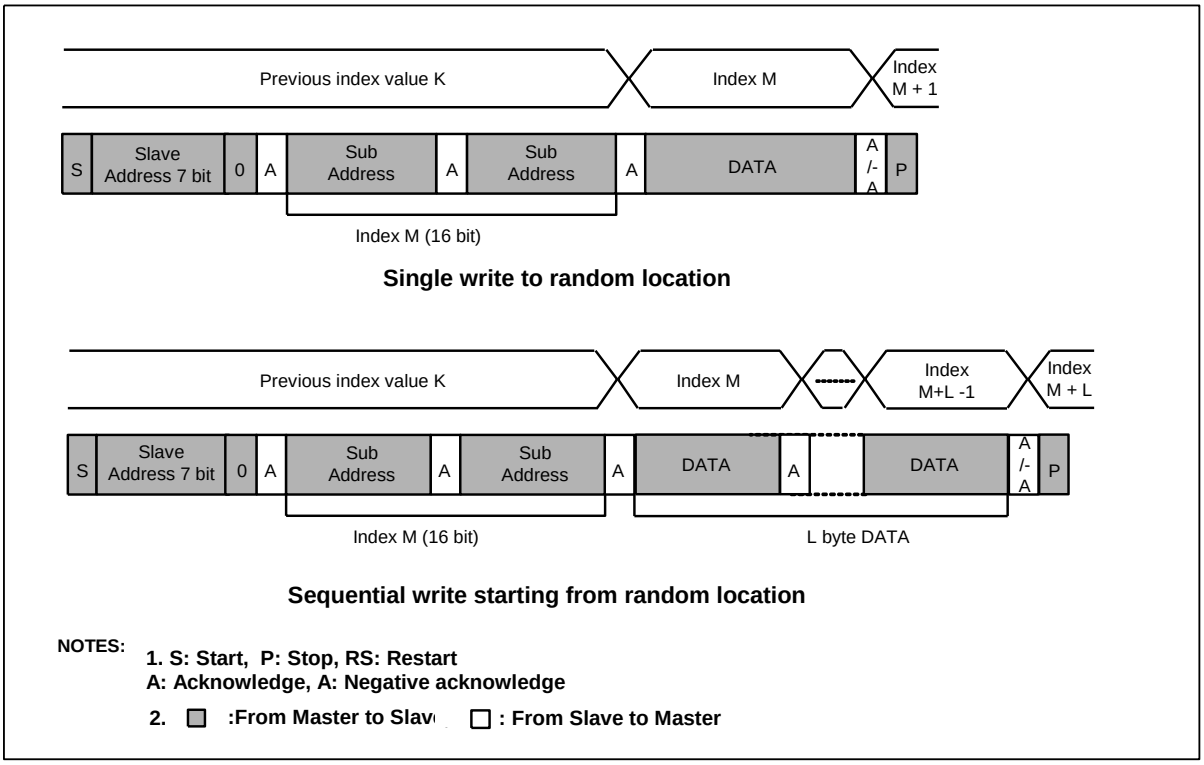


Figure 7 CCI Write Operation

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

3.2 CCI Timing

Table 6, Table 7 and Table 8 describe the timing specification of each I2C mode, standard, fast, and fast mode plus, respectively. And Figure 8, Figure 9, Figure 10 and Figure 11 illustrate the timing diagram of each I2C mode, basic, write, read, and read multiple, respectively.

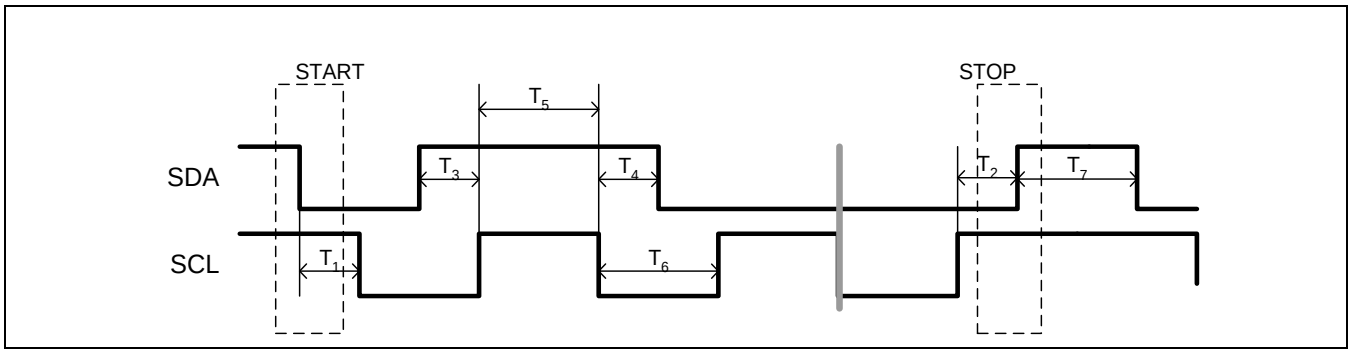


Figure 8 Timing Diagram of CCI

Table 6 I2C Standard Mode Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	-	0	100	kHz
Hold time for start condition	T_1	4.0	-	µs
Setup time for stop condition	T_2	4.0	-	
Data setup time	T_3	250	-	ns
Data hold time	T_4	0	3.45	µs
High period of the SCL clock	T_5	4.0	-	
Low period of the SCL clock	T_6	4.7	-	
Bus free time between stop and start conditions	T_7	4.7	-	
Rise time for both SDA and SCL signals	-	-	1000	ns
Fall time for both SDA and SCL signals	-	-	300	
Capacitive load for each bus line	CB	-	400	pF

Table 7 I2C Fast Mode Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	-	0	400	kHz
Hold time for start condition	T_1	0.6	-	µs
Setup time for stop condition	T_2	0.6	-	
Data setup time with external clock (MCLK) above 15 MHz	T_3	0.6	-	µs
Data hold time	T_4	0	0.9	

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

High period of the SCL clock	T_5	0.6	-	µs
Low period of the SCL clock	T_6	1.3	-	
Bus free time between stop and start conditions	T_7	1.3	-	
Rise time for both SDA and SCL signals	-	-	300	ns
Fall time for both SDA and SCL signals	-	-	300	
Capacitive load for each bus line	CB	-	400	pF

Table 8 I2C Fast Mode+ Timing Specifications

Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	-	0	1000	kHz
Hold time for start condition	T_1	0.26	-	µs
Setup time for stop condition	T_2	0.26	-	
Data setup time, external clock (MCLK) above 15 MHz	T_3	50	-	ns
Data hold time	T_4	0	-	µs
High period of the SCL clock	T_5	0.26	-	
Low period of the SCL clock	T_6	0.5	-	
Bus free time between stop and start conditions	T_7	0.5	-	
Rise time for both SDA and SCL signals	-	-	120	ns
Fall time for both SDA and SCL signals	-	-	120	
Capacitive load for each bus line	CB	-	550	pF

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

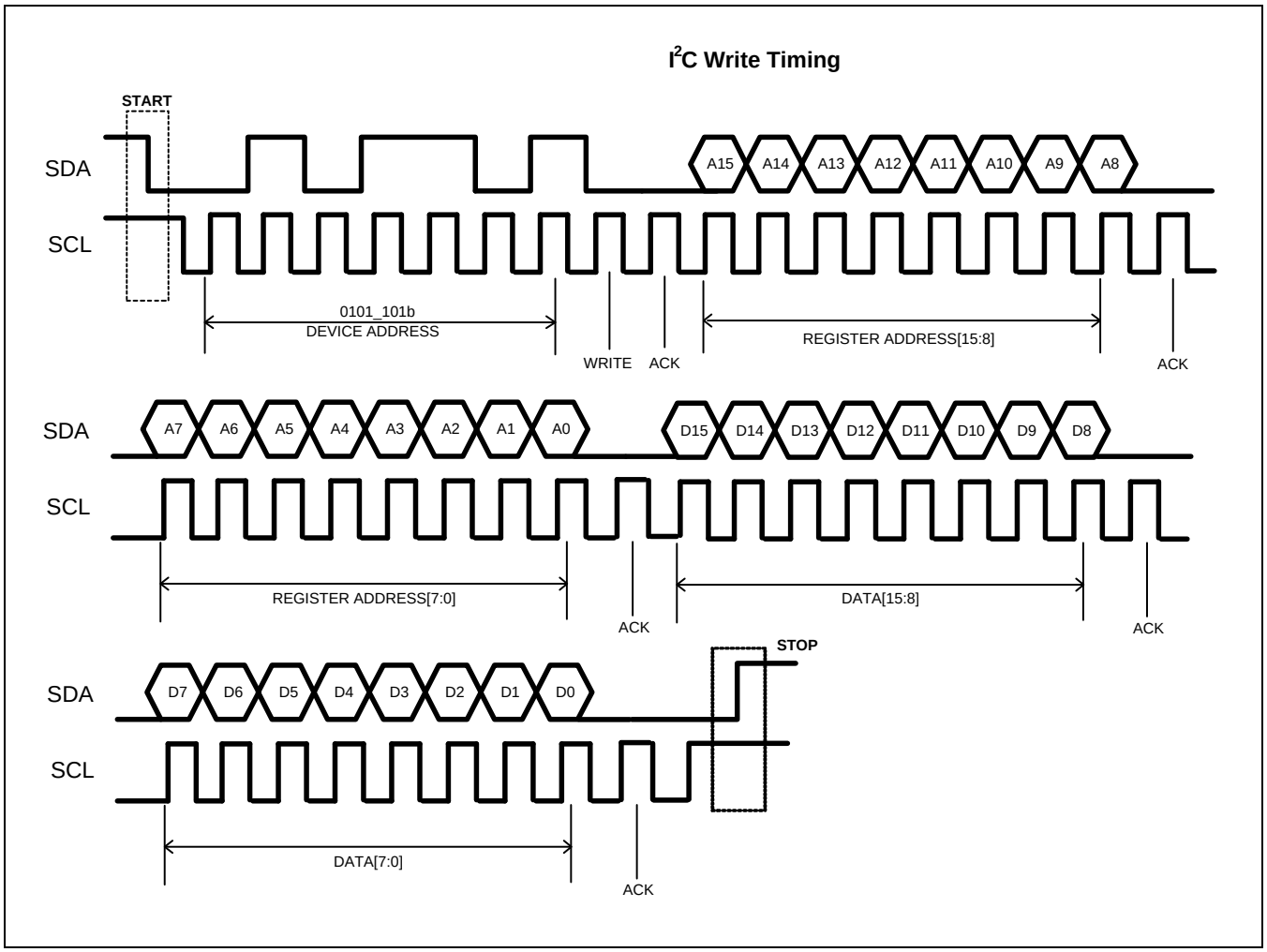


Figure 9 CCI Write Timing Example

NOTE: The device address can be changed by pin configuration of I2C_SEL, as described in the pad description.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

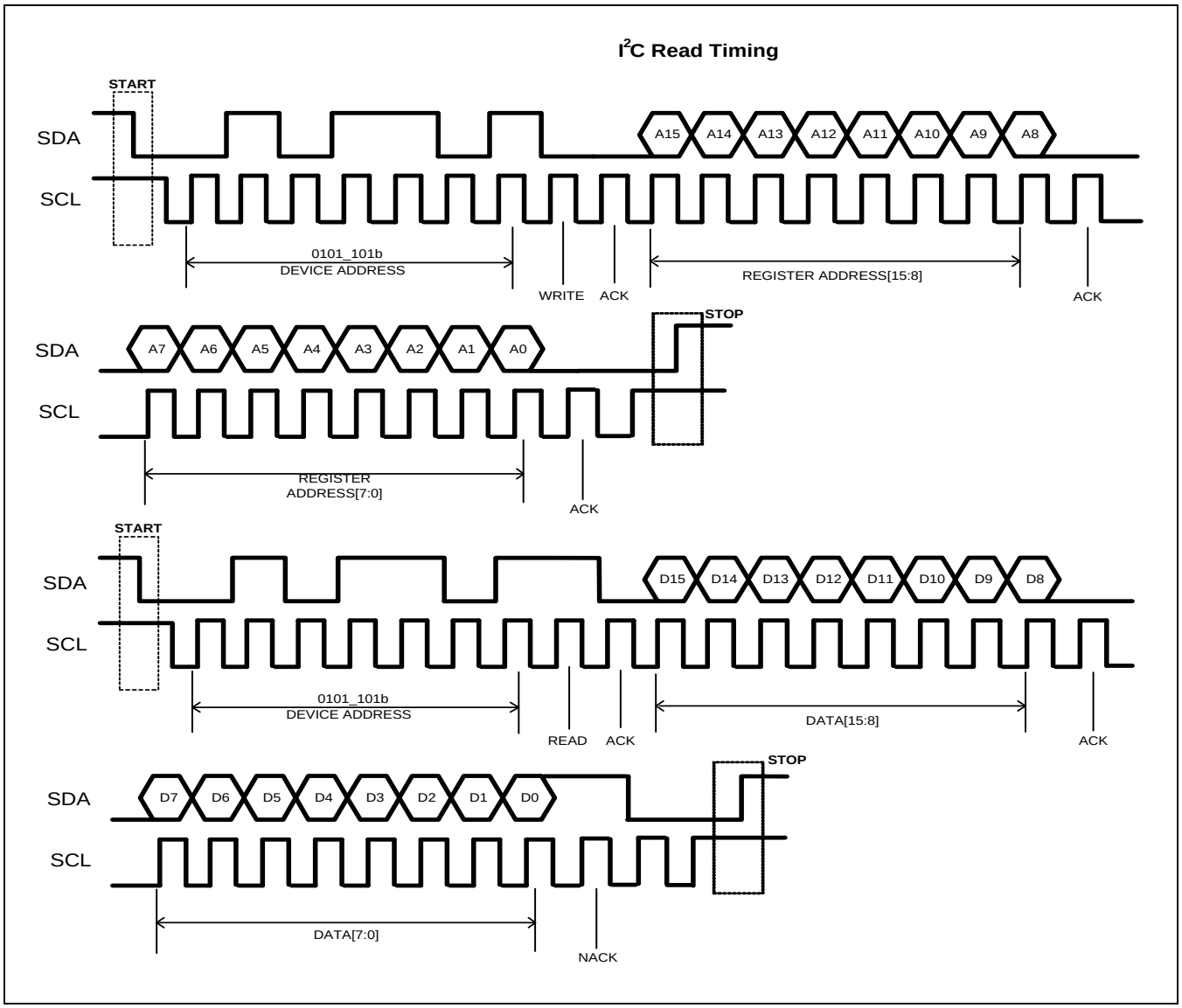


Figure 10 CCI Read Timing Example

NOTE: The device address can be changed by the pin configuration of I2C_SEL, as described in the pad description.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

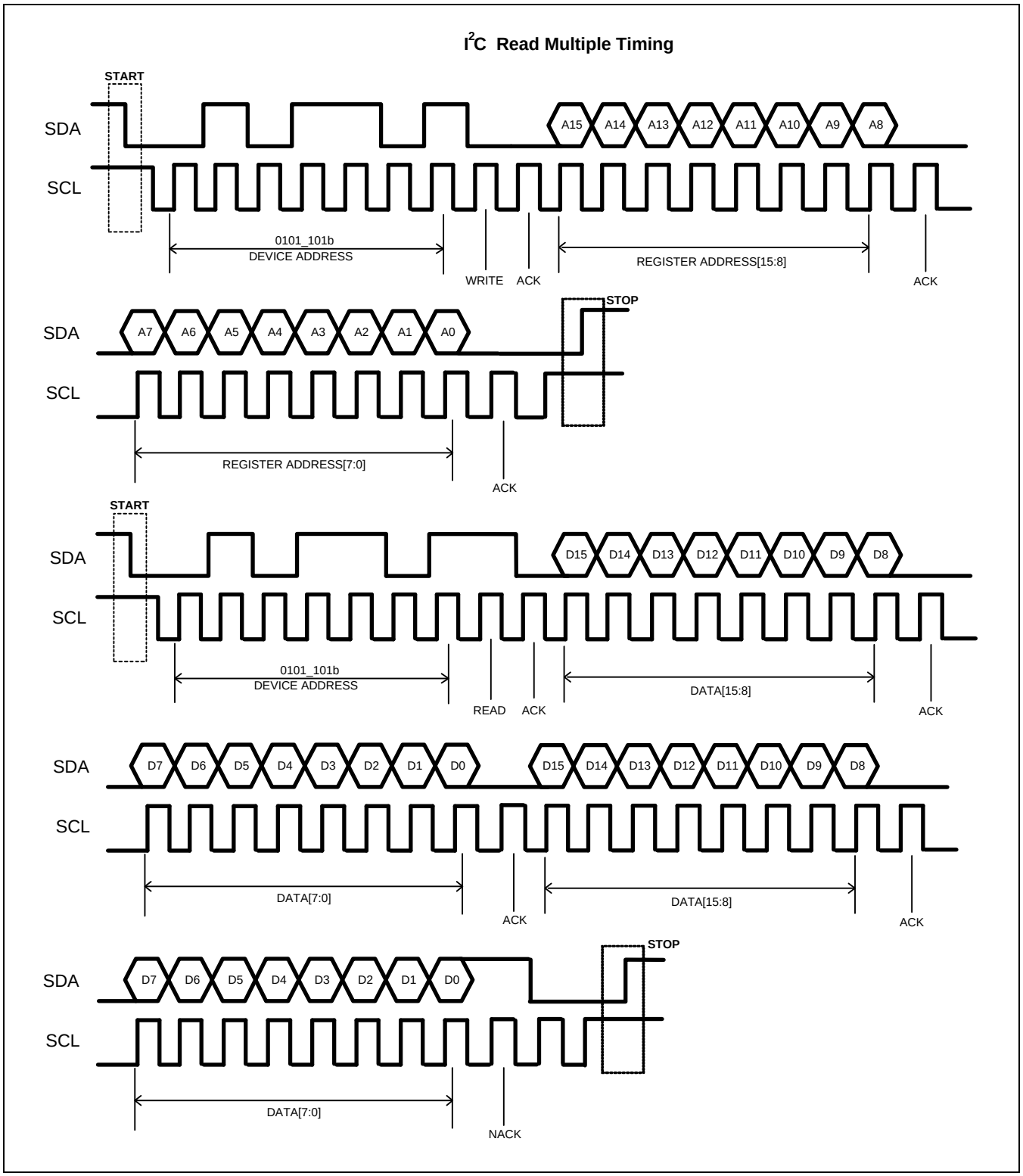


Figure 11 CCI Read Multiple Timing Example

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

3.3 CCI Slave ID

It is possible to configure up to eight I2C slave addresses using I2C_SEL3 ~ I2C_SEL0 pins.

Table 9 I2C ID Address (I2C_SEL3 ~ I2C_SEL0 Pad)

I2C_SEL[3:0]	Slave Address (7-bit + Read Mode)	Slave Address (7-bit + Write Mode)	Comment
0	0010_0001b/21h	0010_0000b/20h	Address 1
1	0011_0001b/31h	0011_0000b/30h	Address 2
2	0010_1001b/29h	0010_1000b/28h	Address 3
3	0011_1001b/39h	0011_1000b/38h	Address 4
4	0110_0001b/61h	0110_0000b/60h	Address 5
5	1010_0001b/A1h	1010_0000b/A0h	Address 6
6	1100_1001b/C9h	1100_1000b/C8h	Address 7
7	1110_1011b/EBh	1110_1010b/EAh	Address 8

3.4 2nd CCI

The S5K3B6SA can support 2nd CCI support. 2nd CCI must have to share I2C lanes and GND.

Table 10 Lane Mode Control Registers

Index	Reset value	Bits	Register name	Type	Description
0x104E	0x0000	[0:0]	sys_i2c_address_sw_ctrl	RW	Enable to set ‘Slave ID’ of I2C by register
0x1050	0x0020	[7:0]	sys_i2c_address_control	RW	Set ‘Slave ID’ of I2C (Only available when sys_i2c_address_control is set to high)
0x1052	0x0000	[1:0]	sys_2nd_i2c_if_control	RW	2nd cci interface control [0] : enable 2nd CCI interface, [1] : enable ACK for 2nd CCI interface
0x1054	0x006C	[7:0]	sys_2nd_i2c_address_control	RW	Set 2nd ‘Slave ID’ of 2nd I2C (Only available when sys_second_i2c_address_control is set to high)

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

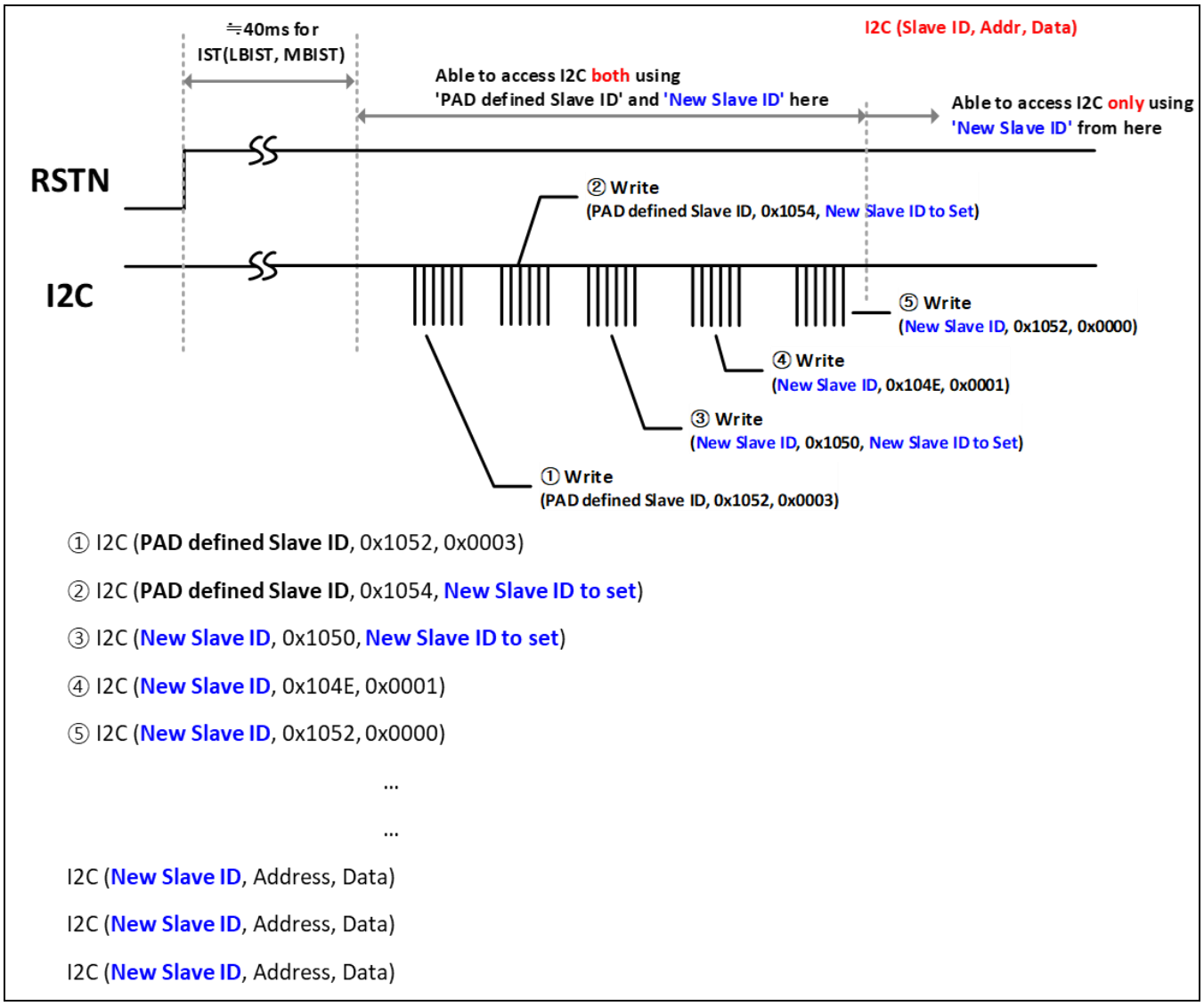


Figure 12 I2C slave ID change via register

3.5 Register Update Timing

3.5.1 Update time.

S5K3B6SA support 3 types of registers for sensor operation. Description for each type of register is following table.

Table 11 Update time description of registers.

Type	Description
RW	Update register right after I2C write. (E.g. : Analog control registers which is not control during data stream.)

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

RW1	Update register at next frame of I2C write time. (E.g. : Integration time ,frame size, HDR , Output interface control registers.)
RW1s	Shadow register of RW1 register – Update register at next frame of I2C write time, RW1S Shadow register applied frame output after 2 frames from register write by I2C.
RW2	Update register at 2 frames later from I2C write time. (E.g. : Analog Gain and Digital Gain registers for the purpose of synchronize with integration time.)
RW2S	Shadow register of RW2 register – RW2S Shadow register applied frame output after 2 frames from register write by I2C.

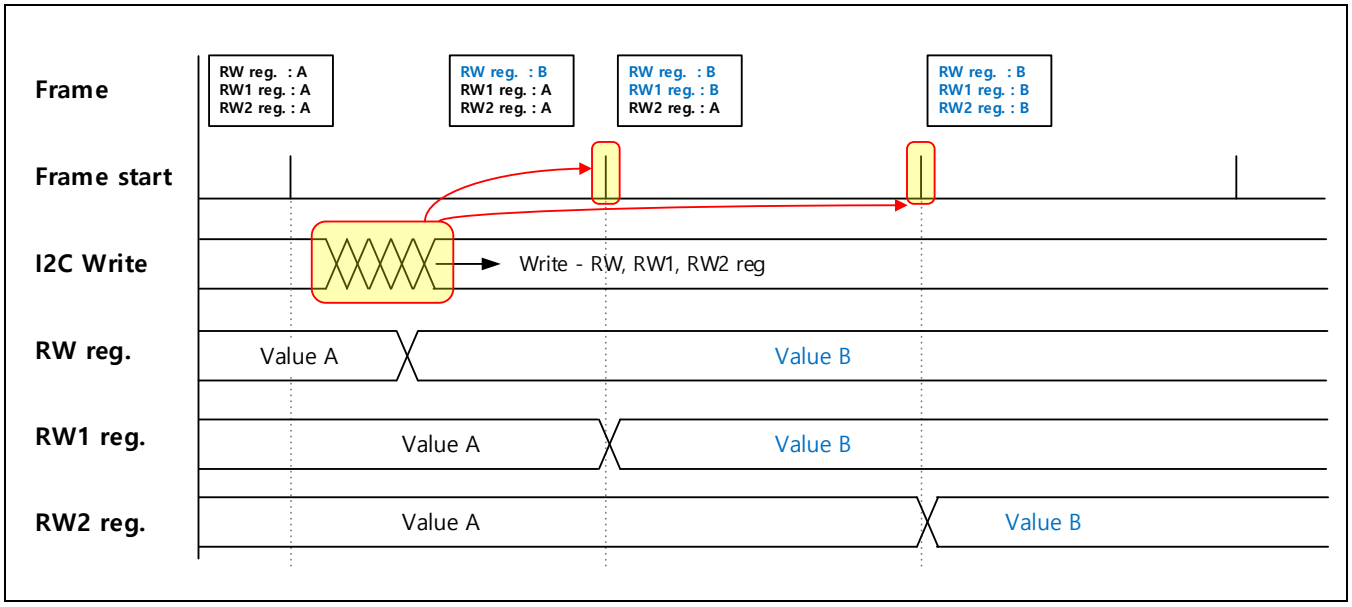


Figure 13 Update timing according to the type of register.

Guide: 1. It is recommended that I2C write should start at MIPI frame start packet

3.5.2 Update timing of AE related registers.

S5K3B6SA has registers to control amount of exposure to get image that has a proper exposure value. Update timing of AE related registers can be referred following diagram.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

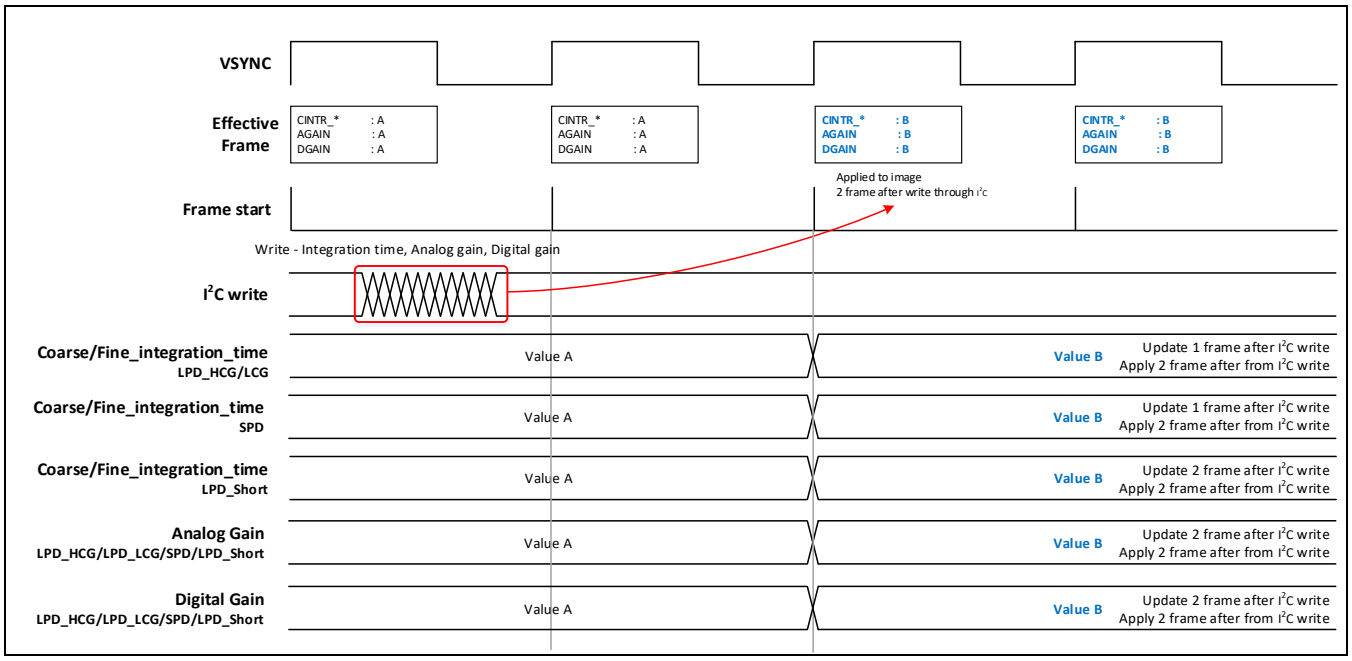


Figure 14 Timing diagram of AE related registers update time

3.5.3 Context Switch

3.5.3.1 Manual control

Context applied frame output after 2 frame from register write by I2C.

Table 12 Manual control of context switch

Index	Reset value	Bits	Register name	Type	Description
0x3016	0x0000	[0]	tgr_shadow_auto_en	RW	1’b1 : Enable automatic control of context switch function. (Controlled by “tgr_shadow_pattern”) 1’b0 : Enable manual control of context switch function. (Controlled by “tgr_shadow_select”)
0x3018	0x0001	[0]	tgr_shadow_enable	RW	1’b1 : Enable control context switching. 1’b0 : Disable control context switching.
0x301C	0x0000	[0]	tgr_ms_shadow_select	RW1	Select which context will be applied. (1’d0 : Context0, 1’d1 : Context1)

NOTE: For manual control of context switching, the registers set to (0x3016[0] = 1d'0, 0x3018[0] = 1d'1, 0x301C = 1d'0/1d'1)

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

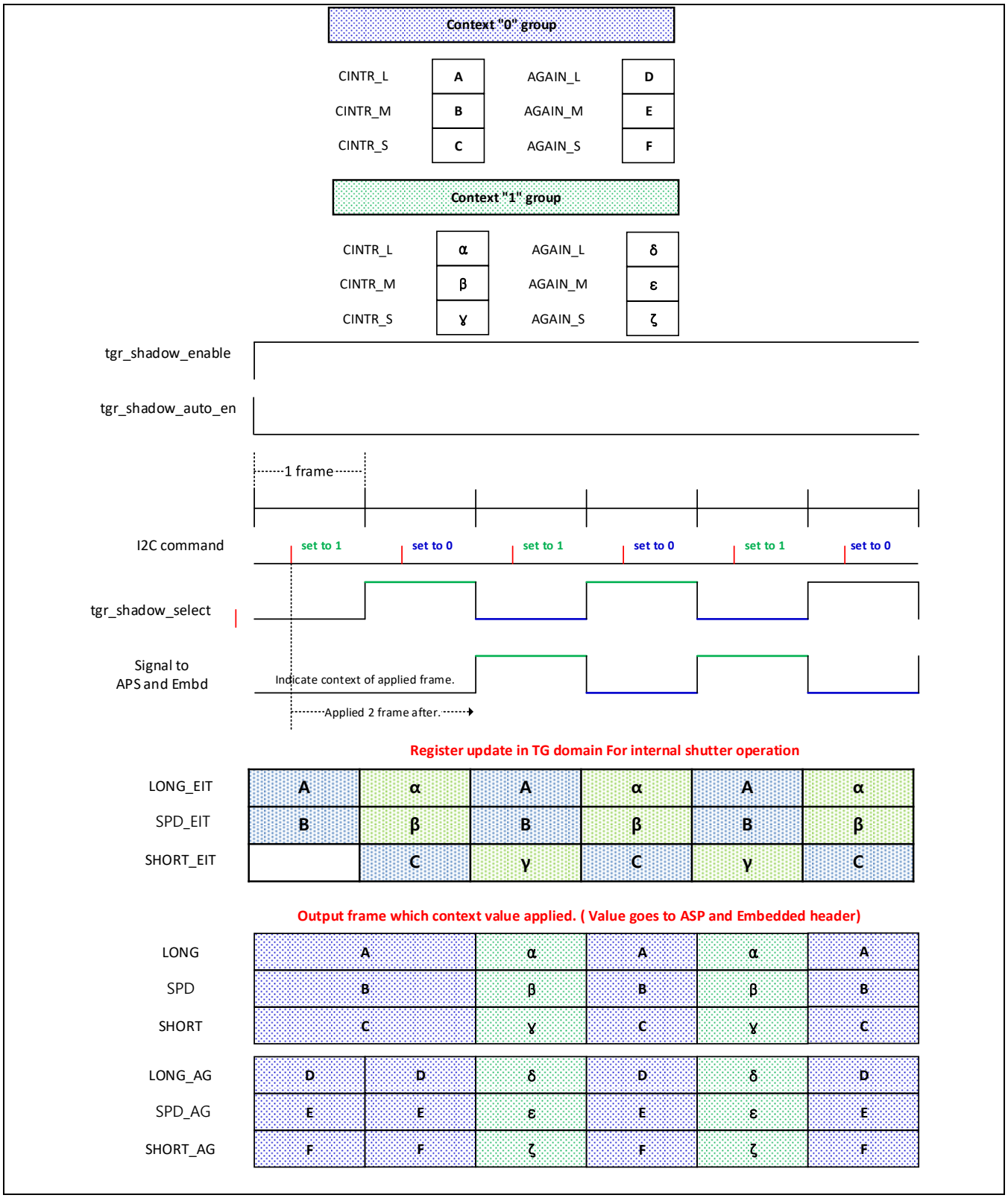


Figure 15 Manual control of context switching

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

3.5.3.2 Automatic control

Context controlled by "tgr_shadow_pattern" register.

Table 13 Automatic control of context switch

Index	Reset value	Bits	Register name	Type	Description
0x3016	0x0000	[0]	tgr_shadow_auto_en	RW	1'b1 : Enable automatic control of context switch function. (Controlled by "tgr_shadow_pattern") 1'b0 : Enable manual control of context switch function. (Controlled by "tgr_shadow_select")
0x3018	0x0001	[0]	tgr_shadow_enable	RW	1'b1 : Enable control context switching. 1'b0 : Disable control context switching.
0x3020	0x0000	[15:0]	Tgr_shadow_pattern	RW	Select frame's shadow pattern. (0: Original, 1 : Shadow) Frame change every 16 cycle refer to the register bit.

NOTE: For manual control of context switching, the registers set to (0x3016[0] = 1d'1, 0x3018[0] = 1d'1, 0x3020 = Set)

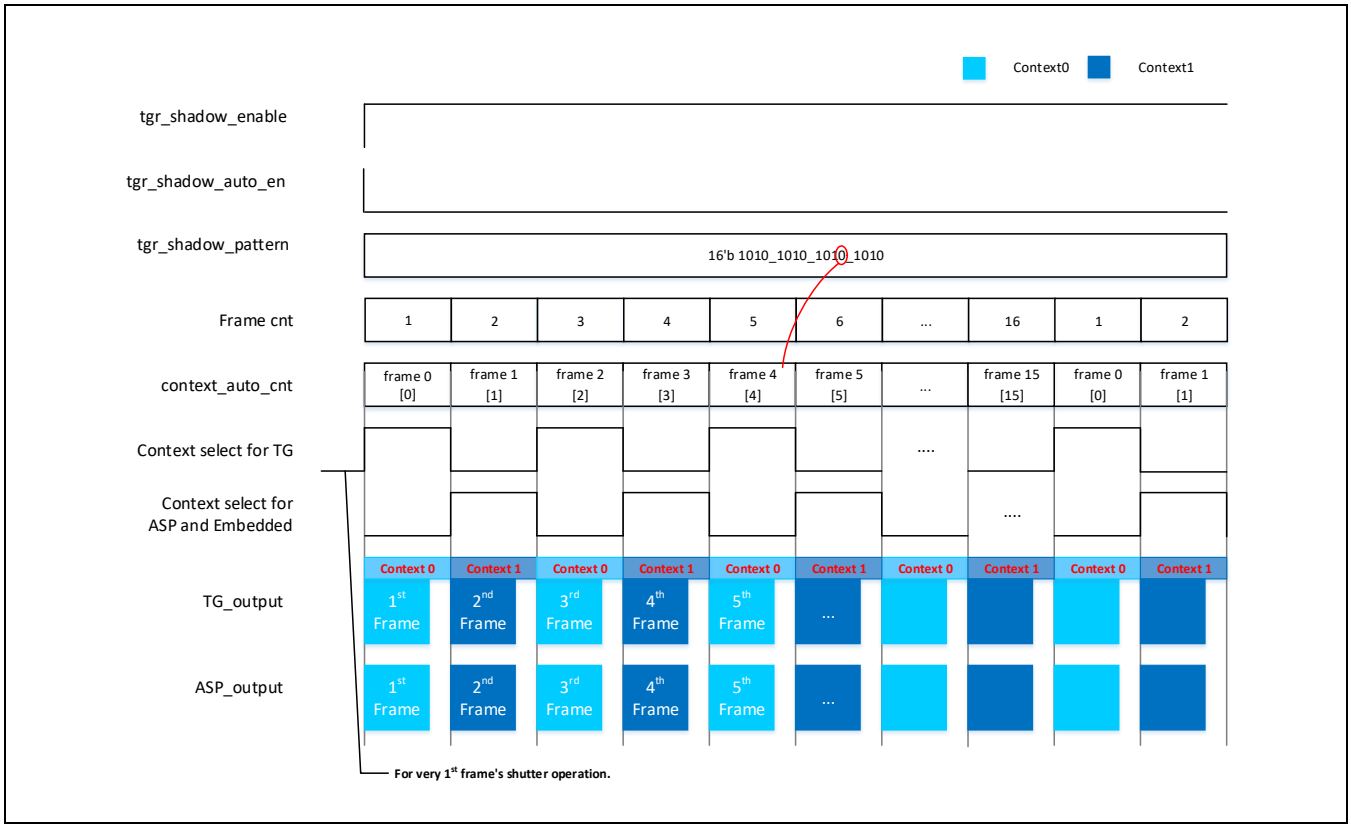


Figure 16 Automatic control of context switching

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

3.5.3.3 Update time of register in Context group

Register in each context group will be apply at the first selected context which after 2 frame.

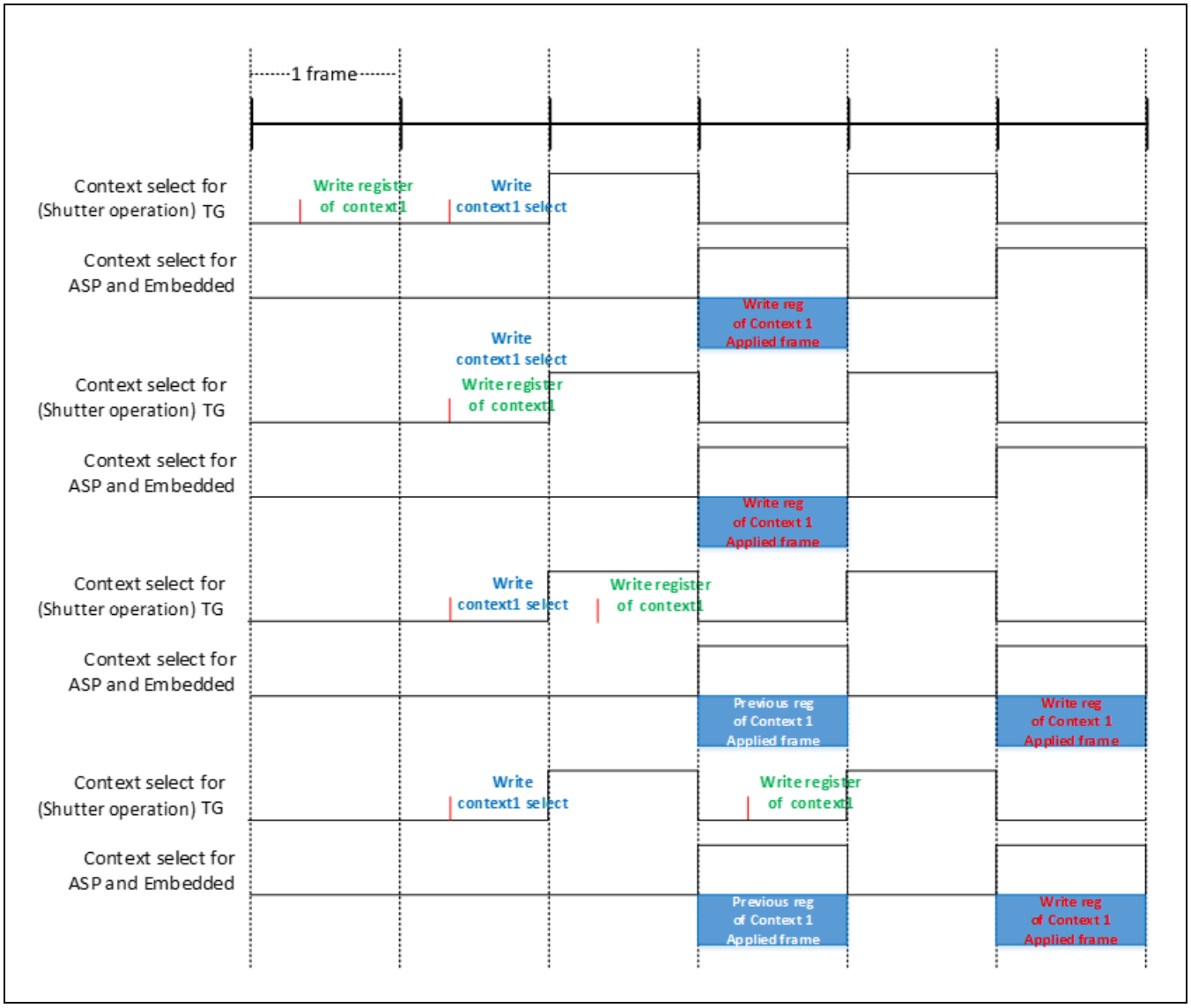


Figure 17 Update time by context selecting

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4 Imager information

4.1 Pixel Array Information and limitation

The effective pixel array is 1920(H)×1536(V). Although, the total pixel array includes different types of pixels, pixel array can be read from (8, 48) to (1927, 1583). The pixels are covered by red, green and blue color filters arranged in RGB Bayer structure. Micro-lens is placed on the top of the color filters to increase the fill factor for higher sensitivity. *Figure 19* is a layout of units for various CFA(Color Filter Array)s.

There is a limitation when set start point and readout area of physical pixel array due to readout scheme and H/W limitation of S5K3B6SA. Start point of Horizontal array and number of readout columns should be multiple of 8 and start point of vertical array and number of readout lines should be multiple of 4. This limitation is also applied when using crop function in horizontal but not in vertical. Set for cropping of vertical lines could be even or odd number and no need to set value as multiple of 4.

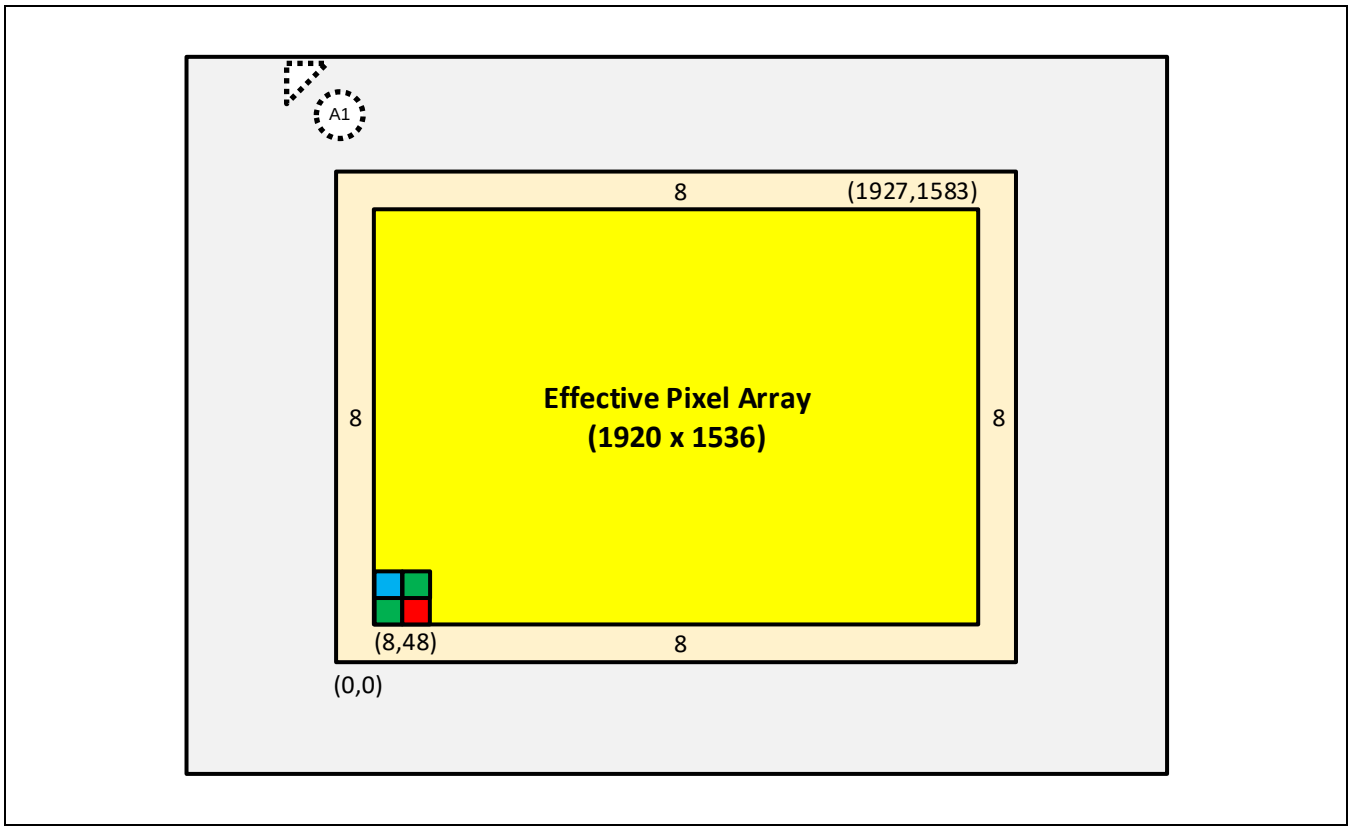


Figure 18 Pixel Array Information

NOTE: For top view of the chip, vertically flips the illustrated image.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

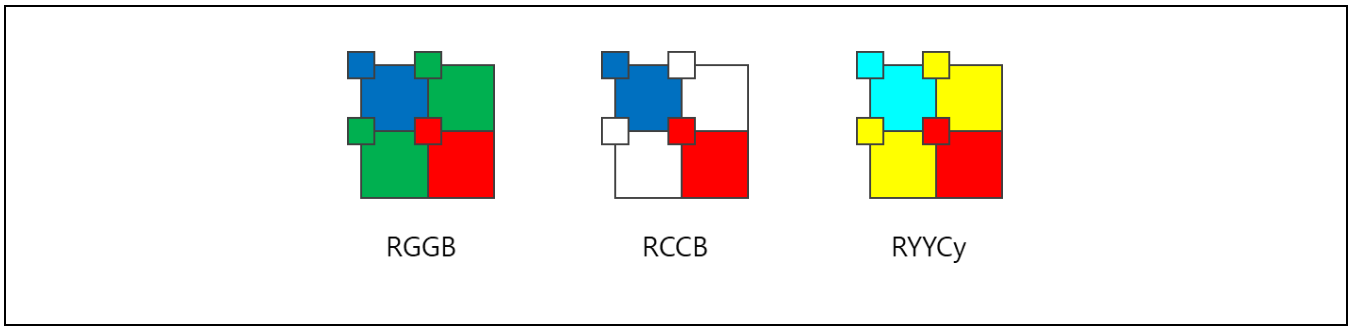


Figure 19 Unit pixel array information

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.2 Imager Classification

4.2.1 Imager Characteristics

All relevant registers for Imager Revision (0x0000 to 0x0008) and interpretation of each register are provided by writing the OTP. *Table 14* describes the relevant registers for imager revision and interpretation of each register. In the case of the ES2 sample of 3B6, the value of 0x0008 becomes 01_010_000b.

Table 14 Relevant registers for imager revision

Address	Field Name	Access	Field Position	Reset Value	Description
0x0002	Sys_model_id	RO	[15:0]	0x3026	Model ID
0x0004	sys_color_filter_array	RO	[3:0]	0x0	- RGGB : 0x0 - RCCG : 0x1 - RCCB : 0x2 - BW : 0x3 - RYYCy : 0x4 - RGBW : 0x5
0x0006	sys_cra	RO	[7:0]	0x03	0x03 : CRA 17°
0x0008	sys_revision_number	RO	[7:6]	0b01	- AS : 00 - ES : 01 - QS : 10 - MP : 11
			[5:3]	0b010	Major revision
			[2:0]	0b000	Minor revision

4.2.2 Chip ID

To distinguish each chip, user can read CHIP ID value set output on system register or Embedded header written in OTP

Table 15 Chip ID

OTP Address <PAGE0>	Sys. Reg. Address (example)	Embedded position by byte [dec] <Header>	Description
0x5A36 [7:0]	0x0014 [7:0]	17	CHIP_ID_0
0x5A36 [15:8]	0x0014 [15:8]	18	CHIP_ID_1
0x5A38 [7:0]	0x0016 [7:0]	19	CHIP_ID_2
0x5A38 [15:8]	0x0016 [15:8]	20	CHIP_ID_3
0x5A3A [7:0]	0x0018 [7:0]	21	CHIP_ID_4
0x5A3A [15:8]	0x0018 [15:8]	22	CHIP_ID_5
0x5A3C [7:0]	0x001A [7:0]	23	CHIP_ID_6
0x5A3C [15:8]	0x001A [15:8]	24	CHIP_ID_7

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5 HDR calibration

S5K3B6SA outputs merged each 4 streams like as *Figure 20* from the Split PD structure which has physically different sensitivity. On merged output, there are 3 transition points and each stream is merged by each normalization factor. For linear final output, all the normalized signals have to be aligned with the same slope, for this reason, HDR calibration is mandatory.

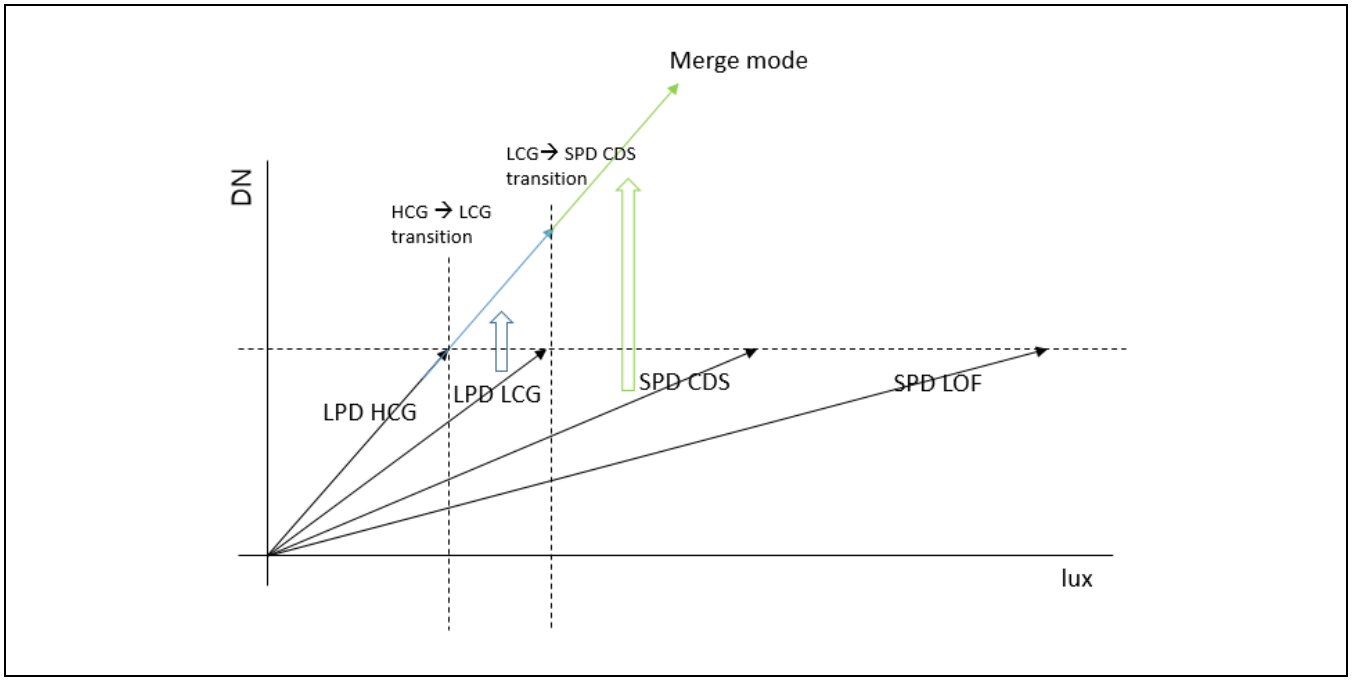


Figure 20 HDR merged output

HDR calibration is needed for two factors, one is the deviation of sensitivity from fab process between each sample and the other is the deviation of sensitivity from different optical condition, e.g. lens on the application. The sample deviation from fab process, i.e. Wafer level calibration is proceeded in Wafer level test, and the HDR ratio data is written in NVM memory. For the deviation from optical condition, checking real HDR ratio and extracting calibration factor is needed with the particular lens. If the lens to be use is determined, send it to the FAE to check the calibration condition.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

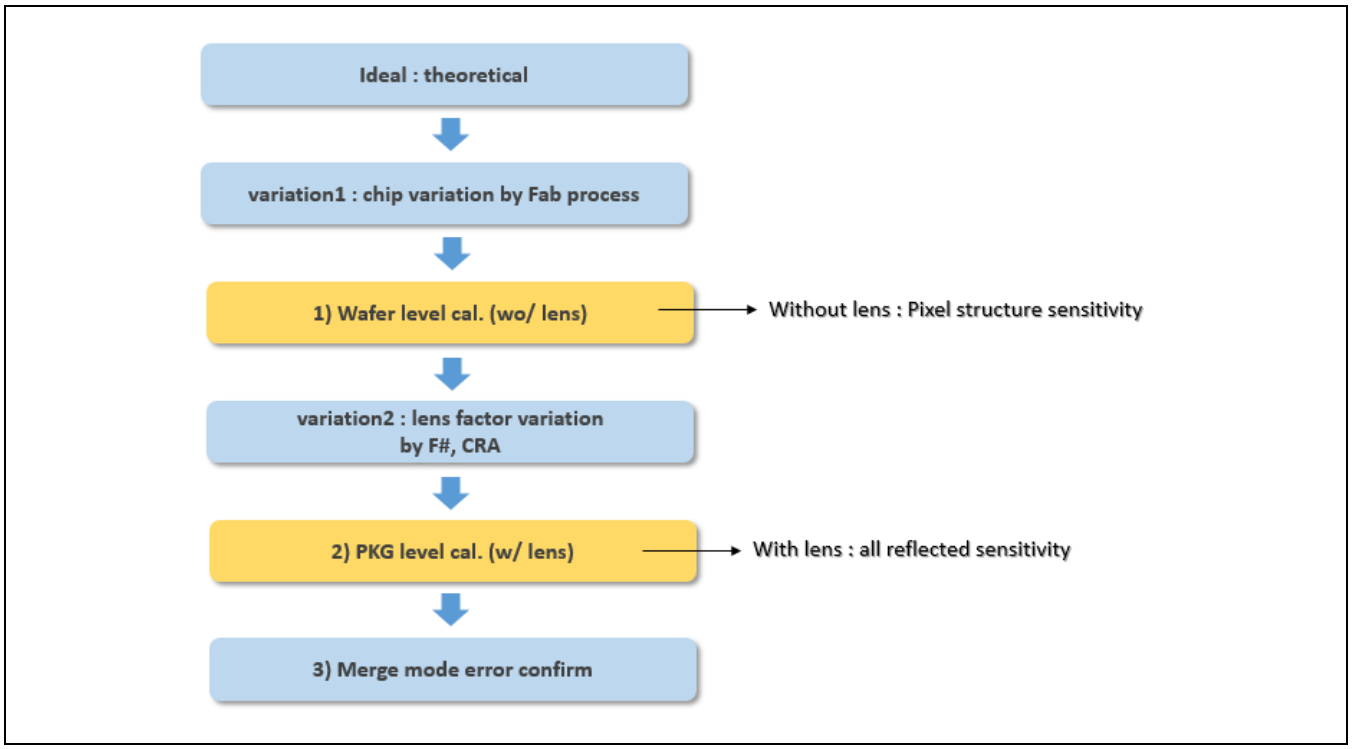
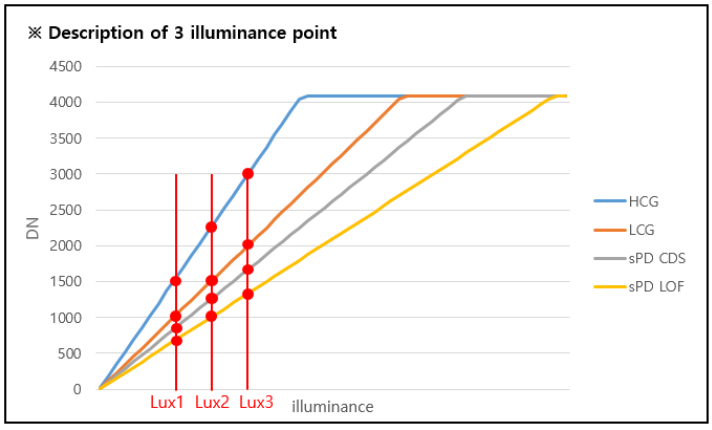


Figure 21 HDR Calibration Process

5.1.1 Calibration Method

In the real application optical condition (i.e. with the application lens), extract real HDR ratio between HCG, LCG and SPD_LOF stream in the condition from each linear output stream. Since there are high deviation of sensitivity between LPD and SPD streams, exposure time control (much lower exposure time for LPD and much higher exposure time for SPD) is recommended for locating all the streams in the linear region.

Basically, calculating HDR ratio from 3 illuminance points. In the ideal case, the HDR ratios from the 3 points have the same value, but for minimizing the deviation, using average value of HDR ratio from 3 points.



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Figure 22 Sampling points for calculating HDR ratio

There are system gain ratios measured and written in Wafer level test process, for all color channels between each stream in NVM memory. By referring to this data, channel gain for compensating the difference of HDR ratio between color channels can be calculated.

Table 16 HDR ratio in OTP

Register Address <OTP PAGE1>	Register Value (example)	Description (example: RCCG)
0x5A04	0x0533	Sys. Gain ratio of HCG : SPD LOF R
0x5A06	0x0138	Sys. Gain ratio of LCG : SPD LOF R
0x5A08	0x053E	Sys. Gain ratio of HCG : SPD LOF Gr
0x5A0A	0x0137	Sys. Gain ratio of LCG : SPD LOF Gr
0x5A0C	0x053B	Sys. Gain ratio of HCG : SPD LOF Gb
0x5A0E	0x0137	Sys. Gain ratio of LCG : SPD LOF Gb
0x5A10	0x051A	Sys. Gain ratio of HCG : SPD LOF B
0x5A12	0x0136	Sys. Gain ratio of LCG : SPD LOF B

NOTE: If user want to be guided the HDR calibration, contact to FAE.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6 Functional Description

This chapter includes the following sections:

- Functional Block Diagram
- PLL and clock generator
- Operation mode
- Slave mode
- Frame readout format
- Image orientation (H/V)
- Frame rate control
- Integration time control (Electronic shutter control)
- Gain control
- Auto dark level correction
- Image output format

S5K3B6SA provides several readout modes and specific functions for video systems and for the required variety of clocks. Two Phase-Locked Loops (PLLs) are embedded in the sensor. One generates clocks for systems, and the other one generates clocks for high-speed serial interfaces. The master input clock range is between 6 MHz and 64 MHz.

6.1 Functional Block Diagram

S5K3B6SA is composed of an APS array, 12-bit ADCs, an embedded Timing Generator (TG), pixel data process blocks, OTP, two Phase-Locked Loop of PLL1 and PLL2, Automotive Signal Processing block which support interleaving, Merge and Compress for HDR data and interface blocks.

The following figure illustrates the functional block diagram of sensor.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

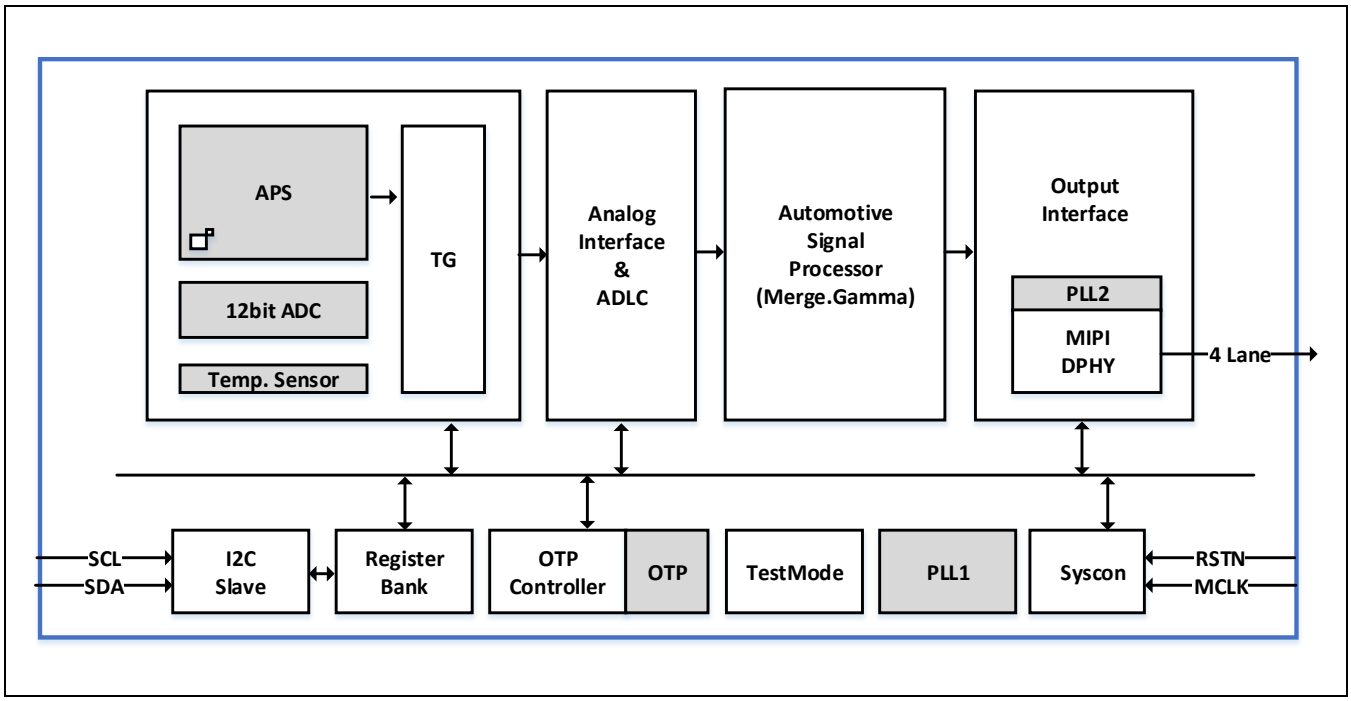


Figure 23 Function Block Diagram

6.2 PLL and Clock Generator

S5K3B6SA contains two PLLs and a clock generator, which generate all the necessary video timing, output pixel clocks, and interface serializing clocks.

By setting the divide-ratio for pre PLL clock divider (`pll_sys_p and pll_int_p`), PLL multiplier (`pll_sys_m and pll_int_m`), and post scaler (`pll_sys_s and pll_int_s`) appropriately, User can get the necessary PLL output clocks (`PLL_OUT1 and PLL_OUT2`).

PLL can handle any external input clock in the range of 6 MHz to 64 MHz (27 MHz is default), and synthesize PLL_OUT1 between 40 MHz to 2500 MHz and PLL_OUT2 between 70.3125 MHz to 4500 MHz. For proper PLL operation, PLL input clock (`PLL_IN1 and PLL_IN2`) should be in the range of 6 MHz to 12 MHz. All PLL programming should be performed during software standby mode for stable system operation.

MIPI output is Double Data Rate (DDR) interface that uses MIPI word clock. Hence, the MIPI word clock is 1/16th of the bit clock. Following figure illustrates the overall clock tree structure.

Interface PLL(PLL2) of S5K3B6SA is able to support SSCG(Spread Spectrum Clock Generation) to avoid EMI issues.

Recommend input frequency for valid InSysTem MBIST, LBIST operation.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

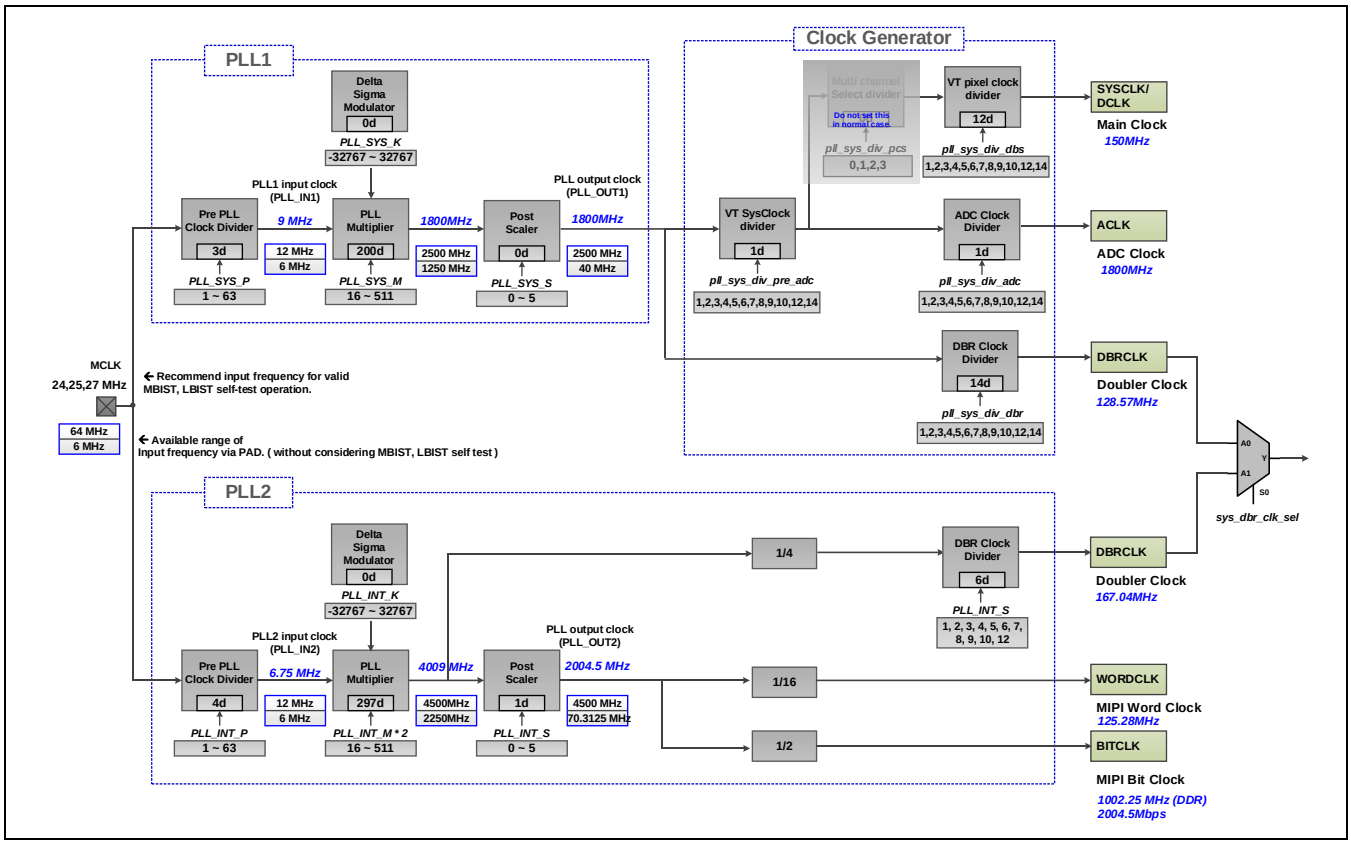


Figure 24 PLL and Clock Generator

NOTE: All the register setting values are represented in decimal unit.

6.2.1 Clock Relationships

The host provides the external input clock (with values varying between 6-64MHz), in addition to setting dividers and multiplier in order to get the required video timing and output pixel clocks.

Use the following equations to synthesize all the necessary frequencies.

PLL1 Input Clock = External Input Clock (MCLK)/pll_sys_p (6 MHz to 12 MHz)
PLL2 Input Clock = External Input Clock (MCLK)/pll_int_p (6 MHz to 12 MHz)
PLL_OUT1 = (PLL_IN1×(pll_sys_m+pll_sys_k/65536))/2^pll_sys_s (40 MHz to 2500 MHz)
PLL_OUT2 = (PLL_IN2×(pll_int_m+pll_int_k/65536)×2)/2^pll_int_s
(70.3125MHz to 4500 MHz, Bit clock divided by 2 for DDR)
1 ≤ pll_*_p ≤ 63 : 16 ≤ pll_*_m ≤ 511 : 0 ≤ pll_*_s ≤ 5 : -32767 ≤ pll_*_k ≤ 32767

$$F_{ref1} = MCLK \div sys_P, \quad F_{VCO1} = F_{ref} \times \mathrm{sys_M} + \left(\frac{sys_K}{2^{16}}\right), \quad F_{PLL_OUT1} = F_{VCO1} \div 2^{sys_S}$$

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$$sys_clk = F_{PLL_OUT1} \div \text{sys_div_pre_adc} \div \text{sys_div_dbs}$$

NOTE: For more information about the PLL and clock system control, contact to FAE.

Table 17 PLL Component Output Frequency

Parameter	Min.	Typ.	Max.	Unit	Remarks
Input Frequency Range	6	–	64	MHz	External input clock frequency range
Reference Frequency Range	6	–	12	MHz	Output of pre divider (F_{ref1})
VCO Frequency Range	1250	–	2500	MHz	Output of PLL multiplier VCO oscillation range (F_{VCO1})
PLL Output Frequency Range	40	–	2500	MHz	Output of PLL post scaler (F_{PLL_OUT1}) Minimum value is just for test. (S ≤ 5)

NOTE: For more information about the PLL and clock system control, contact to FAE.

Example of main PLL calculation:

((27M[external clock] / 3[pre-divider]) × (200[multiplier] + (0[DSM] / 65536)) / 1[2^0post scalar] = 1800M[PLL output]

Table 18 System PLL and Clock divider Set-up Capability Registers

Index	Reset value	Bits	Register name	Type	Description
0x101C	0x0003	[5:0]	sys_pll_sys_P	RW	System PLL pre-divider
0x101E	0x00C8	[9:0]	sys_pll_sys_M	RW	System PLL multiplier
0x1020	0x0000	[2:0]	sys_pll_sys_S	RW	System PLL Scaler
0x1022	0x0000	[15:0]	sys_pll_sys_K	RW	Value of 16bit DSM. [15] is signed bit. Range of value is -32767 ≤ k ≤ 32767.
0x1026	0x0011	[3:0]	sys_pll_sys_div_pre_adc	RW	Division ratio of ADC clock divider. Pre divide ADC and DBUS clock.
0x1028	0x1EC1	[3:0]	sys_pll_sys_div_adc	RW	Division ratio of ADC clock divider. Post divide ADC clock.
		[7:4]	sys_pll_sys_div_dbs	RW	Division ratio of DBS clock divider. Post divide DBUS and System clock.
		[11:8]	sys_pll_sys_div_dbr	RW	Division ratio of DBR clock divider.
		[15:12]	Sys_pll_sys_div_op	RW	Division ratio of OP clock divider.

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$$F_{ref2} = MCLK \div int_P, \quad F_{VCO2} = F_{ref2} \times 2 \times (\mathrm{int_M} + \left(\frac{int_K}{2^{16}}\right)), \quad F_{PLL_OUT2} = F_{VCO2} \div 2^{int_S}$$

NOTE: PLL_INT_CLK is the output clock of PLL2. Data_Rate = F_{PLL_OUT2}

NOTE:

Table 19 PLL Component Output Frequency

Parameter	Min.	Typ.	Max.	Unit	Remarks
Input Frequency Range	6	–	64	MHz	External input clock frequency range
Reference Frequency Range	6	–	12	MHz	Output of pre divider (F_{ref2})
VCO Frequency Range	2250	–	4500	MHz	Output of PLL multiplier VCO oscillation range (F_{VCO2})
PLL Output Frequency Range	70.3125	–	4500	MHz	Output of PLL post scaler (F_{PLL_OUT2}) Minimum value is just for test. (S ≤ 5)

NOTE: For more information about the PLL and clock system control, contact to FAE.

Example of calculation of PLL2 for MIPI:

((27M[external clock] / 4[pre-divider]) × (297[multiplier] × 2 + (0[delta sigma modulator] / 65536)) / 2^1[post scalar] = 2004.75M[PLL output]

Table 20 Output PLL and Clock divider Set-up Capability Registers

Index	Reset value	Bits	Register name	Type	Description
0x102A	0x0004	[5:0]	sys_pll_int_P	RW	Interface PLL pre-divider
0x102C	0x0129	[9:0]	sys_pll_int_M	RW	Interface PLL multiplier
0x102E	0x0001	[2:0]	sys_pll_int_S	RW	Interface PLL Scaler (F_{PLL_OUT2})
0x103A	0x0000	[15:0]	sys_pll_int_K	RW	Interface PLL control registers Value of 16bit DSM. [15] is signed bit. Range of value is -32767 ≤ k ≤ 32767.
0x103E	0x0018	[7:0]	sys_pll_int_ctl4	RW	Interface PLL control registers
		[1:1]	DBR_CLK_EN	RW	
		[5:2]	DIV_DBR	RW	

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.2.2 Spread Spectrum Clock Generation (SSCG)

By register control through I2C, features such as modulation frequency, modulation rate and spectrum method of SSCG can be modified.

Table 21 Registers for System PLL (PLL1)

Index	Reset value	Bits	Register name	Type	Description
0x1010	0x0000	[0:0]	sys_pll_sys_sscg_en	RW	1: to enable pin for dithered mode (active high)
0x1012	0x8100	[7:0]	sys_pll_sys_MFR	RW	0≤mfr≤255
		[13:8]	sys_pll_sys_MRR	RW	0≤mrr≤63
		[15:14]	sys_pll_sys_sel_pf	RW	00:down spread, 01:up spread, 1x : center spread

Table 22 Registers for Interface PLL (PLL2)

Index	Reset value	Bits	Register name	Type	Description
0x1036	0x0080	[15]	sys_pll_int_ctl0	WR	SSCG_EN
0x1038	0x0000	[7:0]	sys_pll_int_MFR	RW	0≤mfr≤255
		[13:8]	sys_pll_int_MRR	RW	0≤mrr≤63
		[15:14]	sys_int_sys_sel_pf	RW	00:down spread, 01:up spread, 1x : center spread

Equation for each of modulation frequency and modulation rate is as follows:

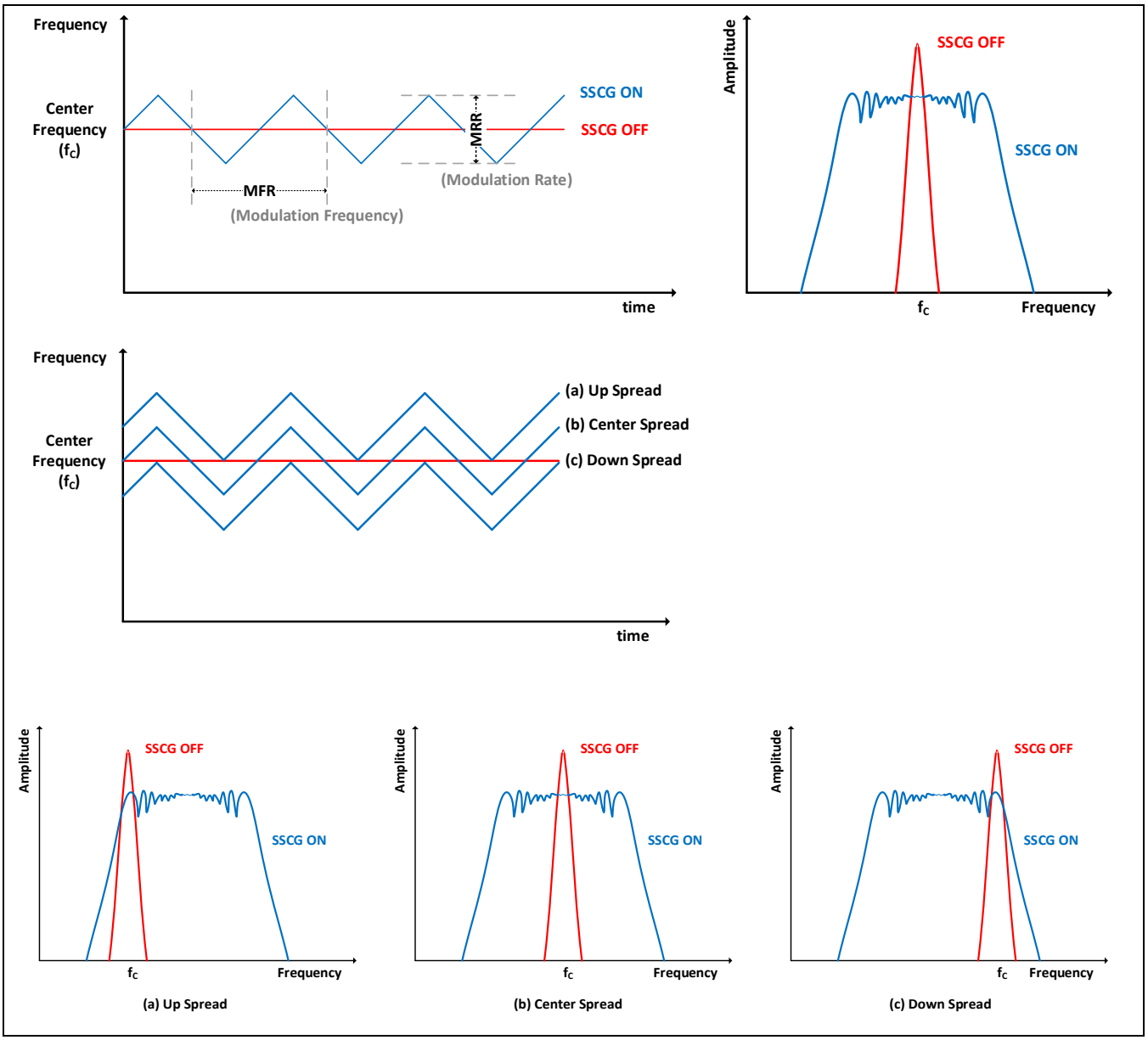
Modulation Frequency [Hz]:

$$\frac{\mathrm{MCLK} * 10\text{^}6}{p * mfr * 2\text{^}5}$$

Modulation Rate [%] :

$$\frac{\mathrm{mfr} * \mathrm{mrr}}{m * 2\text{^}6} * 100$$

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6.3 Operation Modes

S5K3B6SA can support a wide range of operation modes.

Operation mode is function of several mode parameters such as:

- Interface bandwidth
- Requested sensor output size and sensor operation mode
- CIS output size and number of bits per pixel
- Required Vertical blank time

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

- Power consumption limitations e.g. use single PLL, limited MIPI lanes, system lower power modes.

S5K3B6SA supports the operation modes as described in below Table along with their speed and output size.

Table 23 Typical Operation Modes

Mode	Interface	Output format	ADC [Bit]	Frame Rate [fps]	Bit Rate (Per Lane)	H-Pixel [ea]	V-Pixel [ea]	Comment
24b Combined		RAW12*2 *3)	12	60	1280	1936	1552	
20b Combined and Compress		RAW20	12	60	1120	1936	1552	Intra-scene DCG*2) and Lateral overflow with LFM (HDR4)
16b Combined and Compress		RAW16	12	60	960	1936	1552	
14 Combined and Compress		RAW14	12	60	800	1936	1552	
12 Combined and Compress		RAW12	12	60	640	1936	1552	
Line Re-order		RAW12	12	45	2080	1936	6208	
Pixel Re-order		RAW12	12	45	1920	7744	1552	

NOTE: 1. To improve the performance, the values of operation modes, output size, frame rate, and so on can be changed.

NOTE: 2. Sensor readout same pixel twice with difference conversion gain in frame.

3. The 24b Combined frame output in the RAW12 format where the horizontal pixel size doubled due to there is no defined output format of RAW24 in MIPI alliance.

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6.4 Slave mode

6.4.1 General Description

S5K3B6SA supports 'DUAL SYNC' mechanism to synchronize multiple image sensors. The synchronized data streaming mode refer to external VSYNC (FSYNC) signal so called 'slave mode'.

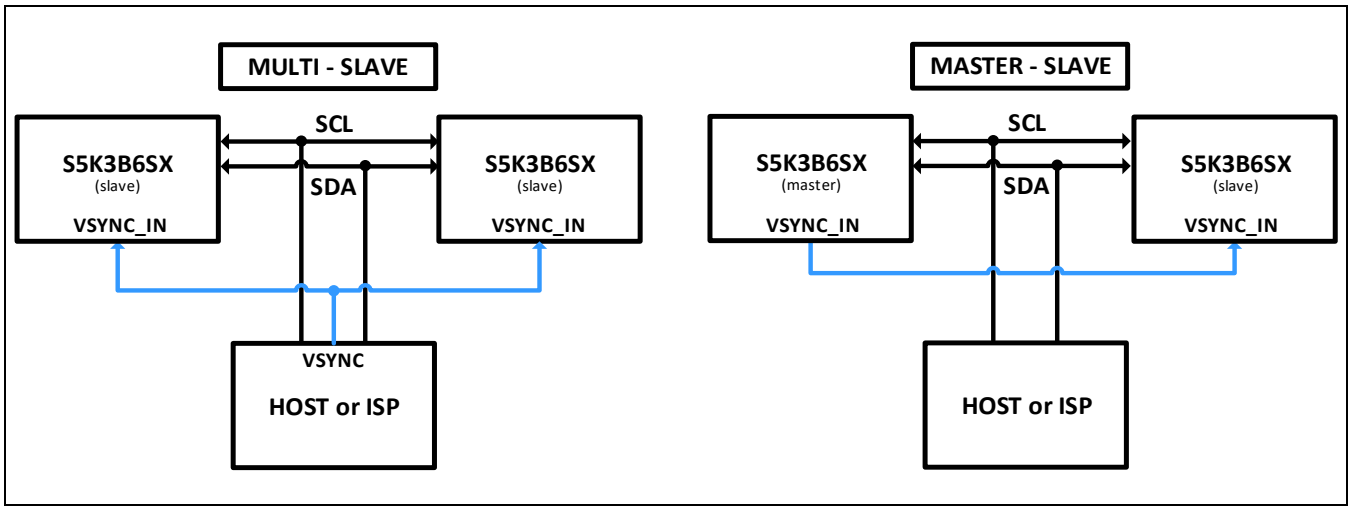
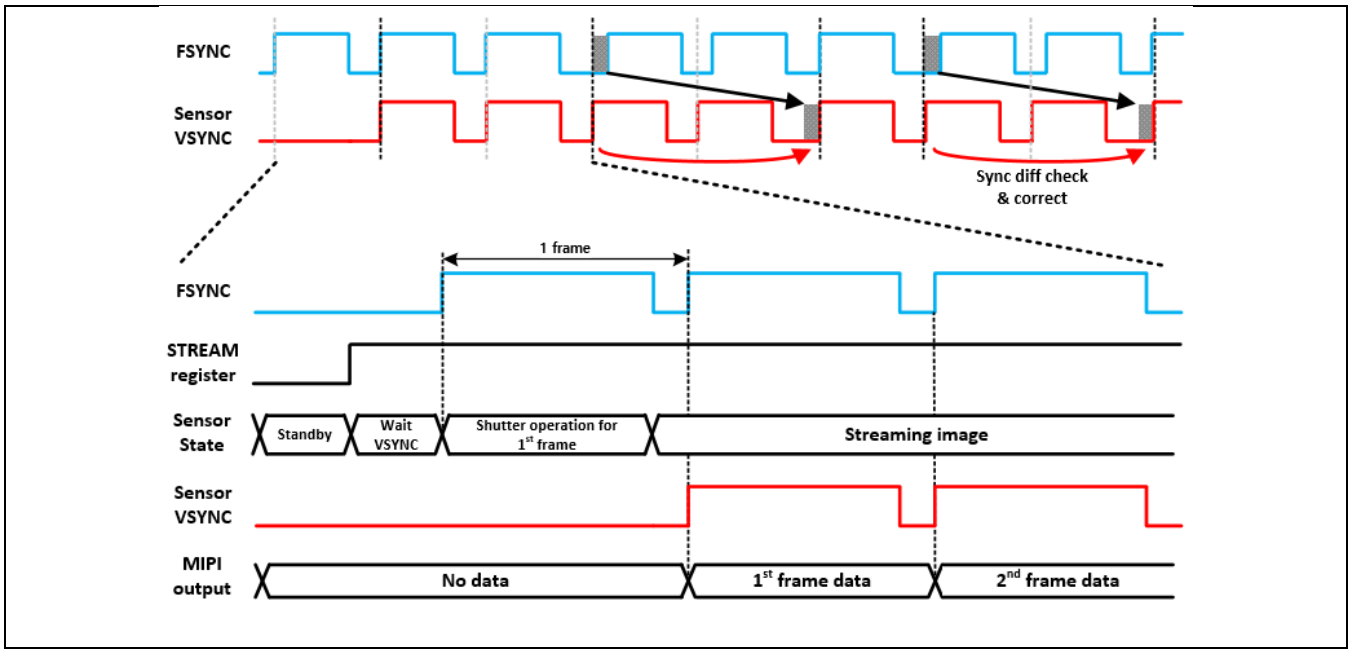


Figure 25 Multiple Camera System

This allows to align end of exposure of first row to the external trigger. Sensor outputs data 1 frame after from 1st External VSYNC(FSYNC) because of time for 1st frame's shutter operation like *Figure 26*.

When sensor is received first 'external VSYNC(FSYNC)', Dual sync mechanism can take up to 4 frames for synchronizing External VSYNC and Sensor VSYNC. Frames for synchronizing can be changed according to position of FSYNC and Sensor VSYNC.



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Figure 26 Concept of DUAL SYNC Mechanism

When the difference between external VSYNC and sensor VSYNC(Aligned to MIPI transmission) is larger than 'max_diff_line_num(0x3404) * 1H', the sensor compensates difference between them by adjusting length of the next frame. The length of the next frame is adjusted by the sensor internal adjustment unit, so Latency time (the delay between FSYNC and start of sensor output) is not fixed if the cycle of FSYNC and frame rate of sensor are not exactly same. In HDR4 readout mode(LPD HCG, LPD LCG, SPD CDS, SPD LOF), Latency could be minimum 0H to maximum 5H time. For example, Latency time has a maximum delay of 48.38us based on HDR4 60fps set-file (can be modified according to set-file settings). It depends on the difference between FSYNC and Sensor VSYNC.

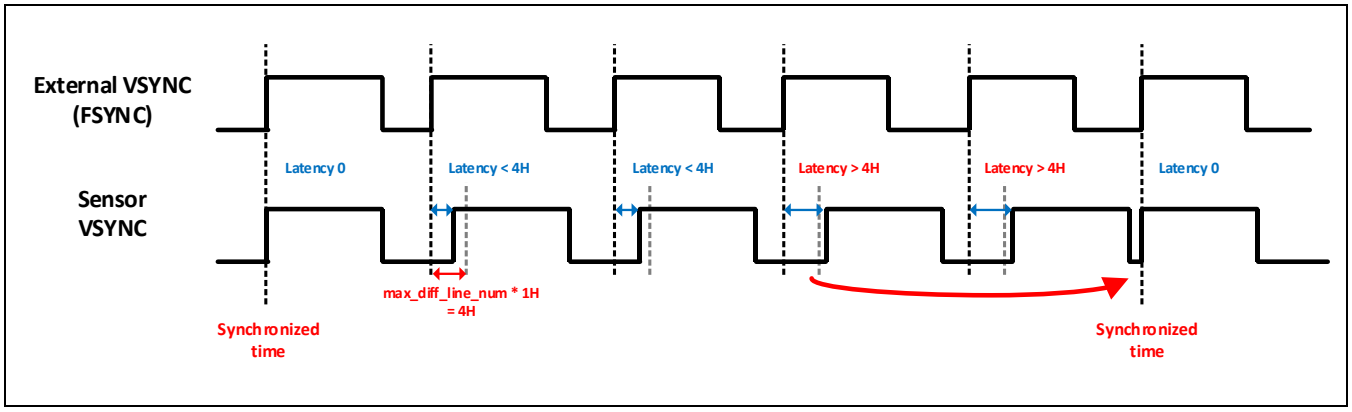


Figure 27 DUAL SYNC Continuous Mode

6.4.2 Synchronization modes

The VSYNC frequency can be calculated using the following formula:

$$VSYNC[Hz] = \frac{\text{SYS_CLK}}{\text{frame length lines} \times \text{line length pck}}$$

6.4.2.1 Continuous Mode

Like as *Figure 27* Sensor can synchronize only if the difference between FSYNC and sensor internal frame rate(Sensor VSYNC) are bigger than max_diff_line_num(0x3404) * 1H to synchronize data output reference to FSYNC. The latency (the gap between FSYNC and start of sensor output) could be differ frame by frame within range of minimum 0H to maximum 5H time.

6.4.2.2 Trigger Mode

Sensor supports 'Trigger mode' as an option of FSYNC mechanism(DUAL SYNC Mechanism). 1st shutter operation starts with first external trigger signal. (refer to rising edge of 1st FSYNC trigger signal) After triggering,

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sensor is free running with the frame rate which is set by registers. FSYNC Signal is ignored for synchronizing with sensor – except 1st FSYNC trigger signal.

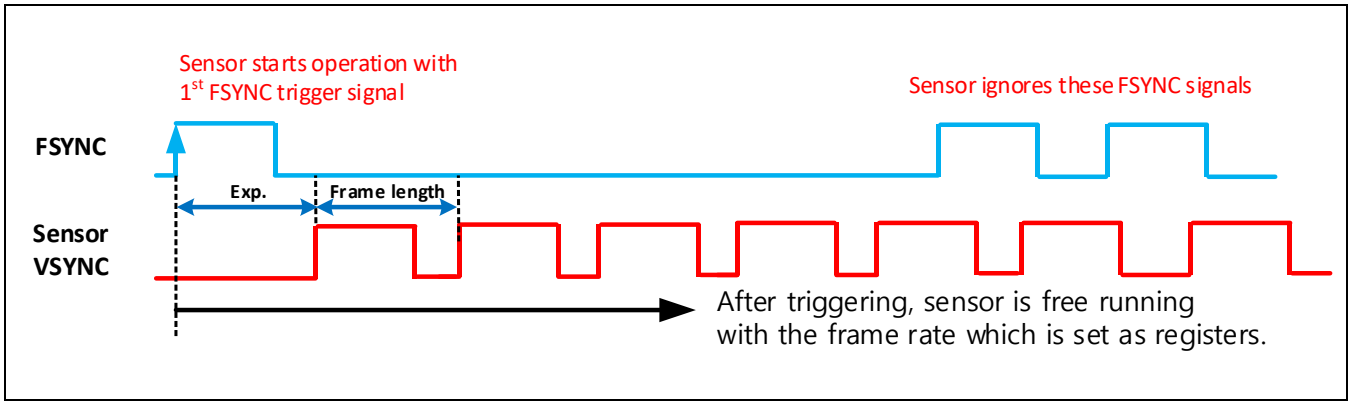


Figure 28 DUAL SYNC Trigger Mode

6.4.3 Related Registers

The following table informs registers about Slave mode. The related registers must be set before streaming on command.(0x0100 : 0x0001)

Table 24 Registers of Dual Synchronization

Index	Reset value	Bits	Register name	Type	Description
0x3400	0x0000	[0:0]	dual_sync_en	RW1	1:dynamic mode enable
	0x0000	[1:1]	slave_mode_en	RW1	1:slave mode enable
0x3402	0x0002	[1:0]	dual_sync_auto_mode	RW1	2'b00 : Auto off mode 2'b01 : Semi-Auto on mode 2'b10 : Auto on mode
0x3404	0x0003	[15:0]	max_diff_line_num	RW1	Threshold for synchronizing start. If difference is bigger than max_diff_line_num(0x3404) * 1H, synchronizing starts.
0x3112	0x0000	[0:0]	tgr_slave_option2	RW1	[0] 1'b1 Trigger both edge of hsync when sensor slave mode.
0x313A	0x000A	[15:0]	tgr_atx_pulse_width	RW1	Setting for dual sync slave mode
0x105E	0x0000	[0:0]	sys_r_vsync_in_oe	RW	Pad setting for VSYNC_IN 1b'0 : Input (for Slave mode) 1b'1 : Output (for Master mode)

6.4.4 Use cases

Slave, Continuous mode with max latency 4H

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

s34000003 //Enable dual sync slave mode s313A0000 //Setting for dual sync slave mode s31120000 //Trigger mode enable s34040004 //Threshold for sync. start - changeable(0~5) s105E0000 //Pad setting for VSYNC_IN
Slave, Trigger mode
s34000003 //Enable dual sync slave mode s313A0000 //Setting for dual sync slave mode s31120030 //Trigger mode enable

6.5 Frame readout format

S5K3B6SA outputs up to 4 ea different (LPD HCG, LPD LCG, SPD-CDS, and SPD-LOF) frames through the line-interleaved HDR operation. During the HDR readout operation, the 1-row readout time of each exposed frame is tuned, and then the data of LPD HCG, LPD LCG, SPD-CDS, and SPD-LOF frame is out in every 4-row readout time period. This sensor can also operate in Intra-scene DCG with LFM mode.

6.5.1 Data Readout Frame Format

S5K3B6SA supports various data types and frame formats. Bellow figure illustrates two kinds of output frame format. In line interleaving, LPD HCG, LPD LCG, SPD-CDS, and SPD-LOF row line data output in sequential order at every one row data output time. But in pixel reorder, LPD HCG, LPD LCG, SPD-CDS, and SPD-LOF data aligned and output in sequential pixel interleaving. In this case, whole pixel data in the same row is output over four times of the one row data output time in line interleaving.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

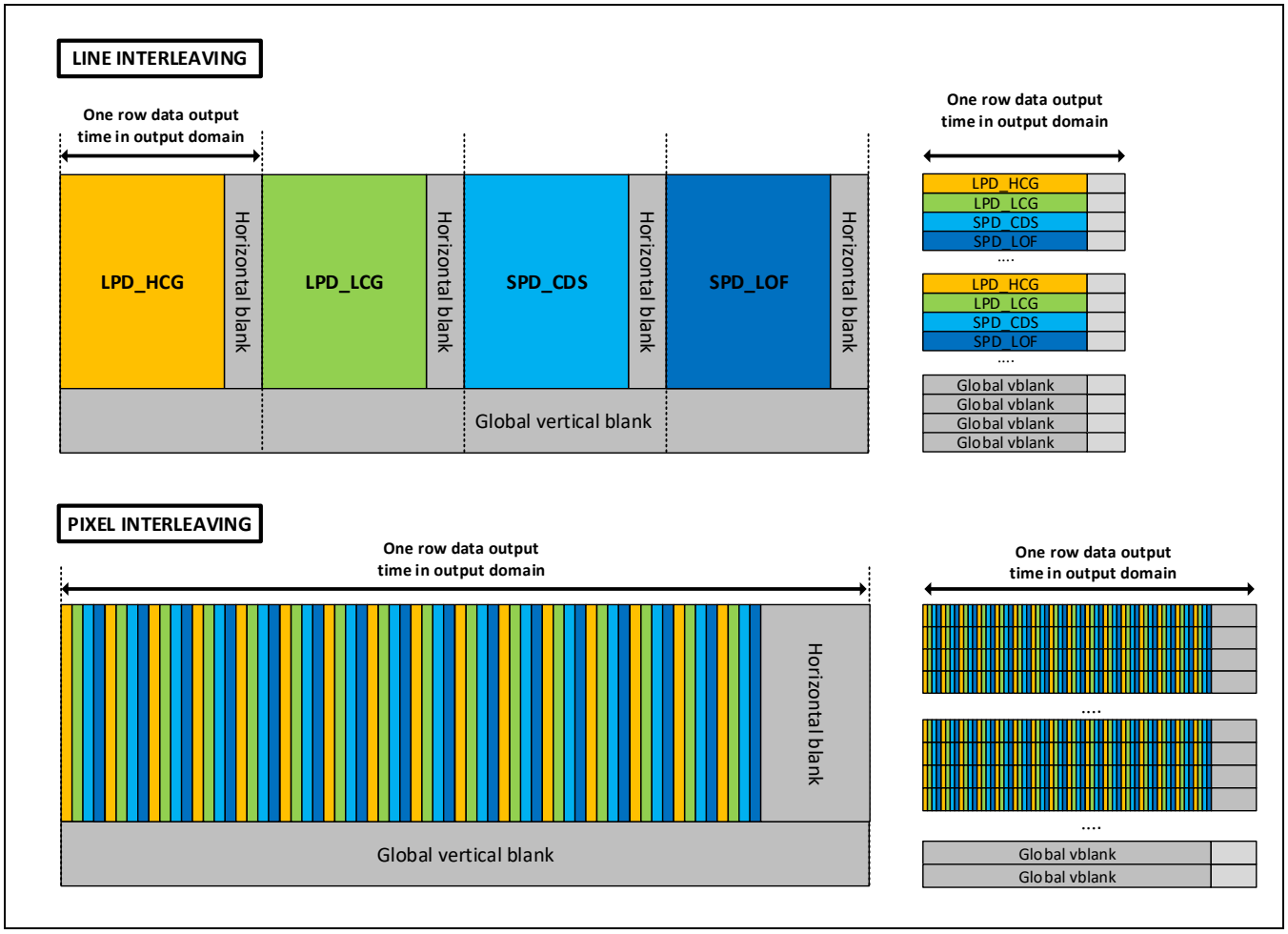


Figure 29 Line/Pixel Interleaved HDR Output Frame Format (Intra scene DCG without LFM)

6.5.2 Data Combine

To reduce the amount of data transmission, S5K3B6SA can output combined 24-bit data instead of 4x12 bit raw data. Four 12-bit data are normalized to 24 bit by using the internal algorithm.

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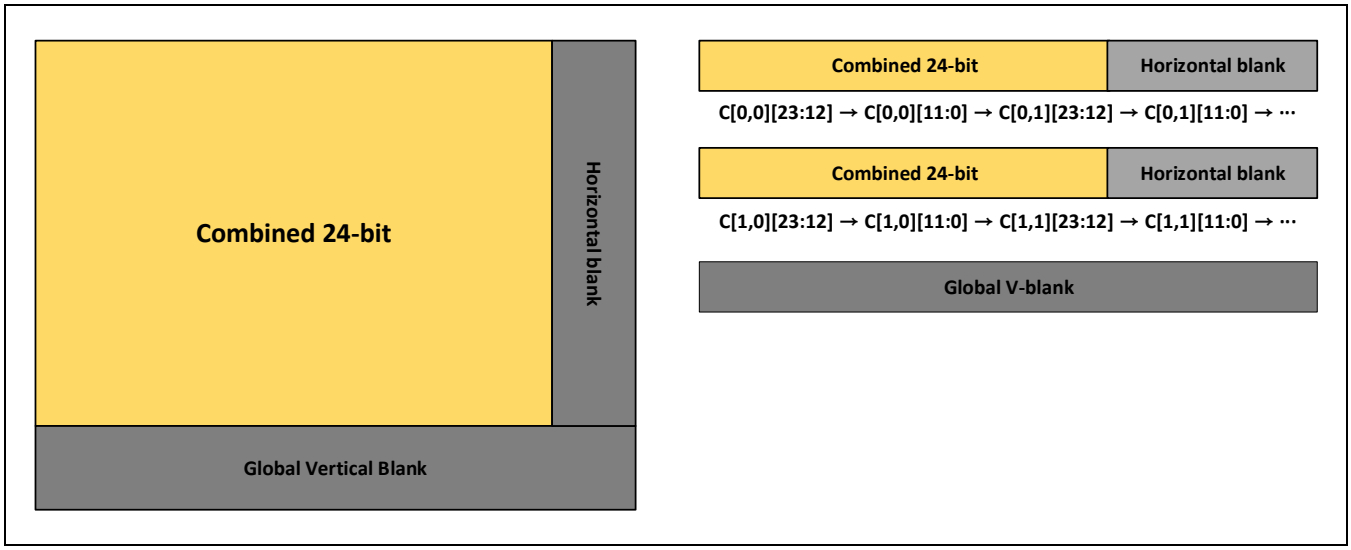
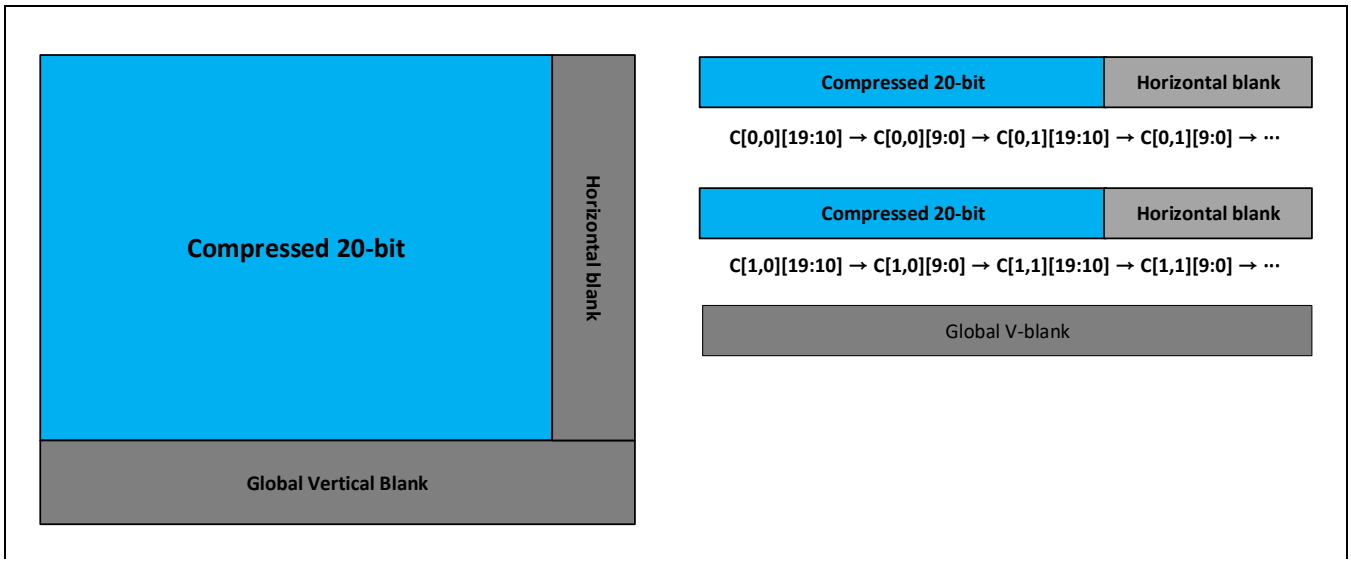


Figure 30 Combined Data Output Frame Format (Intra-scene DCG without LFM mode)

6.5.3 Data Compress

S5K3B6SA supports 20/16/14/12-bit compressed data output mode. The piecewise linear compression compresses the 24-bit combined data to 20/16/14/12-bit data with 32 knee-points. These knee-points are controlled by registers.



Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

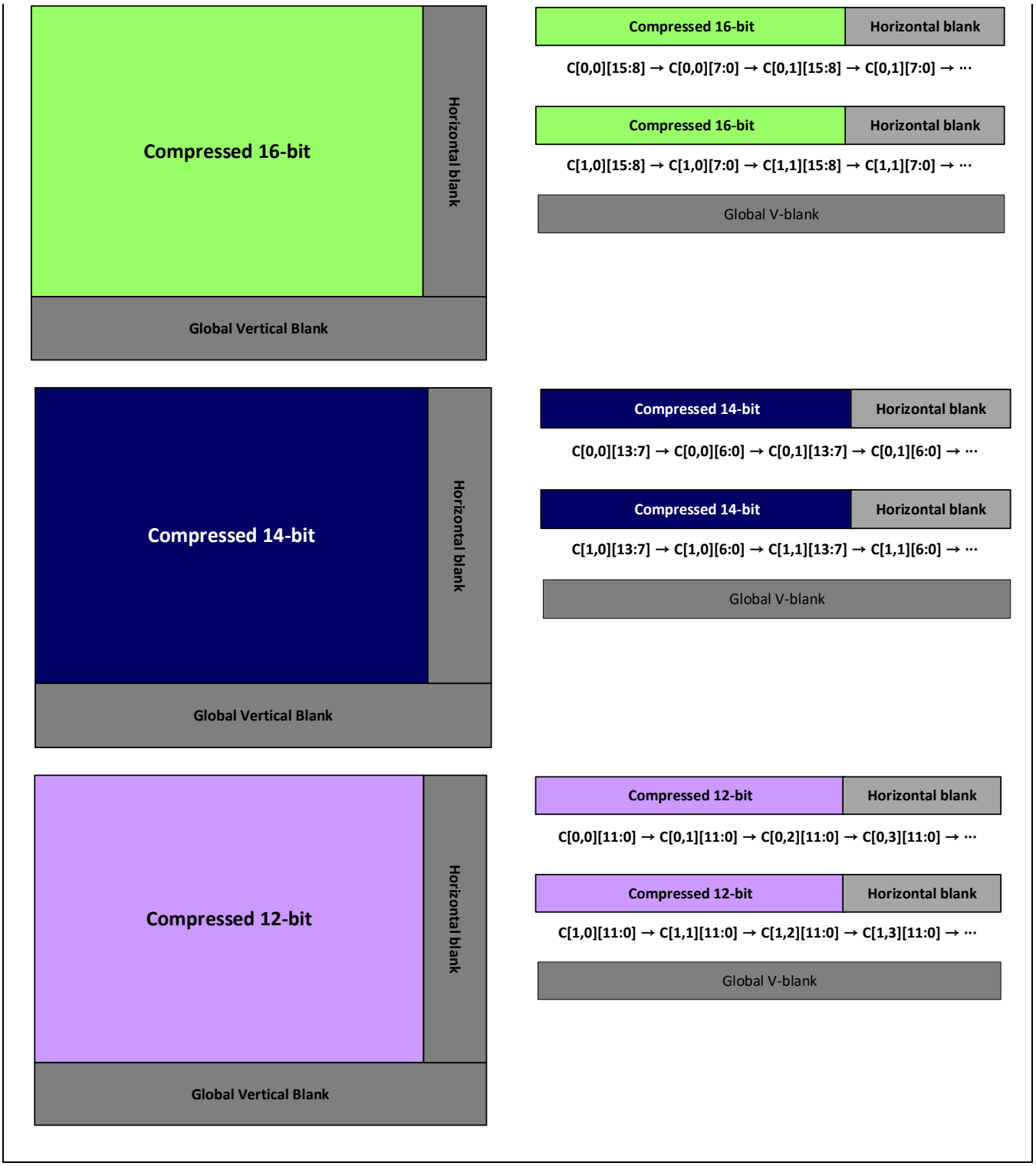


Figure 31 20/16/14/12-Bit Compressed Data Output Format (Intra-scene DCG without LFM mode) Frame rate control register limitation

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.5.4 Intra-frame HDR Settings

S5K3B6SA outputs up to 4ea different (LPD-HCG, LPD-LCG, SPD-CDS and SPD-LOF) frames through the line-interleaved HDR operation. HDR4 mode supports the maximized dynamic range after the recombination in an internal function or in external SoC. During the HDR readout operation, the 1-row readout time of each exposed frame is tuned, and then the data of LPD-HCG, LPD-LCG, SPD-CDS and SPD-LOF frame is out in every 4-row readout time period. This sensor can also operate in HDR3 mode and single exposure mode.

S5K3B6SA supports various data types and frame formats. The following chapters illustrate output frame formats with different row address data of LPD-HCG, LPD-LCG, SPD-CDS and SPD-LOF frames as both HDR3 mode and HDR4 mode. Plus, S5K3B6SA able to output RAW12 data which is not used in HDR process and able to output RAW8 typed defect data from original LPD-HCG/LCG and SPD-CDS/LOF frame and detection data from merged frame.

6.5.5 HDR3 mode

This mode outputs merge and compress 3 different frame. (LPD-HCG, LPD-LCG, SPD-LOF) and is able to output 1 frame which is merge and compressed, and also able to output 3 different frame by re-order mode. In S5K3B6SA, sensor able to output RAW12 data which is not used in HDR process and able to output RAW8 typed defect data from original LPD-HCG/LCG and SPD frame and detection data from merged frame.

6.5.6 HDR4 mode

This mode outputs merge and compress 4 different frame as LPD HCG, LPD LCG, SPD-CDS, and SPD-LOF is able to 1 frame which is merge and compressed, and also able to output 4 different frame by re-order mode. In S5K3B6SA, sensor able to output RAW12 data which is not used in HDR process and able to output RAW8 typed defect data from original LPD-HCG/LCG and SPD-CDS/LOF frame and detection data from merged frame.

6.6 Image orientation

The pixel data are normally readout from left to right in horizontal direction and from top to bottom in vertical direction. By changing the mirror/flip mode, the readout order is reversed and the resulting image is flipped as a mirror image. Pixel data is then read out from right to left in horizontal mirror mode and from bottom to top in vertical flip mode. The horizontal mirror and the vertical flip modes are programmed by the image orientation register.

The sensor module supports four possible pixel readout orders according to the directions of pixel readout and line readout: standard readout, horizontally mirrored readout, vertically mirrored readout, horizontally mirrored and vertically flipped readout.

The registers are as follows:

Table 25 Image Orientation Register

Index	Reset value	Bits	Register name	Type	Description
0x325C	0x0000	[0]	tgr_h_desc5_orient	RW1	Enable horizontal mirror for Active
0x323C	0x0000	[0]	tgr_h_desc3_orient	RW1	Enable horizontal mirror for FOB

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

0x32EC	0x0000	[0]	tgr_v_desc8_orient	RW1	Enable vertical flip

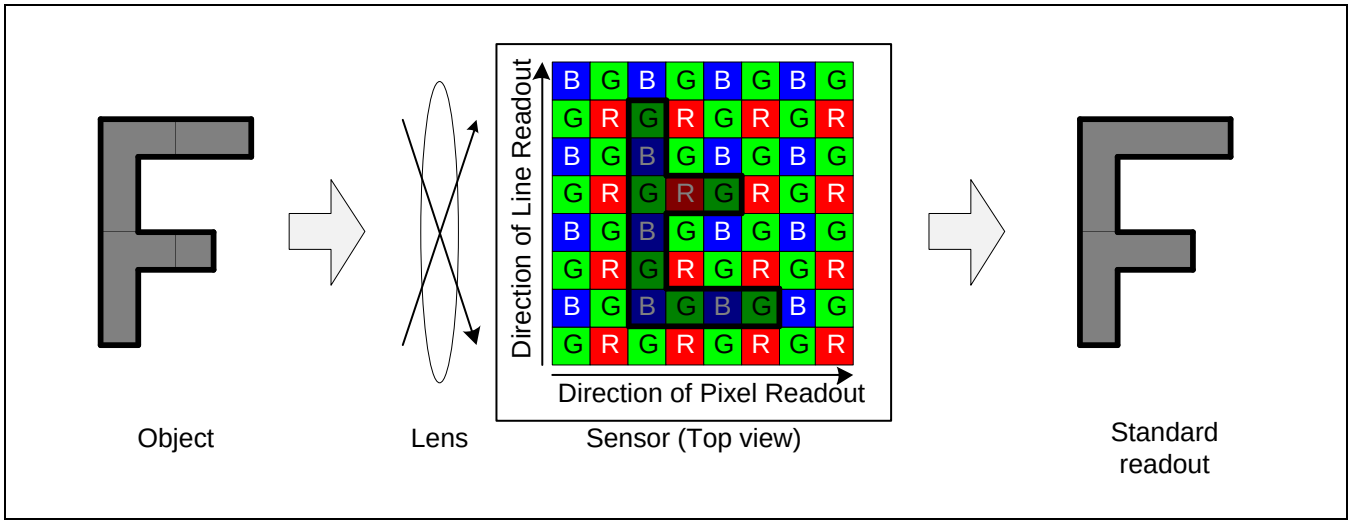


Figure 32 Object, Sensor and displayed image in Standard Readout

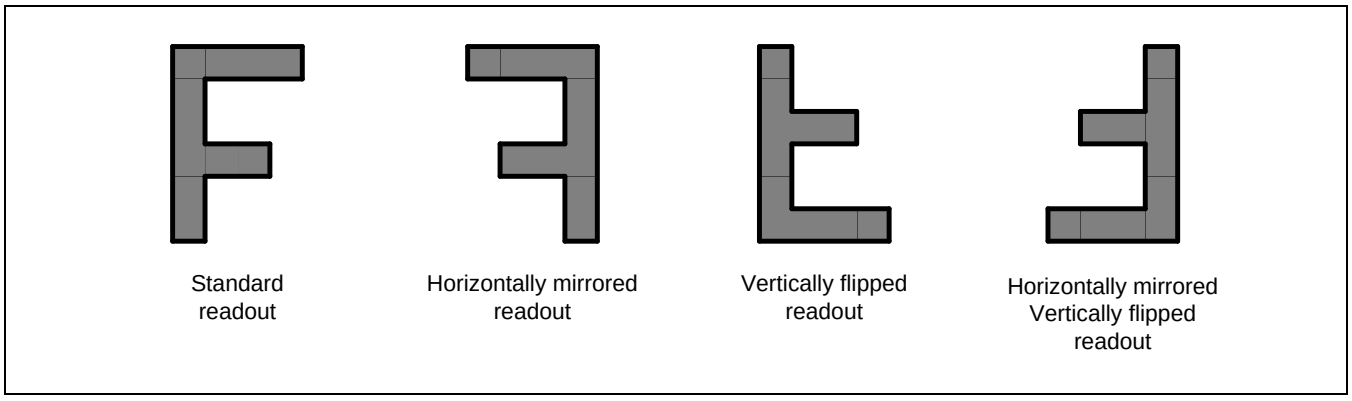


Figure 33 Four readout orders and displayed images

6.7 Frame rate control

Change the size of the virtual frame to change the line rate and frame rate. The number of clocks for horizontal readout and number of lines for vertical readout that consist of the virtual frame are controlled by registers. The horizontal and vertical blanking times (horizontal blanking time: # of clocks for horizontal readout columns, vertical blanking time: # of lines for vertical readout lines) must meet the system requirements.

Use the following equation to calculate the frame rate.

$$\text{frame rate(system domain)} = \frac{\text{sys_clk}}{\text{tgr line length pck} * \text{tgr frame length lines}}$$

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

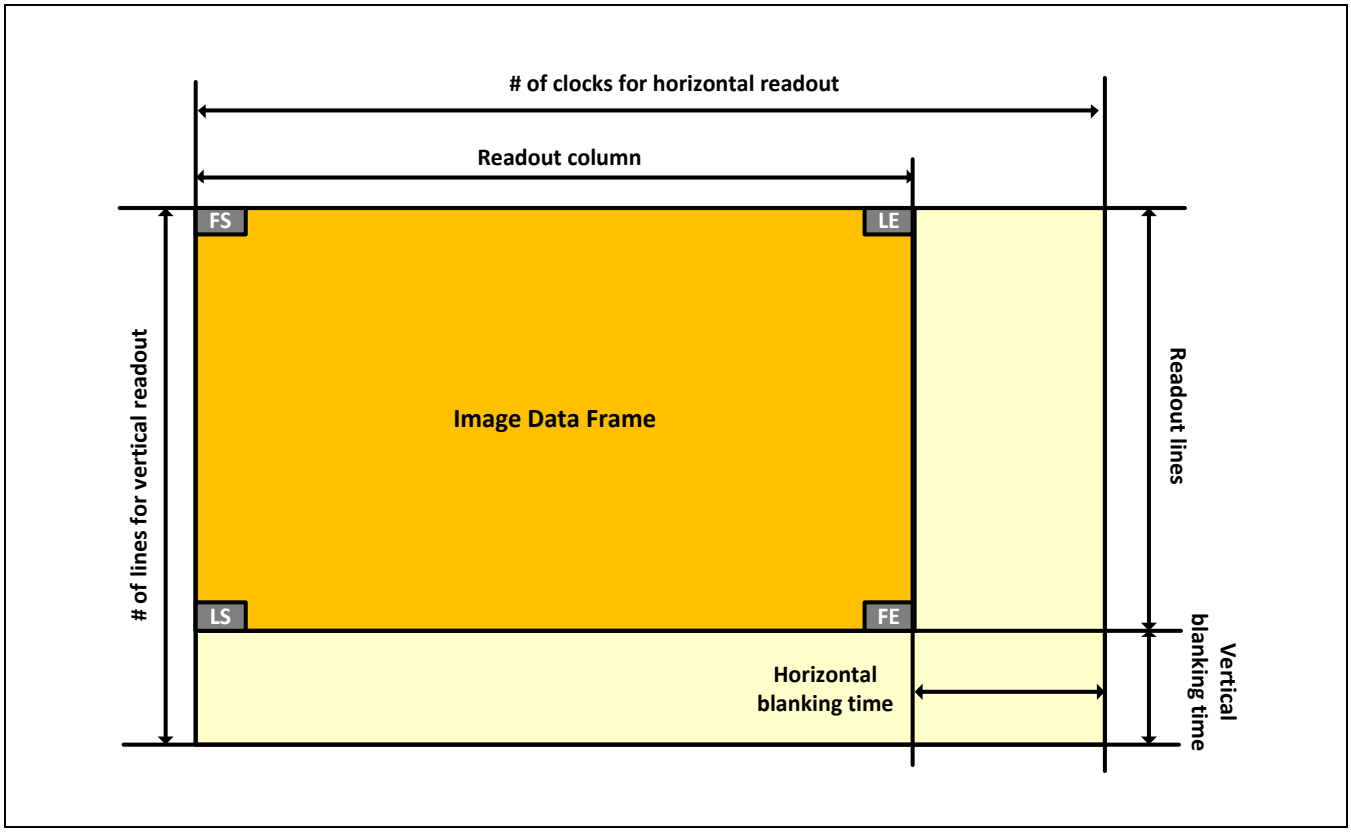


Figure 34 Example format of virtual frame

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.7.1 Input size

The addressed region of the horizontal pixel data output is controlled by the *tgr_h_desc5_addr_start, tgr_h_desc5_addr_witdth* registers, and that of the vertical pixel data output is controlled by the *tgr_v_desc8_addr_start, tgr_v_desc8_addr_width* registers, respectively.

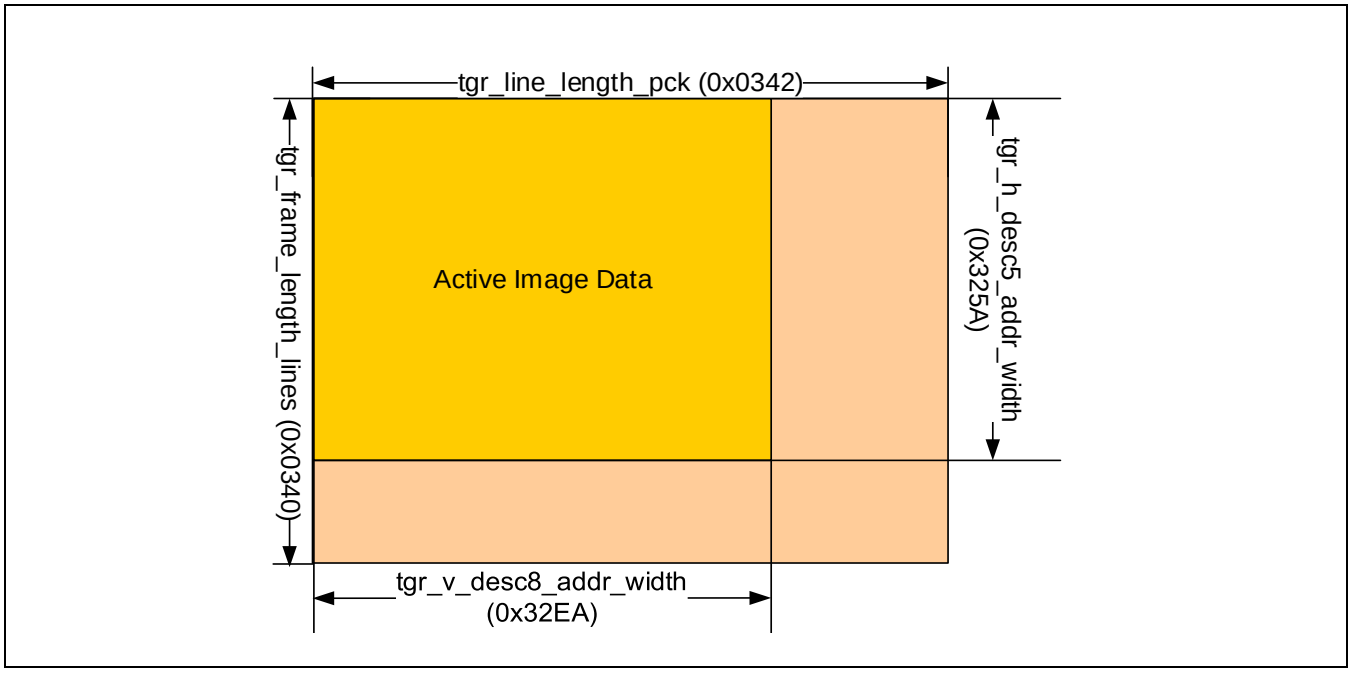


Figure 35 Controlling Input Size

Table 26 Window of Interest register

Index	Reset value	Bits	Register name	Type	Description
0x3258	0x0000	[12:0]	tgr_h_desc5_addr_start	RW1	Horizontal start address
0x325A	0x0790	[12:0]	tgr_h_desc5_addr_width	RW1	Horizontal readout column number.
0x32E8	0x0028	[12:0]	tgr_v_desc8_addr_start	RW1	Vertical start address
0x32EA	0x0610	[12:0]	tgr_v_desc8_addr_width	RW1	Vertical readout line number.

NOTE: If user need to control output size of image, user should control not only this input size registers but also OUTIF and ASP registers related to output size.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.7.2 Frame rate control register limitation

Frame_length_lines and line_length_pck register have limits to be used as follows.

Table 27 Frame rate control registers.

Index	Reset value	Bits	Register name	Type	Description
0x0340	0x0594	[15:0]	tgr_frame_length_lines	RW1	Total frame vertical width in 1 Horizontal time resolution. When write, register is updated on the next frame start.
0x0342	0x06D0	[15:0]	tgr_line_length_pck	RW1	Total frame horizontal width in TG clock cycles. When write, register is updated on the next frame start. 1H(Horizontal) time line_length_pck.

NOTE: tgr_frame_length_lines should be setted differently on each mode - single mode x1, HDR3 mode x3, HDR4 mode x4.

6.8 Integration time control (Electronic Shutter Control)

The pixel integration time is controlled by the shutter operation. During the shutter operation, the amount of integration time is determined by the column step integration time control register (fine_integration_time) and the line step integration time control register (coarse_integration_time). Use the following formula to calculate the total integration time of the sensor. It is common to control the exposure time ratio of LPD and SPD 1:1.

$$\text{Total integration time} = \frac{(\text{tgr_line_length_pck} * \text{coarse_integration_time}) + \text{fine_integration_time}}{\text{sys_clk}}$$

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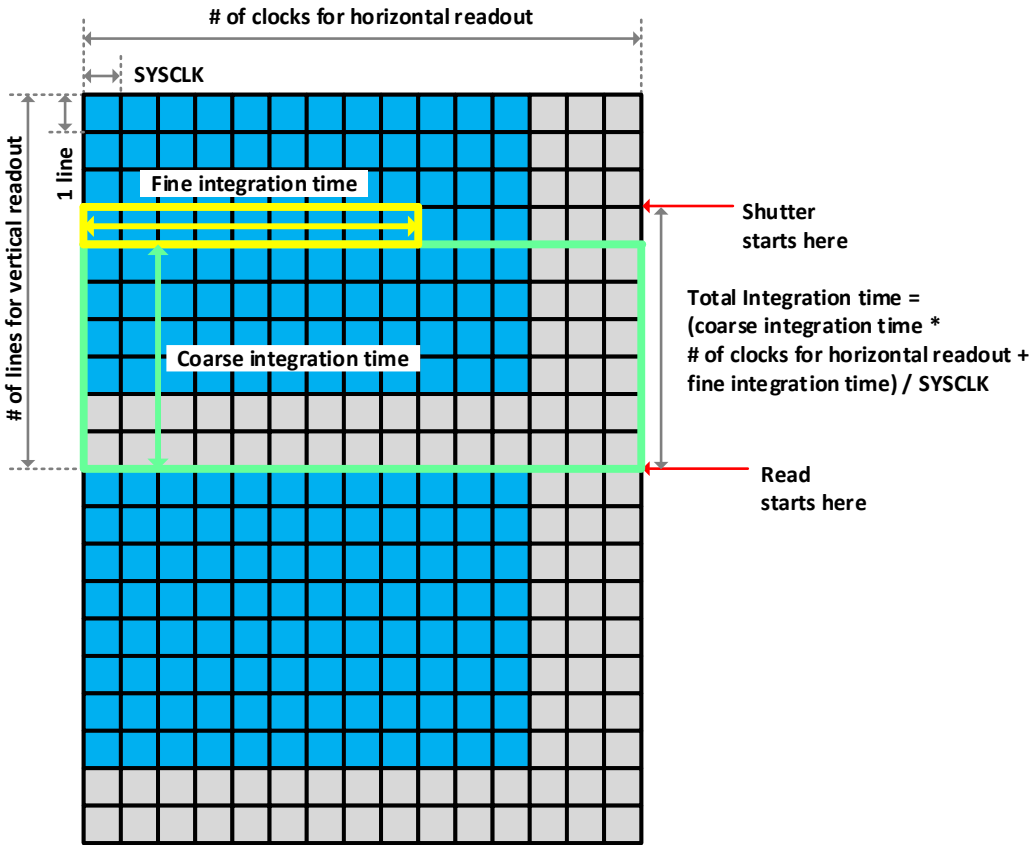


Figure 36 Example of Integration time

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Table 28 Integration time control registers

Index	Reset value	Bits	Register name	Type	Description
0x0202	0x0465	[15:0]	tgr_coarse_integration_time	RW1	Integration time control register that has 1H time unit.
0x0216	0x0465	[15:0]	tgr_coarse_integration_time_spd	RW1	Line unit integration time control for SPD frame at HDR3, HDR4.
0x0218	0x0009	[15:0]	tgr_coarse_integration_time_short	RW2	Line unit integration time control for short exposure frame at HDR4.
0x0200	0x02A9	[15:0]	tgr_fine_integration_time	RW1	Integration time control register that has 1clock unit.
0x0300	0x009B	[15:0]	tgr_fine_integration_time_spd	RW1	Clock unit integration time control for SPD frame at HDR3, HDR4.
0x0400	0x02A9	[15:0]	tgr_fine_integration_time_short	RW2	Clock unit integration time control for short exposure frame at HDR4.

NOTE: *If user need to control fine integration time, other settings should be changed with, so user should confirm with supplier.*

Table 29 Range of Integration time registers at HDR3 operation

Addr	Register name	fine integration time < TXS*		fine integration time >= TXS*	
		min	max	min	max
0x0202	tgr_coarse_integration_time	6	FLL-6	5	FLL-7
0x0216	tgr_coarse_integration_time _spd	6	FLL-6	5	FLL-7
* Setting the value as tgr_coarse_integration_time = 3n+2, tgr_coarse_integration_time_spd = 3n+2 is recommended. * tgr_coarse_integration_time cannot be tgr_coarse_integration_time_spd – 2 * TXS : Time from start of horizontal counter to rising edge of TX					

NOTE: Generally use the right case 'fine integration time >= TXS', if the left case 'fine integration time < TXS' used, the notice will be notified.

Table 30 Range of Integration time registers at HDR4 (SPD LOF) operation

Addr	Register name	fine integration time < TXS*		fine integration time >= TXS*	
		min	max	min	max
0x0202	tgr_coarse_integration_time	7	FLL - 9	6	FLL - 10

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

0x0216	tgr_coarse_integration_time _spd	7	FLL - 9	6	FLL - 10
* Setting the value as tgr_coarse_integration_time = 4n+2, tgr_coarse_integration_time_spd = 4n+2 is recommended. * tgr_coarse_integration_time cannot be tgr_coarse_integration_time_spd – 2 * TXS : Time from start of horizontal counter to rising edge of TX					

NOTE: Generally use the right case 'fine integration time >= TXS', if the left case 'fine integration time < TXS' used, the notice will be notified.

6.8.1 LFM(LED Flicker Mitigation)

LED traffic lights, road signs and car lights are commonly used in recent years. These LED light sources flicker with a frequency of 90Hz or higher. While the flickering is not distinguished by the human vision system it poses a problem to camera sensors that use short exposure time. The reason is that the short exposure window could be when the LED light is off. For automotive applications this may lead to misdetection of traffic lights, car lights or road signs and therefore is handled by the sensor.

In order to avoid such conditions, the flicker light mitigation operates as follows:

The S5K3B6SA consists of a split PD structure - Large and Small PD. The main purpose of SPD is LFM. SPD could be set to a longer exposure time than LPD because of extremely low sensitivity and high FWC. This enables the sensor to detect LED light well. For example, when the frequency of light is 90Hz and exposure time of SPD is longer than 11ms, the sensor can detect LED signal perfectly.

In order to use the SPD data correctly, the level of discrepancy between the signals is calculated during merger operation. For pixels with large discrepancy the merger outputs the "selector value": a de-noised version of the maximum over the all normalized signals (instead of the simple merger as described in previous section). As the long exposure (either in SPD and\or LPD) captures the flicker light because it sums photons over a larger flicker wave period, the "selector value" will contain it.

Table 31 Integration time of SPD control registers

Index	Reset value	Bits	Register name	Type	Description
0x0202	0x0465	[15:0]	tgr_coarse_integration_time	RW1	Line unit integration time control for LPD frame at HDR3, HDR4.
0x0216	0x0465	[15:0]	tgr_coarse_integration_time_spd	RW1	Line unit integration time control for SPD frame at HDR3, HDR4.

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6.9 Gain control

6.9.1 Analog gain control

User can control the gain of pixel signal by analog gain control register(`analogue_gain_code_global`). User can only control the analog gain of LPD HCG and LPD short frame, and use the following equation to control the analog gain.

$$\textbf{Analog gain} = \textbf{Analogue gain register}[13:0]/32$$

Range of analog gain is designed from x1 up to x16

Coarse gain step : x2, Fine gain step : 1/32(@ coarse gain 2^n ~ 2^n+1, n = 0,1,2,3)

The register value can be controlled as following table (default value is 0x20)

Table 32 Analog Gain Examples

Gain Value	Analog_gain_code_XXX Register Value
X1	0x0020
X2	0x0040
X4	0x0080
X8	0x0100
X16	0x0200

Table 33 Analog Gain Control Registers

Index	Reset value	Bits	Register name	Type	Description
0x0204	0x0020	[13:0]	tgr_analogue_gain_code_long_hcg	RW2	Global Gain or Separate gain for LPD HCG frame.
0x020A	0x0020	[13:0]	tgr_analogue_gain_code_long_lcg	RW2	Analog gain for LPD LCG frame
0x0206	0x0020	[13:0]	tgr_analogue_gain_code_middle	RW2	Analog gain for SPD frame
0x0208	0x0020	[13:0]	tgr_analogue_gain_code_short	RW2	Separate gain for VS frame or SPD-CDS at HDR4 mode

NOTE: In S5K3B6SA, analog gain is global; there is no per-channel gain. $in = \frac{x}{32}$

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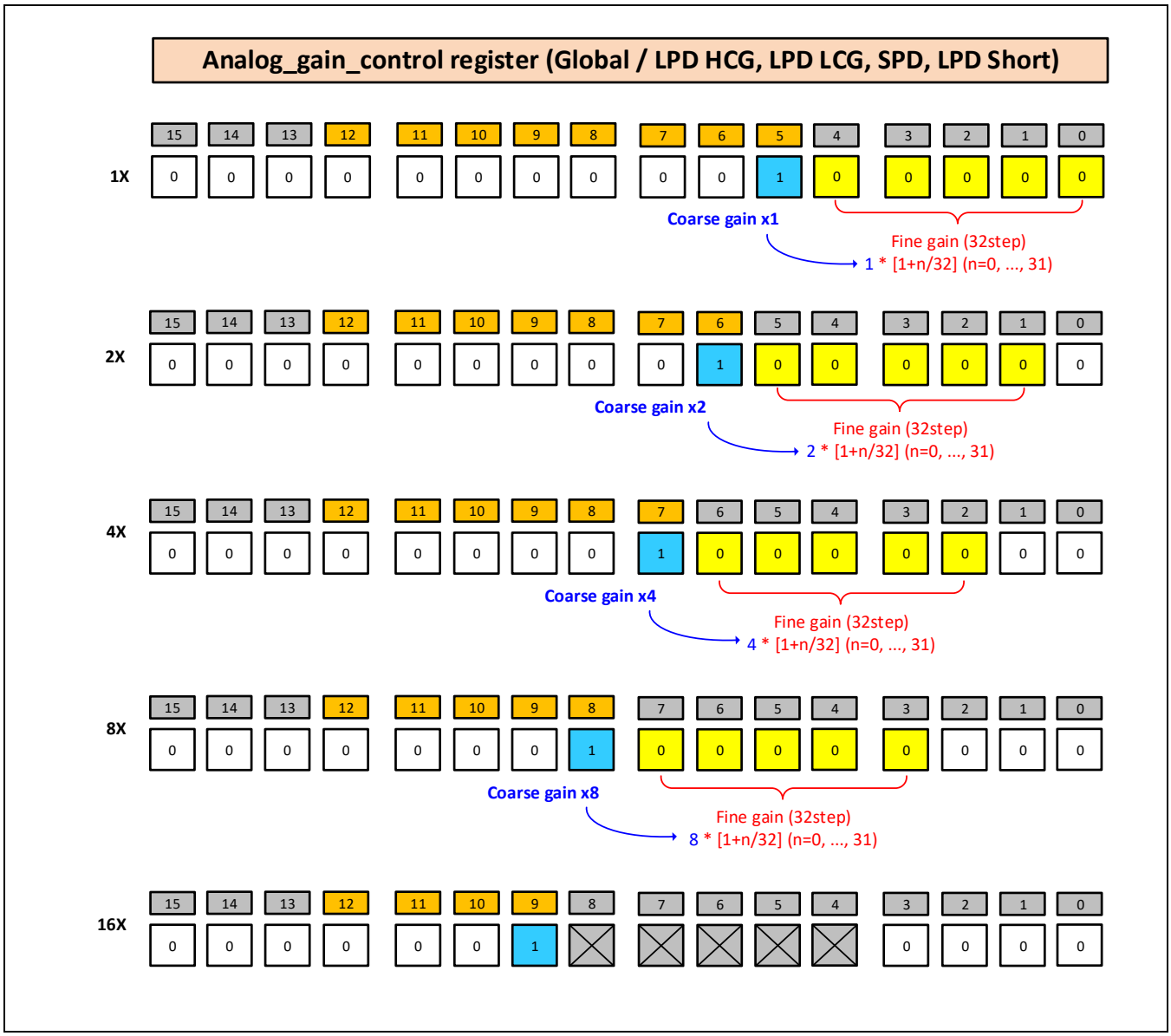


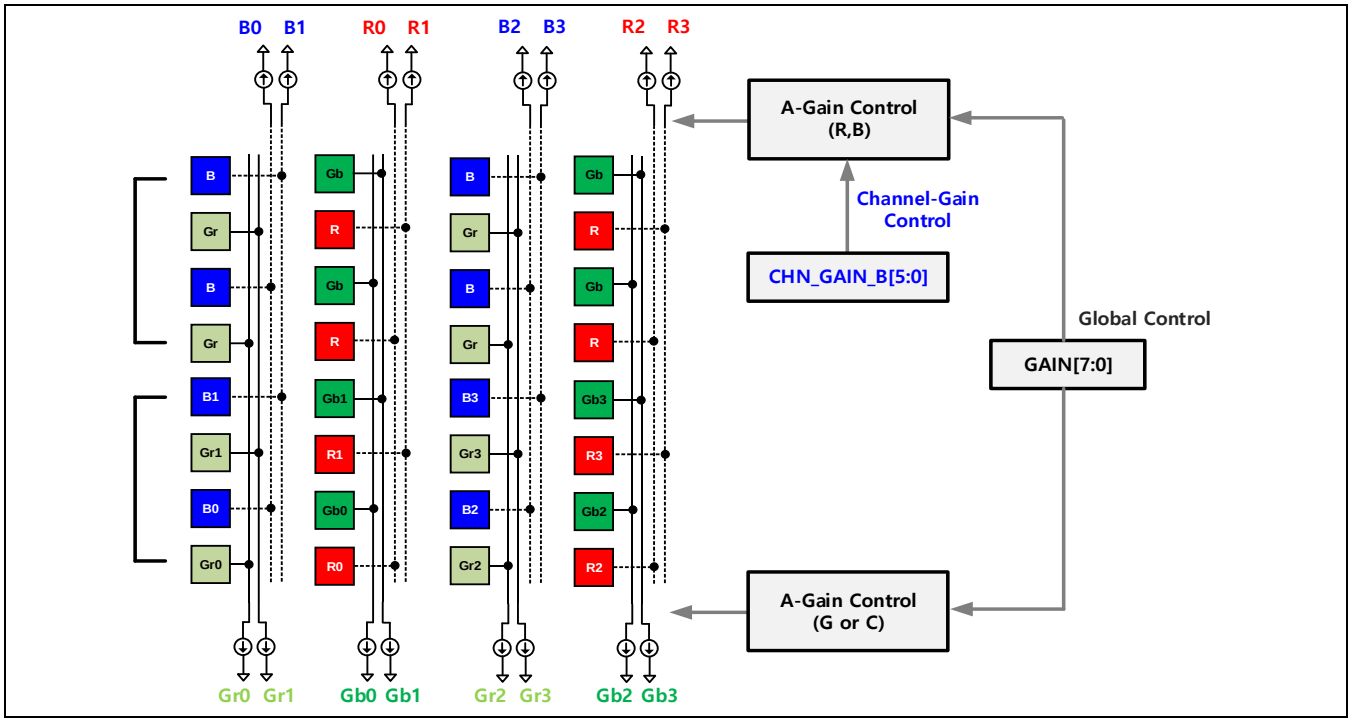
Figure 37 Example of analog gain control

6.9.2 channel gain

3b6 sensor can control channel gain to apply gain on Bottom ADC – for color pixel (R,B). Channel gain is applied on entire frame. Max. gain : x2. Fine gain is adjusted by 32 steps only when coarse gain is x1.

Index	Reset value	Bits	Register name	Type	Description
0x0230	0x0	[4:0]	Tgr_channel_gain_fine	RW2	Gain=Fine_gain[4:0]/32
0x0230	0b0	[5]	tgr_channel_gain coarse	RW2	Gain=2^Coarse gain[5]

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6.9.3 Digital gain control

User can use the digital gain control register to control the gain of pixel signal for each channel or each exposure type. Use the following equation to control the digital gain.

$$\text{Digital gain} = \text{Digital gain register}[15:0] / 256$$

Range of digital gain is designed from x1 up to x256.

Digital gain control step: 1/256

The register value can be controlled as following table (default value is 0x100). The maximum value is 0xFFFF and the value of between min and max can be controlled any bit.

Table 34 Digital Gain Examples

Gain Value	Digital_gain_code_XXX Register Value
X1	0x0100
X255.99	0xFFFF

Table 35 Digital Gain Control Registers

Index	Reset value	Bits	Register name	Type	Description

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

0x8500	0x0001	[0]	msm_asp0_msm_bypass	RW1	bypass block
0x8504	0x0001	[0]	msm_asp0_msm_residual_dark_en	RW1	enable calculation of residual dark output registers
0x850E	0x0000	[15:0]	msm_asp0_msm_pedestal_long_high	RW1	input pedestal value (LPD HCG)
0x8510	0x0000	[15:0]	msm_asp0_msm_pedestal_long_low	RW1	input pedestal value (LPD LCG)
0x8512	0x0000	[15:0]	msm_asp0_msm_pedestal_medium	RW1	input pedestal value (SPD-LOF)
0x8514	0x0000	[15:0]	msm_asp0_msm_pedestal_short	RW1	input pedestal value (SPD-CDS)
0x0232	0x0100	[15:0]	tgr_digital_gain_long_hcg	RW2	Separate msm digital gain of LPD HCG frame. (integer 8 bit / fraction 8 bit)
0x0234	0x0100	[15:0]	tgr_digital_gain_long_lcg	RW2	Separate msm digital gain of LPD LCG frame. (integer 8 bit / fraction 8 bit)
0x0236	0x0100	[15:0]	tgr_digital_gain_middle	RW2	Separate msm digital gain of SPD-LOF frame. (integer 8 bit / fraction 8 bit)
0x0238	0x0100	[15:0]	tgr_digital_gain_short	RW2	Separate msm digital gain of VS frame or SPD-CDS at HDR4 mode (integer 8 bit / fraction 8 bit)

NOTE: In S5K3B6SA, digital gain is global; there are gain controlled per-stream and per-channel(refer to chapter12). $in = \frac{x}{256}$

Before setting the digital gain registers, the value of some registers must be set. (0x8500[0] = 1'd0, 0x8504[0] = 1'd0, 0x840E~0x8514 = 100h)

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Figure 38 Example of digital gain control

6.9.4 Gain control in HDR mode

In HDR mode, if apply_factor_on is 1, the digital gain of SPD_LOF is controlled. If it is 0, the digital gain of LPD_HCG is controlled.

Below is an example when apply_factor_on is 1.

register	gain x1	gain x2	gain x4	gain x8	gain x16	gain x32
0x0236	x1	x2	x4	x8	x16	x32

NOTE: 0x0204(analog gain of LPD_HCG) affects the low-light noise and can be fixed to a specific value.

Below is an example when apply_factor_on is 0.

register	gain x1	gain x2	gain x4	gain x8	gain x16	gain x32
0x0232	x1	x2	x4	x8	x16	x32

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Table 36 Digital Gain Control Registers

Index	Reset value	Bits	Register name	Type	Description
0x8BEA	0x0001	[1]	apply_factor_on_output	RW1	Apply a factor on Jet output to meet the maximal value
0x0232	0x0100	[15:0]	tgr_digital_gain_long_hcg	RW2	Separate msm digital gain of LPD HCG frame. (integer 8 bit / fraction 8 bit)
0x0236	0x0100	[15:0]	tgr_digital_gain_middle	RW2	Separate msm digital gain of SPD-LOF frame. (integer 8 bit / fraction 8 bit)
0x0238	0x0100	[15:0]	tgr_digital_gain_short	RW2	Separate msm digital gain of SPD-CDS frame. (integer 8 bit / fraction 8 bit)

6.10 ISP FE

6.10.1 Introduction

The Pattern Gen handles the optional DTP injection.

Atop interface is the process of setting up images before the processing the data

Pre-BPR and BPR(Bad Pixel Replace) is replace couplet and single defect only in Optical Black area.

The ADLC block corrects the dark level of an active pixel by using the dark level of an optical black pixel.

The following figure illustrates the OB and Active area information of sensor. The compensation is calculating the average of the Optical Black Pixel data and compensating the value for the active pixel value by each frame.

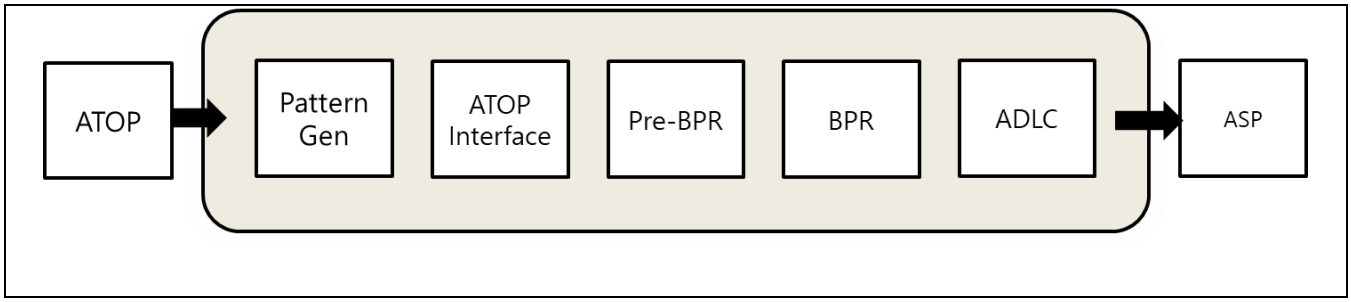


Figure 39 ISPFE block Function Diagram

6.10.2 Digital Test Pattern

This block handles the optional DTP injection. It handles all SMIA patterns, and additional patterns.

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6.10.2.1 Full Frame Deterministic Test Patterns

Two types of full frame deterministic test patterns are defined. Most are Bayer test patterns that more suitable for some tests than real image data and are injected early in the sensor data path. The only exception to this is a test pattern that is intended to test sensor-host link integrity. The data in this pattern is not Bayer data, and it is injected just prior to CSI framing. Use of these full frame test patterns is controlled by the test_pattern_mode parameter. The following table shows all the defined parameter settings and user should disable ADLC function to see test pattern data clearly.

Table 37 set for disable ADLC

Addr	Bits	Name	Set Value (Dec)
0x6002	[12:0]	adc_offset_even	0x0000
0x6004	[12:0]	adc_offset_odd	0x0000
0x600E	[3:0]	atop_if_qec	0x0000
0x6102	[11:0]	data_pedestal_adlc	0x0000
0x6106	[8:0]	adlc_enable	0x0000
0x6166	[8:0]	adlc_enable_m	0x0000
0x61B6	[8:0]	adlc_enable_s	0x0000

Table 38 Main Full Frame Test Pattern Control Parameters

Index	Reset value	Bits	Register name	Type	Description
0x6146	0x0000	[15:0]	ob_data_transfer_addr_min	RW1	Test pattern vertical min address
0x6148	0x0000	[15:0]	ob_data_transfer_addr_max	RW1	Test pattern vertical max address
0x6040	0x0000	[3:0]	dtpgen_test_pattern_hcg	RW1	4'd0 = no pattern 4'd1 = solid 4'd2 = full color bar 4'd3 = fade to gray color bar 4'd4 = PN16 4'd5 = horizontal gradient by address 4'd6 = vertical gradient by address 4'd7 = random + horizontal gradient 4'd8 = random + vertical gradient 4'd9 = random color bar
	0x0000	[7:4]	dtpgen_test_pattern_lcg	RW1	
	0x0000	[11:8]	dtpgen_test_pattern_m	RW1	
	0x0000	[15:12]	dtpgen_test_pattern_s	RW1	
0x6042	0x1FFF	[12:0]	dtpgen_test_data_red	RW1	R pixel value for solid pattern
0x6044	0x0000	[12:0]	dtpgen_test_data_greenR	RW1	Gr pixel value for solid pattern
0x6046	0x0000	[12:0]	dtpgen_test_data_blue	RW1	B pixel value for solid pattern

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0x6048	0x0000	[12:0]	dtpgen_test_data_greenB	RW1	Gb pixel value for solid pattern
0x604A	0x0000	[12:0]	dtpgen_h_cursor_width	RW1	Horizontal cursor width
0x604C	0x0020	[12:0]	dtpgen_h_cursor_position	RW1	Horizontal cursor position
0x604E	0x0000	[12:0]	dtpgen_v_cursor_width	RW1	Vertical cursor width
0x6050	0x0064	[15:0]	dtpgen_v_cursor_position	RW1	Vertical cursor position
0x6146	0x0000	[15:0]	ob_data_transfer_addr_min	RW1	Test pattern vertical min address
0x6148	0x0000	[15:0]	ob_data_transfer_addr_max	RW1	Test pattern vertical max address

6.10.2.2 Solid Color Mode

In the solid color test pattern mode, all pixel data is replaced with fixed Bayer test data that is defined by the four ‘test data color’ parameters, these are described in the following table.

Table 39 Test Data Color Parameters

Index	Reset value	Bits	Register name	Type	Description
0x6042	0x1FFF	[12:0]	dtpgen_test_data_red	RW1	R pixel value for solid pattern
0x6044	0x0000	[12:0]	dtpgen_test_data_greenR	RW1	Gr pixel value for solid pattern
0x6046	0x0000	[12:0]	dtpgen_test_data_blue	RW1	B pixel value for solid pattern
0x6048	0x0000	[12:0]	dtpgen_test_data_greenB	RW1	Gb pixel value for solid pattern

6.10.2.3 100% Color Bars Pattern Mode

In the "100 % color bar" test pattern mode, all pixel data is replaced with a Bayer version of an eight-bar color bar pattern.

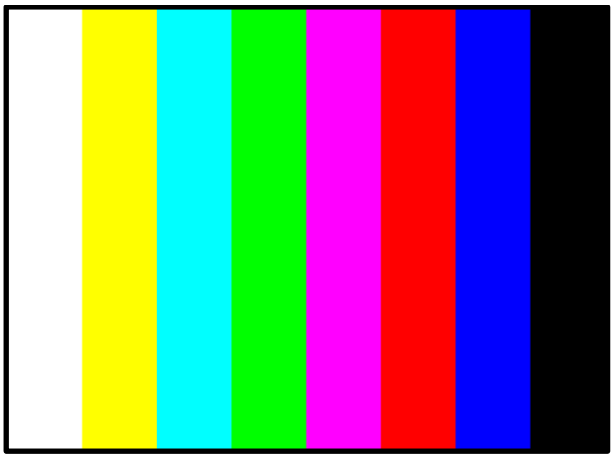


Figure 40 100 % Color Bar Pattern

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

In each bar, all pixels in the same color plane shall have the same value. The values shall be either 0% or 100% of full scale. Hence the colors are composed as follows:

Table 40 Color Composition for Color Bars DTP

Bar Color	R	Gr, Gb	B
White	%100	%100	%100
Yellow	%100	%100	%0
Cyan	%0	%100	%100
Green	%0	%100	%0
Magenta	%100	%0	%100
Red	%100	%0	%0
Blue	%0	%0	100%
Black	%0	%0	%0

6.10.2.4 Pattern Size

Where possible, each bar should occupy exactly 1/8th of the full output image width. The pattern should always occupy the full output image height.

6.10.2.5 "Fade to Grey" Color Bar Mode

In the "fade to grey color bar" test pattern mode, all pixel data is replaced with a color bar that fades vertically from 100 % color bars to mid grey. The "fade to grey" color bar pattern is designed to exercise more of the color space than 100 % bars while still requiring minimal hardware overhead. The following figure gives an indication of the pattern (although the pattern is generated as Bayer data):

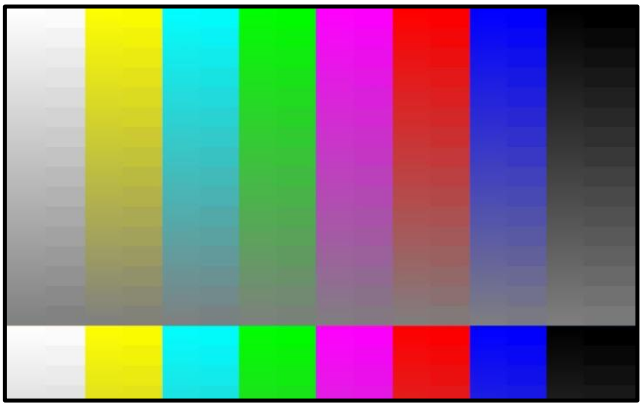


Figure 41 "Fade to Grey" Color Bar Pattern

The pattern is made up of eight vertical bars that fade vertically from one of the 100 % color bar colors towards a mid-grey at the bottom. The bars follow the same order as standard color bars, and each of the bars is sub-divided vertically into a left-hand side that contains a smooth gradient and a right-hand size that contains a quantized version.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

The aim of the quantized portion is to offer areas of flat-field Bayer data that should be large enough to result in known data values even after de-mosaic (independently of the de-mosaic algorithm). To ensure maximum dynamic range in the quantized data, the LSBs of the quantized data should be generated by copying the MSBs of the un-quantized data (rather than forcing them to 0).

6.10.2.6 Pattern Size

Ideally, each of the 8 sub-bars will occupy 1/8th of the full Bayer image width. As the data values of the gradient are most efficiently generated directly from a row counter, arbitrary vertical size is not always convenient for the fade to grey bars. As such, the bar will be divided to 2048 height portions, height of each bar should always be a multiple of 2048 pixels, and the pattern should be allowed to repeat on any additional lines of image.

6.10.2.7 PN16 mode

In the "PN16" test pattern mode, all pixel data is replaced with data from an internally generated 65536-bit pseudo-random 'PN16' sequence. The PN16 test pattern is included to ease testing of sensor-link integrity (measurement of bit error rate etc). The PN16 linear feedback shift register has polynomial X16+X14+X13+X11+1 in Fibonacci type notation (following figure). The PN16 sequence generator should be reset so as to start the sequence in a known state at the first replaced pixel of each frame.

The PN16 reset state shall be 0xFFFF. The sequence shall replace all pixel data in the image output commanded by the host.

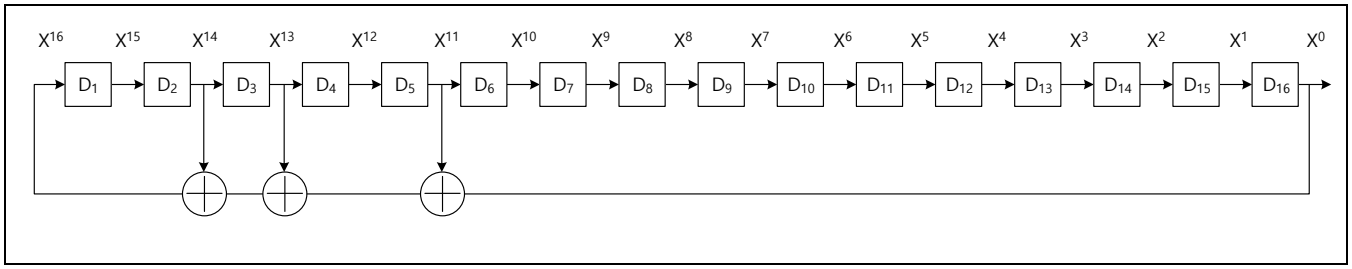


Figure 42 PN16 Linear Feedback shift Registers

6.10.2.8 Random Color Bar

In the "Random color bar" test pattern mode, all pixel data is replaced with random plots of R, G, B White, Black, C, M, Y Color bar test pattern that able to control each horizontal and vertical size. User can control horizontal and vertical size control of color bar via register.

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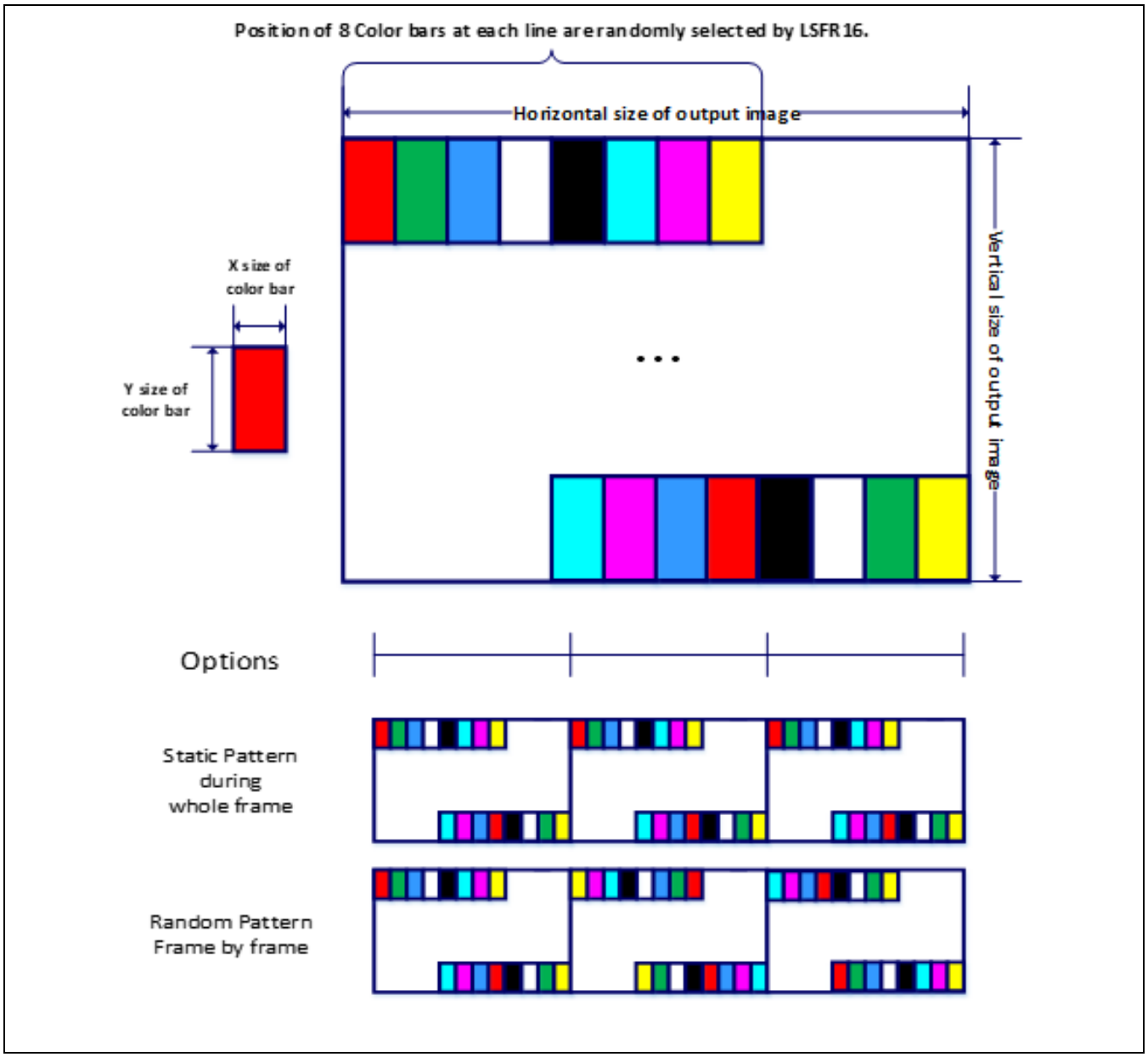


Figure 43 Random color bar

Table 41 Color bar size control

Index	Reset value	Bits	Register name	Type	Description
0x6052	0x0080	[12:0]	dtpgen_random_bar_width	RW1	X-size of one color bar
0x6054	0x0050	[12:0]	dtpgen_random_bar_height	RW1	Y0size of one color bar

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6.10.2.9 Test Cursors

In addition to the generation of full frame deterministic test patterns, the S5K3B6SA sensor can superimpose simple cursors on the image. The cursors are generated by replacing Bayer pixel data with fixed Bayer data within narrow vertical and/or horizontal bands of the image. Injection of the test cursors must be arranged such that the cursors can be superimposed on top of the full frame test patterns as well as array image data. Two cursors are defined, one vertical cursor and one horizontal.



Figure 44 Test cursors

The four parameters described in the following table are used to control the cursors. The position and width of each cursor can be controlled manually. Each cursor can be inhibited by setting its width parameter to zero. The vertical cursor can be put into an automatic position mode, where its position reflects the current value of the sensor frame counter. The vertical cursor moves from left to right in a displayed normal orientation image.

Table 42 Test Cursor Control Registers

Index	Reset value	Bits	Register name	Type	Description
0x604A	0x0000	[12:0]	dtpgen_h_cursor_width	RW1	Horizontal cursor width
0x604C	0x0020	[12:0]	dtpgen_h_cursor_position	RW1	Horizontal cursor position
0x604E	0x0000	[12:0]	dtpgen_v_cursor_width	RW1	Vertical cursor width
0x6050	0x0064	[15:0]	dtpgen_v_cursor_position	RW1	Vertical cursor position

The four registers used to define the output data in solid color mode also define the Bayer data used for the image cursors, so the cursors will not be visible.

6.10.3 ATOPIF (ATOP Interface)

This block is the process of setting up image before the processing the data.

6.10.3.1 ATOPIF Register

Table 43 ATOPIF Register

Index	Reset value	Bits	Register name	Type	Description

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

0x6000	0x0380	[0]	atopif_bypass	RW1	0 - block is enabled 1 - block is bypassed
		[7:5]	adc_res		ADC RES (0:10bit, 1:11bit, 2:12bit, 3:13bit, 4:14bit)
0x600E	0x0007	[0]	qec_on	RW1	dithering on/off (0:off, 1:on)
0x6002	0x0200	[12:0]	adc_offset_even	RW1	Even data offset
0x6004	0x0200	[12:0]	adc_offset_odd	RW1	Odd date offset

6.10.4 ADLC (Auto Dark Level Compensation)

It corrects the dark level generated in the FOB Pixel. Before reading the active pixel, extract and calculate the average of the pixel data of FOB area. Then, the average value of the fob pixel is compensated for the active pixel. Currently, the FADLC MIN 4 is used at the FOB average.

6.10.4.1 ADLC register

Table 44 ADLC Register

Index	Reset value	Bits	Register name	Type	Description
0x6102	0x0100	[11:0]	data_pedestal_adlc	RW1	positive pedestal
0x610E	0b1	[0]	fadlc_filter_en	RW1	0 - fadlc_filter disable 1 - fadlc filter enable
0x6100	0b1	[5]	fadlc_min4	RW1	'd0=all OB data, 'd1=4 min value of OB data
0x6106	0b1	[4]	fadlc_ch_cal	RW1	Select whether to distinguish the color channel when calculating fadlc for HCG/LCG. 'd0=total, 'd1=channel
	0b1	[5]	fadlc_en		fadlc enable (0:off, 1:on)
0x6160	0b1	[5]	fadlc_min4_m	RW1	'd0=all OB data, 'd1=4 min value of OB data
0x6166	0b1	[4]	fadlc_ch_cal_m	RW1	Select whether to distinguish the color channel when calculating fadlc for SPD. 'd0=total, 'd1=channel
	0b1	[5]	fadlc_en_m		fadlc enable (0:off, 1:on)
0x61B0	0b1	[5]	fadlc_min4_s	RW1	'd0=all OB data, 'd1=4 min value of OB data
6x61B6	0b1	[4]	fadlc_ch_cal_s	RW1	Select whether to distinguish the color channel when calculating fadlc for LPD VS. 'd0=total, 'd1=channel
	0b1	[5]	fadlc_en_s		fadlc enable (0:off, 1:on)

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6.10.4.2 ADLC Data Flow

The following shows the data process flow of ADLC. This process compensates for the average of data values in the FOB area from active data.

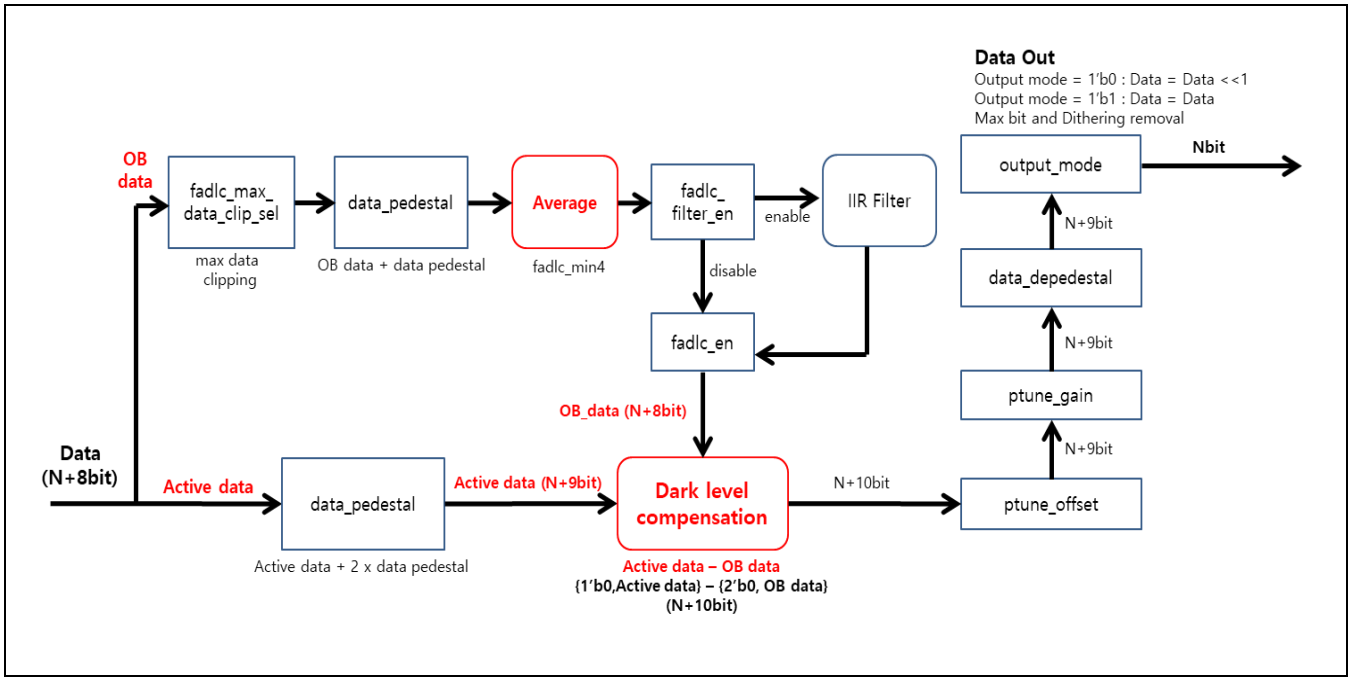


Figure 45 ADLC data Flow

6.11 ASP(Automotive signal processor)

Automotive Signal Processing block which support interleaving, Merge and Compress for HDR data and interface blocks. S5K3B6SA is able to output image with various format through ASP.

Table 45 ASP configuration registers

Index	Reset value	Bits	Register name	Type	Description
0x8000	0x0001	[2:0]	asp_12bit_12bit_ASP_asp_out_mode_sel	RW1	3'd0 : Merger / Compress 3'd1 : Interleaver – No line gap reorder 3'd2 : Interleaver – Line gap reorder 3'd3 : Interleaver – Pixel by pixel 3'd4 : RVM
0x8006	0x0001	[1:0]	asp_12bit_12bit_ASP_asp_gamma_comp_mode	RW1	2'd0 : Compress to 12bit 2'd1 : No compress 20bit (10b/10b) 2'd2 : Compress to 16bit (8b/8b)

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

					2'd3 : Compress to 14bit (7b/7b)
0x8002	0x0000	[2:0]	Asp_12bit_12bit_ASP_asp_debug_sel	RW1	3'd1 : Input data 3'd2 : BP Maker output 3'd3 : BINNING output 3'd4 : De-Interleaved LPD_HCG 3'd5 : De-Interleaved LPD_LCG 3'd6 : De-Interleaved SPD
0x8004	0x0000	[0]	Asp_12bit_12bit_ASP_asp_final_mux_sel	RW1	1'd0 : Image out from 'Out Mode MUX' 1'd1 : Image out from 'Debug MUX'

6.11.1 Crop

User can crop image at horizontal or vertical direction with crop blocks. There are 2 crop blocks in ASP. Crop0 located at the very first of Image pipe line, and crop1 block is at the last of image pipe line.

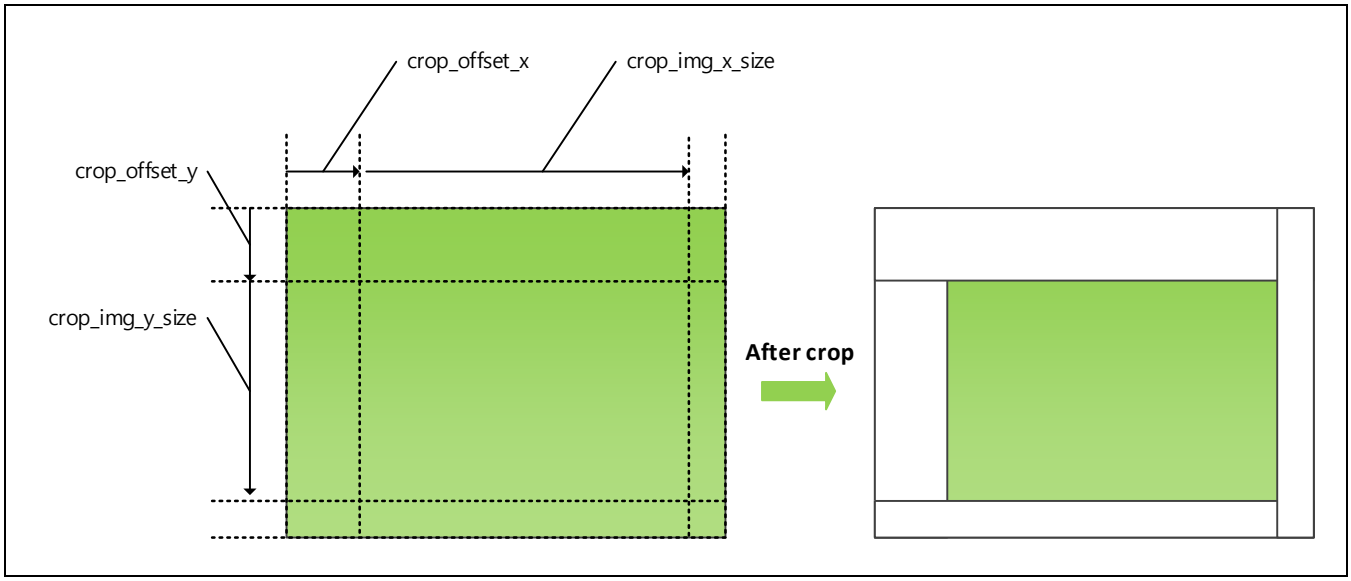


Figure 46 Block diagram of BPC

Table 46 Crop registers

Index	Reset value	Bits	Register name	Type	Description
0x8300	0x0001	[0:0]	crop_start0_crop_bypass	RW1	1'd1 : Bypass crop block. (Disable)
0x8302	0x0000	[13:0]	crop_start0_crop_offset_x	RW1	Horizontal offset to skip after margin.
0x8304	0x0000	[13:0]	crop_start0_crop_offset_y	RW1	Vertical offset to skip after margin.
0x8306	0x0780	[14:0]	crop_start0_crop_img_x_size	RW1	Set output width of horizontal column

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0x8308	0x0438	[14:0]	crop_start0_crop_img_y_size	RW1	Set output width of vertical line
0x830A	0x0000	[3:0]	crop_start0_crop_pad_x	RW1	If x size is not divided by 8. Settted pixel will be padded to the right
0x830C	0x0000	[15:0]	crop_start0_crop_pad_value	RW1	Value of padded data
0x8400	0x0001	[0:0]	crop_start1_crop_bypass	RW1	1'd1 : Bypass crop block. (Disable)
0x8402	0x0000	[13:0]	crop_start1_crop_offset_x	RW1	Horizontal offset to skip after margin.
0x8404	0x0000	[13:0]	crop_start1_crop_offset_y	RW1	Vertical offset to skip after margin.
0x8406	0x0780	[14:0]	crop_start1_crop_img_x_size	RW1	Set output width of horizontal column
0x8408	0x0438	[14:0]	crop_start1_crop_img_y_size	RW1	Set output width of vertical line
0x840A	0x0000	[3:0]	crop_start1_crop_pad_x	RW1	If x size is not divided by 8. Setted pixel will be padded to the right
0x840C	0x0000	[15:0]	crop_start1_crop_pad_value	RW1	Value of padded data

6.11.2 MSM - periodic mismatch correction

This block applies offset and gain to the all input streams according to a fixed periodic pattern (2x2). Input pixels are multiplied by the gain and added an offset value. The block removes the pedestal for internal calculations and adds it back to the output pixels.

Table 47 Periodic Mismatch Parameters

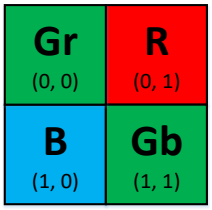
Index	Reset value	Bits	Register name	Type	Description
0x8500	0b1	[0:0]	msm_asp0_msm_bypass	RW1	1'd1 : Bypass MSM block. (Disable)
0x8504	0b1	[0:0]	msm_asp0_msm_residual_dark_en	RW1	enable calculation of residual dark output registers
	0b1	[1:1]	msm_asp0_msm_gain_stream_en	RW1	enable gain per input stream
	0b1	[2:2]	msm_asp0_msm_pedestal_en	RW1	enable input pedestal- remove pedestal before applying gain and add it afterwards
	0b1	[3:3]	msm_asp0_msm_dither_seed_update_en	RW1	enable to update dither seed between frames
	0b1	[4:4]	msm_asp0_msm_dither_start_en	RW1	enable dithering before applying gain
	0b1	[5:5]	msm_asp0_msm_dither_end_en	RW1	enable dithering before removing fractional bits
	0b1	[6:6]	msm_asp0_msm_short_low_periodic_offsets_en	RW1	enable short low periodic offset
	0b1	[7:7]	msm_asp0_msm_short_low_periodic_gains_en	RW1	enable short low periodic gains
	0b1	[8:8]	msm_asp0_msm_short_high_pe	RW1	enable short high periodic offset

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			riodic_offsets_en		
	0b1	[9:9]	msm_asp0_msm_short_high_periodic_gains_en	RW1	enable short high periodic gains
	0b1	[10:10]	msm_asp0_msm_medium_periodic_offsets_en	RW1	enable medium periodic offset
	0b1	[11:11]	msm_asp0_msm_medium_periodic_gains_en	RW1	enable medium periodic gains
	0b1	[12:12]	msm_asp0_msm_long_low_periodic_offsets_en	RW1	enable long low periodic offset
	0b1	[13:13]	msm_asp0_msm_long_low_periodic_gains_en	RW1	2x2 Periodic offsets (format- s13 bits)
	0b1	[14:14]	msm_asp0_msm_long_high_periodic_offsets_en	RW1	enable long high periodic offset)
	0b1	[15:15]	msm_asp0_msm_long_high_periodic_gains_en	RW1	2x2 Periodic offsets (format- s13 bits)
0x850E	0x0000	[11:0]	msm_asp0_msm_pedestal_long_high	RW1	Pedestal value
0x8510	0x0000	[11:0]	msm_asp0_msm_pedestal_long_low	RW1	Pedestal value
0x8512	0x0000	[11:0]	msm_asp0_msm_pedestal_medium	RW1	Pedestal value
0x8514	0x0000	[11:0]	msm_asp0_msm_pedestal_short	RW1	Pedestal value
0x856E	0x0001	[3:0]	msm_asp0_msm_shift	RW1	shifter for periodic gains

6.11.2.1 Diagram of pixel processing at MSM block

Each channel is defined as following:



The naming convention of the registers refers to a 2x2 pattern as following:

xxx_0_0	xxx_0_1
xxx_1_0	xxx_1_1

"XXX" contains either gain or offset + stream name (long_high\long_low\medium\short).

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Applied periodic gain:

Set value / 2048 * (2 ^ (msm_shift – 1))

Applied periodic offset:

If set value > 4095 then "set value - 8192" is applied periodic offset

if set value ≤ 4095 then " Set value" is applied periodic offset

6.11.2.2 Periodic Gain Calculation Pseudo Code

The max and step of periodic gain that can be set are determined according to the shift value. The range of periodic gain is 2^(msm_shift) and the step is 1/(4096/(2^(msm_shift - 1)). The set_value of the initial gain(1x) must be change according to the set value of shift.

Periodic_gain_0_0 = gain_1 * (4096/2^(msm_shift - 1))
Periodic_gain_0_1 = gain_2 * (4096/2^(msm_shift - 1))
Periodic_gain_1_0 = gain_3 * (4096/2^(msm_shift - 1))
Periodic_gain_1_1 = gain_4 * (4096/2^(msm_shift - 1))

Name	Bits	Addr LPD_HCG	Addr LPD_LCG	Addr SPD RS SPD SR	Addr SPD SR LPD_VS	Description
msm_*_periodic_offset_0_0	[12:0]	0x855E	0x854E	0x853E	0x851E	Periodic offset for Gr (format- signed 13 bits)
msm_*_periodic_offset_0_1	[12:0]	0x8560	0x8550	0x8540	0x8520	Periodic offset for R (format- signed 13 bits)
msm_*_periodic_offset_1_0	[12:0]	0x8562	0x8552	0x8542	0x8522	Periodic offset for B (format- signed 13 bits)
msm_*_periodic_offset_1_1	[12:0]	0x8564	0x8554	0x8544	0x8524	Periodic offset for Gb (format- signed 13 bits)
msm_*_periodic_gain_0_0	[11:0]	0x8566	0x8556	0x8546	0x8526	Periodic gain for Gr - multiplier (format- 0.12 bits)
msm_*_periodic_gain_0_1	[11:0]	0x8568	0x8558	0x8548	0x8528	Periodic gain for R - multiplier (format- 0.12 bits)
msm_*_periodic_gain_1_0	[11:0]	0x856A	0x855A	0x854A	0x852A	Periodic gain for B - multiplier (format- 0.12 bits)
msm_*_periodic_gain_1_1	[11:0]	0x856C	0x855C	0x854C	0x852C	Periodic gain for Gb - multiplier (format- 0.12 bits)

Figure 47 Registers for MSM

6.11.3 MSM blocks after Merger

In addition, there are another 2 blocks of MSM- both of them are located after the Merger block. MM after merge applies gain for each channel of CFA

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Table 48 Registers for MSM1

Register Name	Index	Field Name	Type	Bits	Value	Reset Value	Description
msm_1_bypass	0xa400	msm_1_bypass	RW1	[0:0]	0~1	0x0001	0- normal working mode 1- bypass
msm_1_p_gains_2x2	0xa402	msm_1_p_gains_2x2_0_0	RW1	[11:0]	0~4095	0x0800	2x2 periodic gains 0.12 bits
msm_1_p_gains_2x2_1	0xa404	msm_1_p_gains_2x2_0_1	RW1	[11:0]	0~4095	0x0800	2x2 periodic gains 0.12 bits
msm_1_p_gains_2x2_2	0xa406	msm_1_p_gains_2x2_1_0	RW1	[11:0]	0~4095	0x0800	2x2 periodic gains 0.12 bits
msm_1_p_gains_2x2_3	0xa408	msm_1_p_gains_2x2_1_1	RW1	[11:0]	0~4095	0x0800	2x2 periodic gains 0.12 bits
msm_1_p_gains_global	0xa40a	msm_1_p_gains_global	RW1S	[14:0]	0~16384	0x1000	global periodic gains 0.12 bits
msm_1_shift	0xa41c	msm_1_shift	RW1	[2:0]	0~7	0x0001	shifting after multiplication in periodic gains

Table 49 Registers for MSM0

Register Name	Index	Field Name	Type	Bits	Value	Reset Value	Description
msm_0_bypass	0x9a00	msm_0_bypass	RW1	[0:0]	0~1	0x0001	0- normal working mode 1- bypass
msm_0_p_gains_2x2	0x9a02	msm_0_p_gains_2x2_0_0	RW1	[11:0]	0~4095	0x0800	2x2 periodic gains 0.12 bits
msm_0_p_gains_2x2_1	0x9a04	msm_0_p_gains_2x2_0_1	RW1	[11:0]	0~4095	0x0800	2x2 periodic gains 0.12 bits
msm_0_p_gains_2x2_2	0x9a6	msm_0_p_gains_2x2_1_0	RW1	[11:0]	0~4095	0x0800	2x2 periodic gains 0.12 bits
msm_0_p_gains_2x2_3	0x9a8	msm_0_p_gains_2x2_1_1	RW1	[11:0]	0~4095	0x0800	2x2 periodic gains 0.12 bits
msm_0_p_gains_global	0x9aa	msm_0_p_gains_global	RW1S	[14:0]	0~16384	0x1000	global periodic gains 0.12 bits
msm_0_shift	0x9ac	msm_0_shift	RW1	[2:0]	0~7	0x0001	shifting after multiplication in periodic gains

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6.11.4 Radial Gain Control

6.11.4.1 General Description

The goal of the block is to compensate the relative illumination (R.I.) between the LPD and SPD. If radial gain IP isn't operated, color artifact occurs as shown in Fig. 39. Original image on left, pink peripherals are noticeable due to R.I. difference between the LPD and SPD. The compensated image on the right shows that the block has corrected color ratios

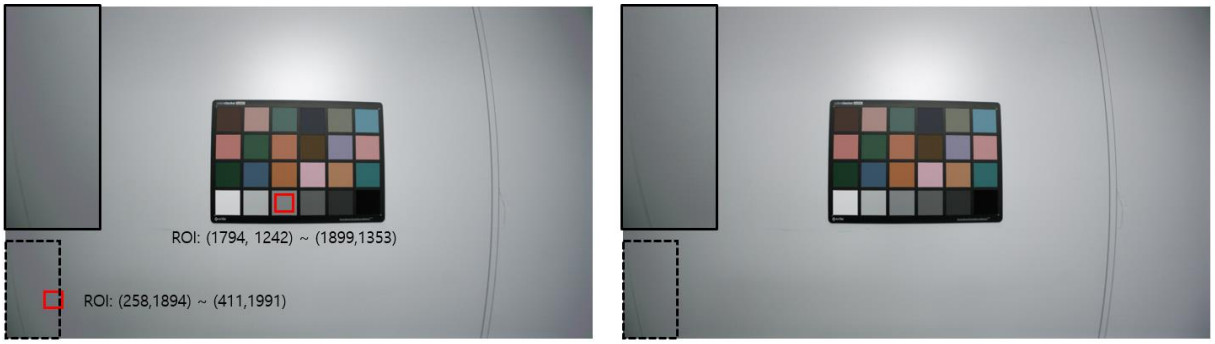


Figure 48 Example of the color artifact

The algorithm consists of three steps.

At first, it sets the optical center and calculates distance map. The distance map is calculated as follow steps.

1) It calculates the square distance from the center to each pixel: $R^2 = (x - x_0)^2 + (y - y_0)^2$

2) It normalizes using maximal square radius and obtains the distance map.

Index	Reset value	Bits	Register name	Type	Description
0xa600	0x2000	[13:0]	radgain_0_center_offset_hor_2	RW	Stream2 = SPD_CDS Horizontal position = center - offset, 8192 is 0 position
0xa602	0x2000	[13:0]	radgain_0_center_offset_ver_2	RW	Stream2 = SPD_CDS Vertical position = center - offset, 8192 is 0 position
0xa654	0x0000	[13:0]	radgain_0_max_radius_2	RW	Stream2 = SPD_LOF Maximum radius stream 3
0xa656	0x2000	[13:0]	radgain_0_center_offset_hor_3	RW	Stream3 = SPD_LOF Horizontal position = center - offset, 8192 is 0 position

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0xa658	0x2000	[13:0]	radgain_0_center_offset_ver_3	RW	Stream3 = SPD_LOF Vertical position = center - offset, 8192 is 0 position
0xa6aa	0x0000	[13:0]	radgain_0_max_radius_3	RW	Stream3 = SPD_LOF Maximum radius stream 3

The second step is the bi-quadratic process. It uses the 10 control parameters (P0,… , P9). The 10 control parameters are calculated from 10 areas as shown in Fig. 40. The gain values of pixels are calculated using the bi-quadratic interpolation. (Similar to cubic Bezier curve).

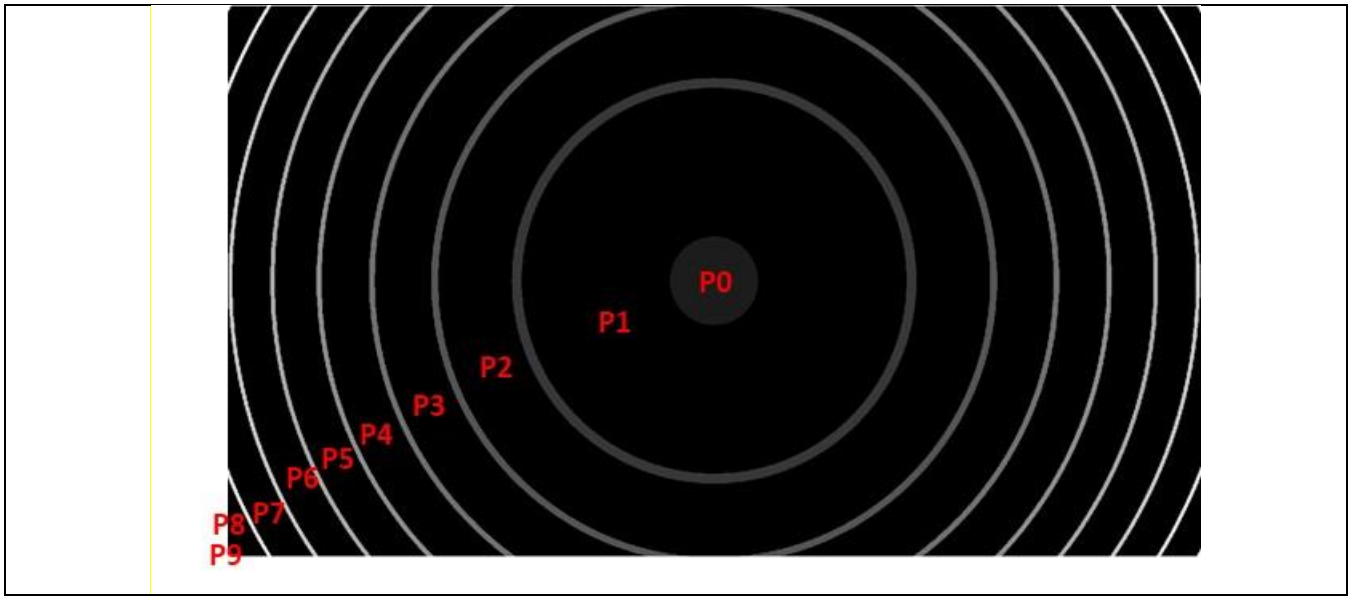


Figure 49 Area indication associated with 10 control parameters

The final step generates the final output image. The final image can be obtained by multiplying the gain values into the input image.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.11.4.2 Related Registers

Table 50 Registers of Radial gain IP

Index	Reset value	Bits	Register name	Type	Description
0xa500	0x0001	[1:0]	radgain_0_bypass	RW1	0 - normal working mode 1- bypass
0xa502 bitwise	0x0001	[0:0]	radgain_0_pedestal_en	RW1	Pedestal enable
	0x0001	[1:1]	radgain_0_stream_gain_en_0		gain enable stream 0
	0x0001	[2:2]	radgain_0_stream_gain_en_1		gain enable stream 1
	0x0001	[3:3]	radgain_0_stream_gain_en_2		gain enable stream 2
	0x0001	[4:4]	radgain_0_stream_gain_en_3		gain enable stream 3
	0x0001	[5:5]	radgain_0_dither_en		Enable dithering

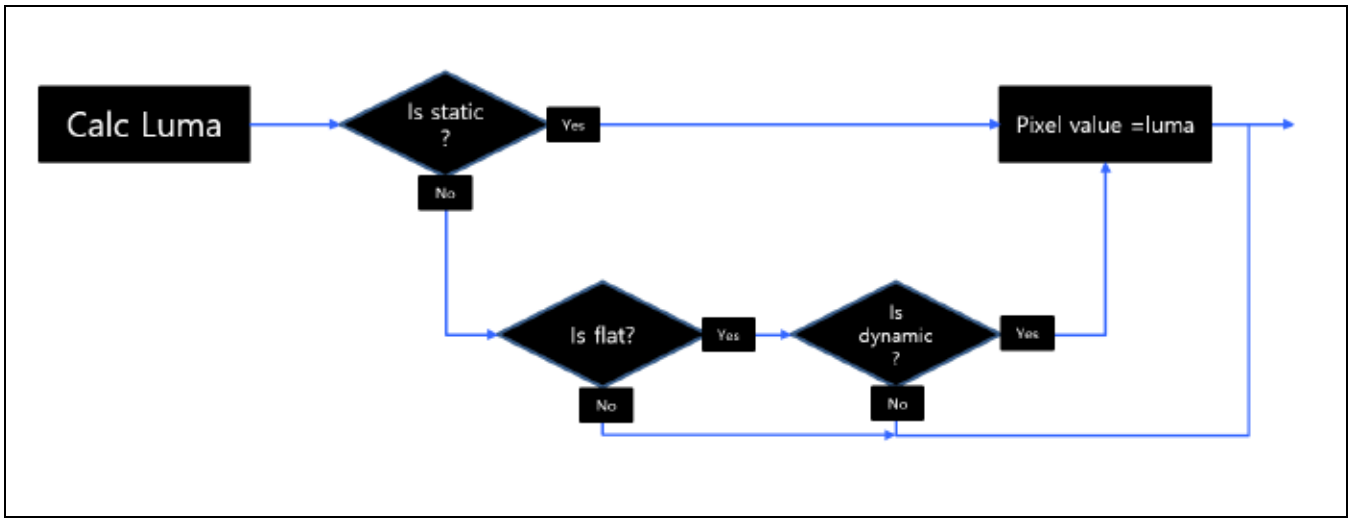
6.11.5 BPC(Bad Pixel Correction)

6.11.5.1 General Description

The block considers a static BP list provided as a single bit map.

The block performs both static BPC and dynamic detection and correction of bad pixels while considering the relevant pixels' values at the vicinity of the central pixel.

The relevant pixels are the pixels of the same type and color that are not marked as static bad pixels.



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Figure 50 Block diagram of BPC

Table 51 Register for Bad Pixel Correction

Index	Reset value	Bits	Name	Description
0x106C	0x2C08	[15:0]	sys_sys_dummy1	Set base axis point of active pixel array.
0x8700	0x0001	[0:0]	bpmlc_asp_bypass	1'd0 : Activate block, 1'd1 : Bypass block
0x8702	0x0004	[5:0]	bpmlc_asp_embedded_line	Set number of embedded lines at the first of active line.
0x8800	0x0001	[0:0]	dspcllc_0_bypass	1'd0 : Activate block, 1'd1 : Bypass block.
0x8804	0x0003	[1:0]	Dspcllc_0_correction_mode	2'd0 : non, 2'd1 : Static, 2'd2 : Dynamic, 2'd3 : Both

6.11.5.2 Static BPC

The input BP map marks bad pixels. This map is obtained from the bp-marker block that reads the coordinates of the BP pattern in a 3x3 local neighborhood from the OTP memory. The block replaces static bad pixel with the average between the minimum and maximum values in the neighborhood with some exceptions. Explained in the next sections.

BP data should write in OTP in a manner of ascending order, first sort in ascending order of vertical coordinate and secondly in horizontal coordinate. Basically, static BP map in OTP memory is written at the Wafer test level in production.

6.11.5.3 Neighborhood Pixels

The values of the pixels in the neighborhood of a central pixel have 2 uses:

- Calculating the value to replace bad pixels (both static and dynamic).
- Identifying dynamic bad pixels.

The pixels of the same exposure type and color that are not marked as static BP are used as neighborhood pixels.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.11.5.4 Dynamic BPC

Dynamic BPC is only applied in flat regions.

In order to identify a flat region, the difference between minimum and maximum values in the pixel's neighborhood is compared to a threshold. The threshold is scaled linearly with respect to the pixel's luma. On the other hand, In order to detect a bad pixel, the difference between the central pixel value and the luma is compared to a threshold. The threshold is scaled with respect to the luma value.

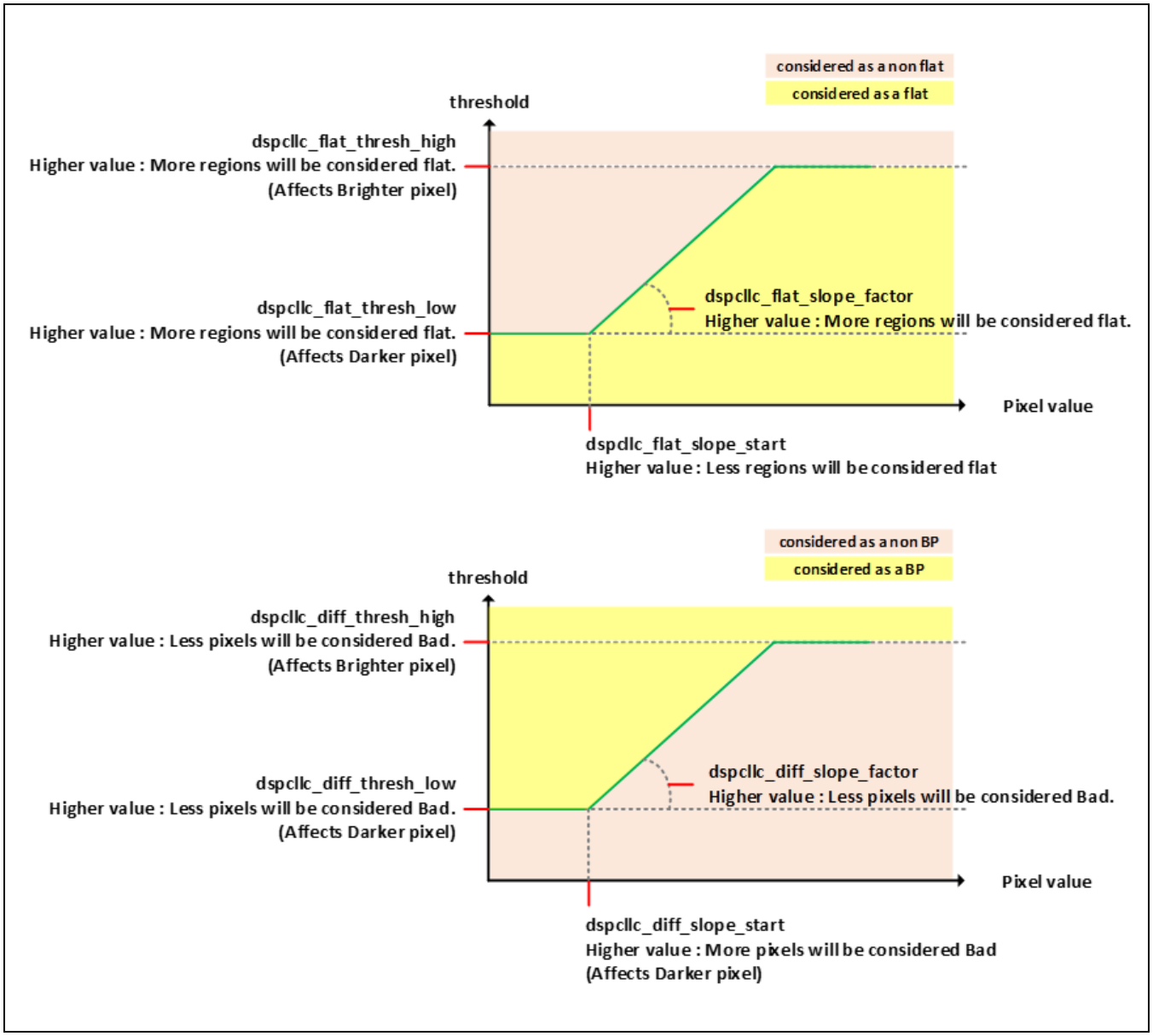


Figure 51 Register definition for flat zone and for bad pixel

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Table 52 Registers for flat region detection

Name	Bits	Index				Set value	Description
		Addr LPD_HCG	Addr LPD_LCG	Addr SPD SR SPD SR	Addr LPD_VS SPD RS		
dspcllc_0_flat_despeckle_en	[12:0]	0x8806[0]	0x8806[4]	0x8806[8]	0x8806[12]	0x1111	1'd1 : Activate operation, 1'd0 : Disable.
dspcllc_0_pedestal_in_#	[11:0]	0x8808	0x881A	0x882C	0x883E	0x0000	Pedestal_in
dspcllc_0_flat_thresh_high_#	[11:0]	0x880A	0x881C	0x882E	0x8840	0x012C	upper bound for the flatness test threshold
dspcllc_0_flat_thresh_low_#	[11:0]	0x880C	0x881E	0x8830	0x8842	0x0014	low flat threshold
dspcllc_0_flat_slope_start_#	[11:0]	0x880E	0x8820	0x8832	0x8844	0x0014	start number for applying scaling to the flat threshold.
dspcllc_0_flat_slope_factor_#	[11:0]	0x8810	0x8822	0x8834	0x8846	0x20D0	factor for scaling to the flat threshold
dspcllc_0_diff_thresh_high_#	[11:0]	0x8812	0x8824	0x8836	0x8848	0x02BC	upper bound for the diff test threshold
dspcllc_0_diff_thresh_low_#	[11:0]	0x8814	0x8826	0x8838	0x884A	0x0007	low diff threshold
dspcllc_0_diff_slope_start_#	[11:0]	0x8816	0x8828	0x883A	0x884C	0x0006	start number for applying scaling to the diff threshold
dspcllc_0_diff_slope_factor_#	[11:0]	0x8818	0x882A	0x883C	0x884E	0x008C	factor for scaling to the diff threshold

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.11.5.5 Use Case

Example setting for using Static BPC	
s106c2C08	//Set basis axis point at active pixel array
s87000000	//bpmlc_bypass //1'd1:Bypass, 1'd0:Enable
s87020000	//bpmlc_embedded_line
s88000000	//dspcllc_bypass //1'd1:Bypass, 1'd0:Enable
s88040001	//dspcllc_0_correction_mode //0-non, 1-static BP, 2-dynamic BP, 3-static and dynamic BP
s88061111	//dspcllc_0_despeckle_low_cost_config //[0]:LPD_HCG, [4]:LPD_LCG, [8]:SPD SR, [12]:LPD_VS / SPD RS
Example setting for using Dynamic BPC	
s88000000	//dspcllc_0_bypass //1'd1:Bypass, 1'd0:Enable
s88020002	//0-empty, 1-static BPs, 2-dynamic BPs, 3-static+dynamic BPs
s88040002	//0-empty, 1-static BPs, 2-dynamic BPs, 3-static+dynamic BPs
s88061111	//dspcllc_0_flat_despeckle_en_0,1,2,3
//LPD_HCG	
s88080100	//dspcllc_0_pedestal_0
s881202BC	//dspcllc_0_diff_thresh_high_0
s88140007	//dspcllc_0_diff_thresh_low_0
s880A012c	//dspcllc_0_flat_thresh_high_0
s880C0014	//dspcllc_0_flat_thresh_low_0
//LPD_LCG	
s881A0100	//dspcllc_0_pedestal_1
s882402BC	//dspcllc_0_diff_thresh_high_1
s88260007	//dspcllc_0_diff_thresh_low_1
s881C042C	//dspcllc_0_flat_thresh_high_1
s881E0014	//dspcllc_0_flat_thresh_low_1
//SPD_RS	
s882C0100	//dspcllc_0_pedestal_2
s883602BC	//dspcllc_0_diff_thresh_high_2
s88380007	//dspcllc_0_diff_thresh_low_2
s882E032C	//dspcllc_0_flat_thresh_high_2
s88300014	//dspcllc_0_flat_thresh_low_2
//SPD_SR	
s883E0100	//dspcllc_0_pedestal_3
s884802BC	//dspcllc_0_diff_thresh_high_3
s884A0007	//dspcllc_0_diff_thresh_low_3
s8840032C	//dspcllc_0_flat_thresh_high_3,
s88420014	//dspcllc_0_flat_thresh_low_3

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.11.6 Phase Correction

6.11.6.1 General Description

The pixel design of S5K3B6SA, separates long and short actual pixel centers. In order to correct the channel misalignment one needs to perform a corrective shift by ½ the pixel size, in both horizontal and vertical directions. This is done using an interpolation filter which is performed on the small photo diode streams. This function is only available at SPD.

The block performs linear correction of the spatial shift arising from the pixel architecture.

The block's inputs are four 12bit streams of different exposures. The block processes only one\two of the input streams, depending on the HDR scheme used (works only on the streams from the small photo diode). Default correction kernel applies ½ pixel shift in both horizontal and vertical directions (up-left) and correction kernel dimensions are 5X5.

The filter is designed using bi-polynomial interpolation (of 4th degree). It uses a 5X5 correction kernel with (1/2, 1/2) shift. The default filter coefficients (in 0.13 precision) are: [140,-1008, 7560, 1680,-180].

Table 53 General control registers for phase correction

Index	Reset value	Bits	Register name	Type	Description
0x8600	0x0001	[0:0]	phasecorrection_0_bypass	RW1	Bypass block operation.
0x8602	0x2000	[13:0]	phasecorrection_0_size_y	RW1	ROI size of Y
0x8604	0x0000	[13:0]	phasecorrection_0_start_y	RW1	Y start Regions
0x8606	0x4000	[14:0]	phasecorrection_0_size_x	RW1	ROI size of X
0x8608	0x0000	[14:0]	phasecorrection_0_start_x_region_0_0	RW1	X start Regions
0x860A	0x0000	[0:0]	phasecorrection_0_bin_horz_en_0_0	RW1	Horizontal filter enable
0x860C	0x0000	[0:0]	phasecorrection_0_bin_vert_en_0_0	RW1	Vertical filter enable
0x8624	0x0000	[11:0]	phasecorrection_in_out_pedestal	RW1	Input and output pedestal
0x8626	0x0000	[0:0]	phasecorrection_0_mirror_horz_en_0_0	RW1	Mirror Horizontal filter
0x8628	0x0000	[0:0]	phasecorrection_0_mirror_vert_en_0_0	RW1	Mirror Vertical filter

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6.11.6.2 Phase Correction Parameters

The values of the filter itself are given using the following registers (two's compliment)

Table 54 Phase correction register

Index	Reset value	Bits	Register name	Type	Description
0x8610	0x008c	[13:0]	phasecorrection_0_coeffs_horz_0_0	RW1	coeffs for horizontal correction. [left2]
0x8612	0x3c10	[13:0]	phasecorrection_0_coeffs_horz_0_1	RW1	coeffs for horizontal correction. [left1]
0x8614	0x1d88	[13:0]	phasecorrection_0_coeffs_horz_0_2	RW1	coeffs for horizontal correction. [center]
0x8616	0x0690	[13:0]	phasecorrection_0_coeffs_horz_0_3	RW1	coeffs for horizontal correction. [right1]
0x8618	0x3f4c	[13:0]	phasecorrection_0_coeffs_horz_0_4	RW1	coeffs for horizontal correction. [right2].
0x861A	0x008c	[13:0]	phasecorrection_0_coeffs_vert_0_0	RW1	coeffs for vertical correction. [above2]
0x861C	0x3c10	[13:0]	phasecorrection_0_coeffs_vert_0_1	RW1	coeffs for vertical correction. [above1]
0x861E	0x1d88	[13:0]	phasecorrection_0_coeffs_vert_0_2	RW1	coeffs for vertical correction. [center]
0x8620	0x0690	[13:0]	phasecorrection_0_coeffs_vert_0_3	RW1	coeffs for vertical correction. [below1]
0x8622	0x3f4c	[13:0]	phasecorrection_0_coeffs_vert_0_4	RW1	coeffs for vertical correction. [below2]

NOTE: Coefficient values range are -8192 to 8191 and sum of whole value should be 8191.

6.11.6.3 Use Case of Phase Correction

<Default>
s86000000 //phasecorrection_0_bypass (bin_corr enable)
s860203B0 //phasecorrection_0_size_y
s86040000 //phasecorrection_0_start_y
s86060508 //phasecorrection_0_size_x
s86080000 //phasecorrection_0_start_x_region
s860A0001 //phasecorrection_0_bin_mode (horizontal_en)
s860C0001 //phasecorrection_0_bin_vert_en
s860E0002 //phasecorrection_0_periodicity_mode
s8610008C //phasecorrection_0_coeffs_horz

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s86123C10 //phasecorrection_0_coeffs_horz_1
s86141D88 //phasecorrection_0_coeffs_horz_2
s86160690 //phasecorrection_0_coeffs_horz_3
s86183F4C //phasecorrection_0_coeffs_horz_4
s861A008C //phasecorrection_0_coeffs_vert
s861C3C10 //phasecorrection_0_coeffs_vert_1
s861E1D88 //phasecorrection_0_coeffs_vert_2
s86200690 //phasecorrection_0_coeffs_vert_3
s86223F4C //phasecorrection_0_coeffs_vert_4
s86240100 //phasecorrection_0_in_out_pedestal

<Phase correction applied>
s86000000 //phasecorrection_0_bypass (bin_corr enable)
s860203B0 //phasecorrection_0_size_y
s86040000 //phasecorrection_0_start_y
s86060508 //phasecorrection_0_size_x
s86080000 //phasecorrection_0_start_x_region
s860A0001 //phasecorrection_0_bin_mode (horizontal_en)
s860C0001 //phasecorrection_0_bin_vert_en
s860E0002 //phasecorrection_0_periodicity_mode
s861002BD //phasecorrection_0_coeffs_horz
s861207C3 //phasecorrection_0_coeffs_horz_1
s86140AFD //phasecorrection_0_coeffs_horz_2
s861607C3 //phasecorrection_0_coeffs_horz_3
s861802BD //phasecorrection_0_coeffs_horz_4
s861A02BD //phasecorrection_0_coeffs_vert
s861C07C3 //phasecorrection_0_coeffs_vert_1
s861E0AFD //phasecorrection_0_coeffs_vert_2
s862007C3 //phasecorrection_0_coeffs_vert_3
s862202BD //phasecorrection_0_coeffs_vert_4
s86240100 //phasecorrection_0_in_out_pedestal

6.11.7 HQDNS - Noise Reduction

6.11.7.1 General Description

The denoising concept is based on the following principles
1. Pattern matching
2. Separate UV and green denoising

The pattern matching technique is very strong technique especially in noisy environment. It based on large area scan and search for pattern similar to that pattern in the central pixel. All the central pixels of the similar patterns are averaged producing the new pixel value.

The following general scheme describes the block architecture:

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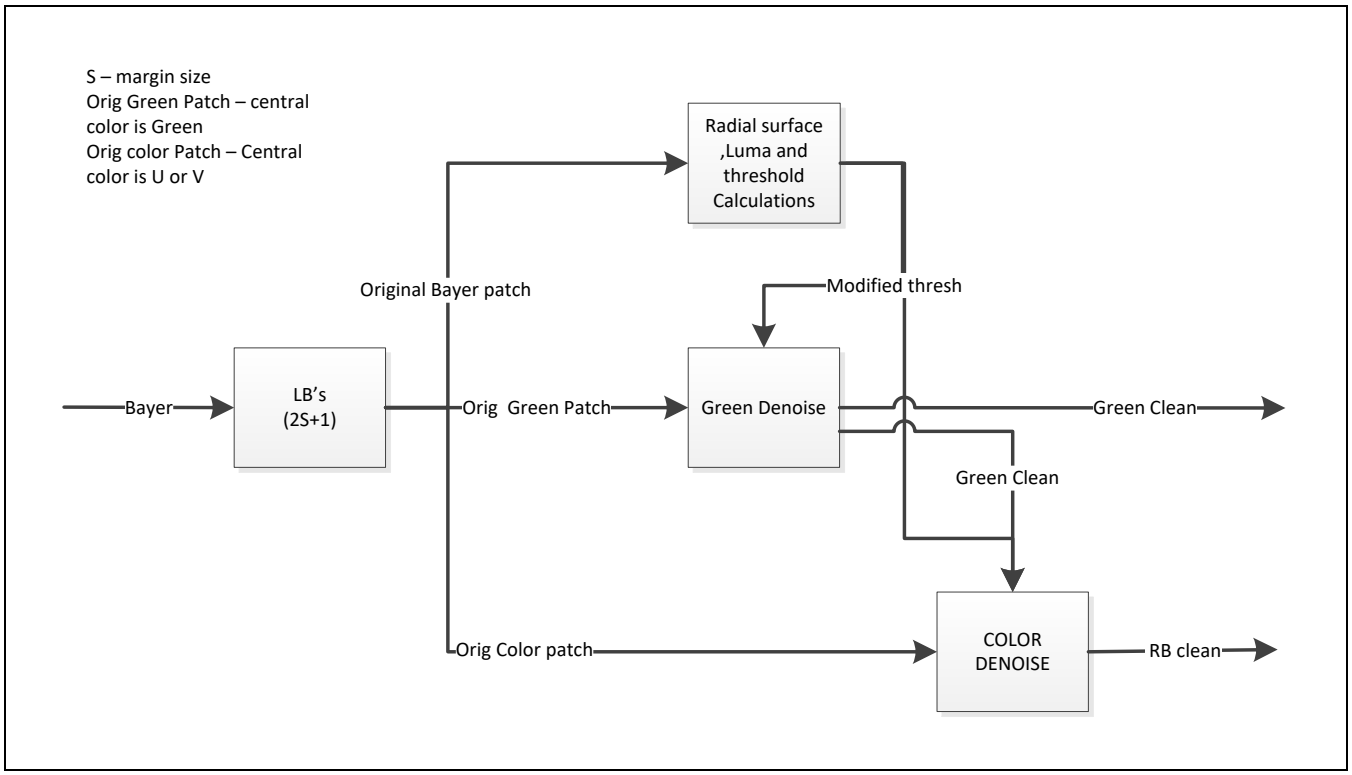
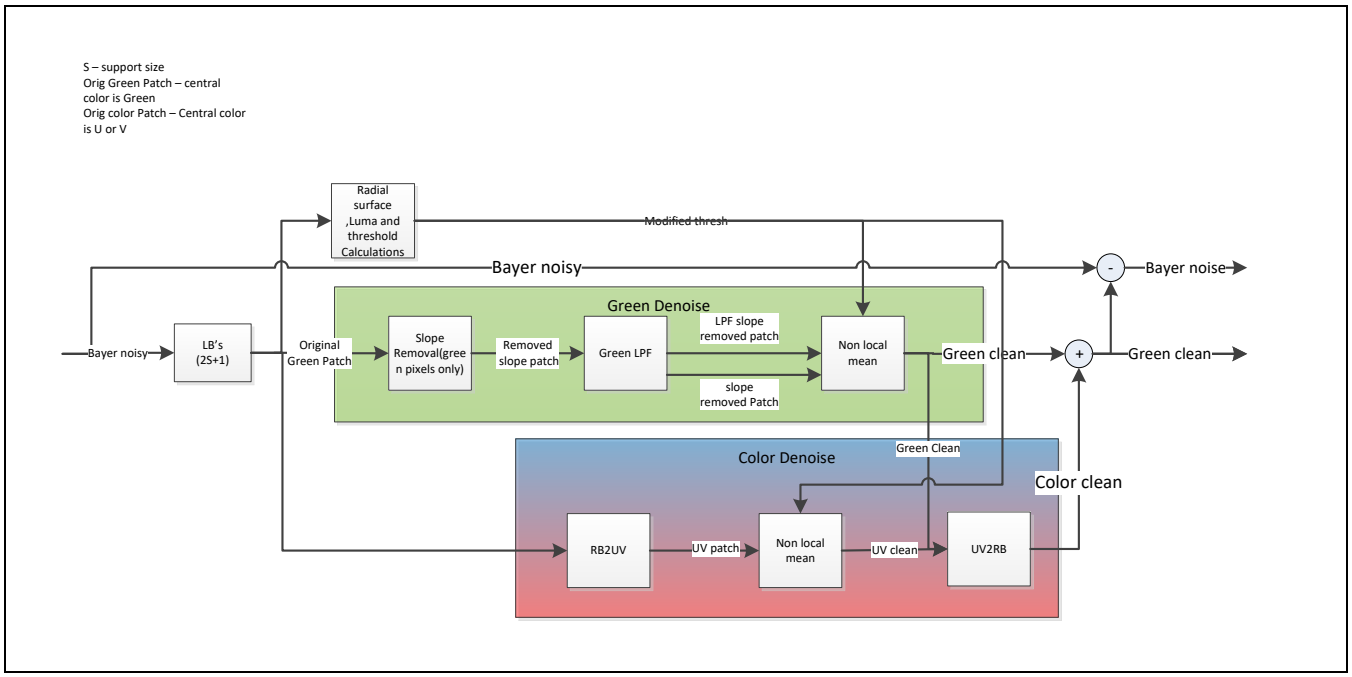


Figure 52 general high level scheme for Denoise block

Each of the sub blocks can be divided to inner sub–blocks. For each main sub–block you can find a detailed explanation in the scheme below:



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Figure 53 sub block scheme for Denoise block

6.11.7.2 Thresholds calculation

6.11.7.2.1 Luma calculation

Calculation of averaged luma is being made. Linear interpolation defined by low luma and high luma threshold register is calculated to each pixel. Result is multiplied by the radial surface being calculated for each coordinates.

Calculation of luma is being done specifically to channel that will be used on the SAD calculation

Luma calculation will be performed on those specific pixels from the central of the support.

Following calculation will be performed with the following Luma weight mask:

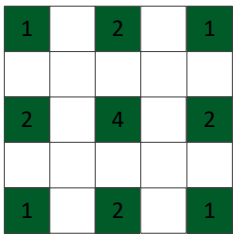
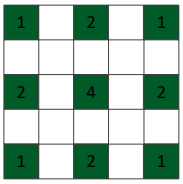


Figure 54 Luma weight mask

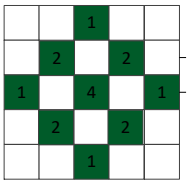
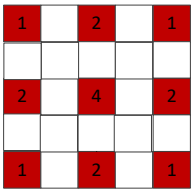
$$Luma = \frac{1}{N} \sum_{i=index \in support} W[i] * support[i]$$

The following lists describe all the luma patterns according to the activated mode of the block

Table 55 Bayer luma masks

Mask number	Central pixel	High frequency pattern	Luma mask	comments
1	Green	off (green_high_resolution_sad_en : 0)		

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2	Green	on (green_high_resolution_sad_en : 1)		
3	UV	Not relevant		

6.11.7.2.2 Radial distance calculation

Radial calculation feature is the option to increase the SAD thresholds as a function of the distance from the center of the image. The procedure goes as follows:

X and y coordinates are being calculated (relevant for crop only):

```
x = _min( x + m_regs.start_x, m_regs.sensor_width );
y = _min( y + m_regs.start_y, m_regs.sensor_height);
```

the distance in each axis from the center of the image is being calculated:

```
cx = x - m_regs.sensor_width/2;
cy = y - m_regs.sensor_height/2;
cx = abs(cx)/64; cy = abs(cy)/64;
```

The total distance is being calculated and shifted so only major distance will have a “radial” influence on thresholds:

```
square_distance = (cx*cx + cy*cy);
```

Finally, factor per coordinate is being calculated (proportional to radial thresholds power register):

```
surface = square_distance*m_regs.radial_thresholds_power/RADIAL_POWER_FRACTION +
          RADIAL_SURFACE_UNITY;
iLimit = _max( RADIAL_SURFACE_UNITY, m_regs.radial_thresholds_limit );
radial_surface = _min( surface, iLimit );
```

6.11.7.2.3 Find thresholds calculation

Both Luma calculation and radial surface calculation are considered for determining final threshold values. Two registers are defined to each exposure ratio and each channel (a total of 12 registers) (threshold_high and threshold_low). The final thresholds are computed using simple linear interpolation method.

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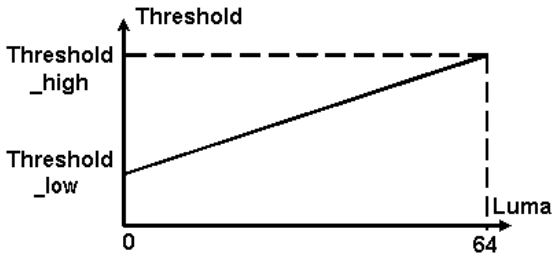


Figure 55 Linear interpolation

$$result = low + (high - low) * luma / max_luma$$

(max_luma>luma>0)

If radial surface is applied, the result is the multiplied by the radial weight.

$$result = result * radial_weigth / RADIAL_SURFACE_UNITY$$

(RADIAL_SURAFCE_UNITY>radial_weigth>0)

Final result is threshold for the pixel in all future SAD results.

6.11.7.3 RB2UV

UV range for chrome pixels is calculated using green median of the color pixel. The computation is done as shown in the following figure and the resulted values are pseudo UV values.

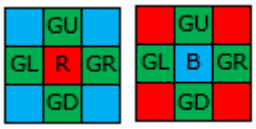


Figure 56 RB2UV

U = R - medium(GU, GD, GL, GR)
V = B - medium(GU, GD, GL, GR)

6.11.7.4 UV2RB

Conversion back to RB range will be the reverse process with one exception. We use corrected value from the denoising

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6.11.7.5 Slope estimation and slope removal

Before performing green denoising, the slope of the signal is removed. The principle of the slope removal is shown in the following figures.

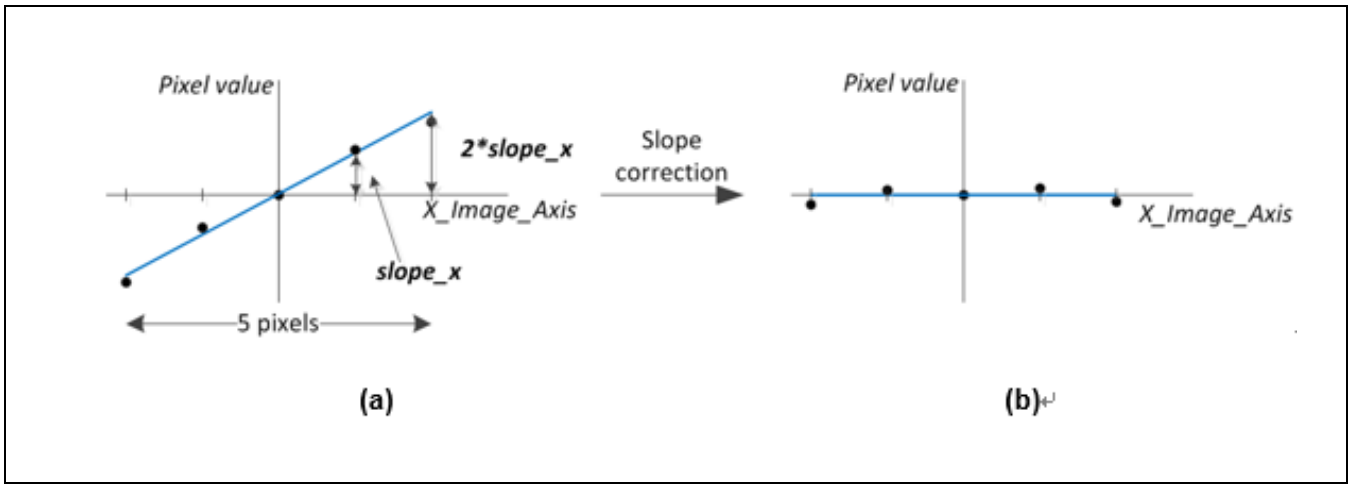


Figure 57 Slope removal principle. (a) before and (b) after slope correction

To compute the slope linear regression computation is applied for both axis. Linear regression attempts to fit the observed data with linear function. The goal of linear regression is to fit line into the observed data:

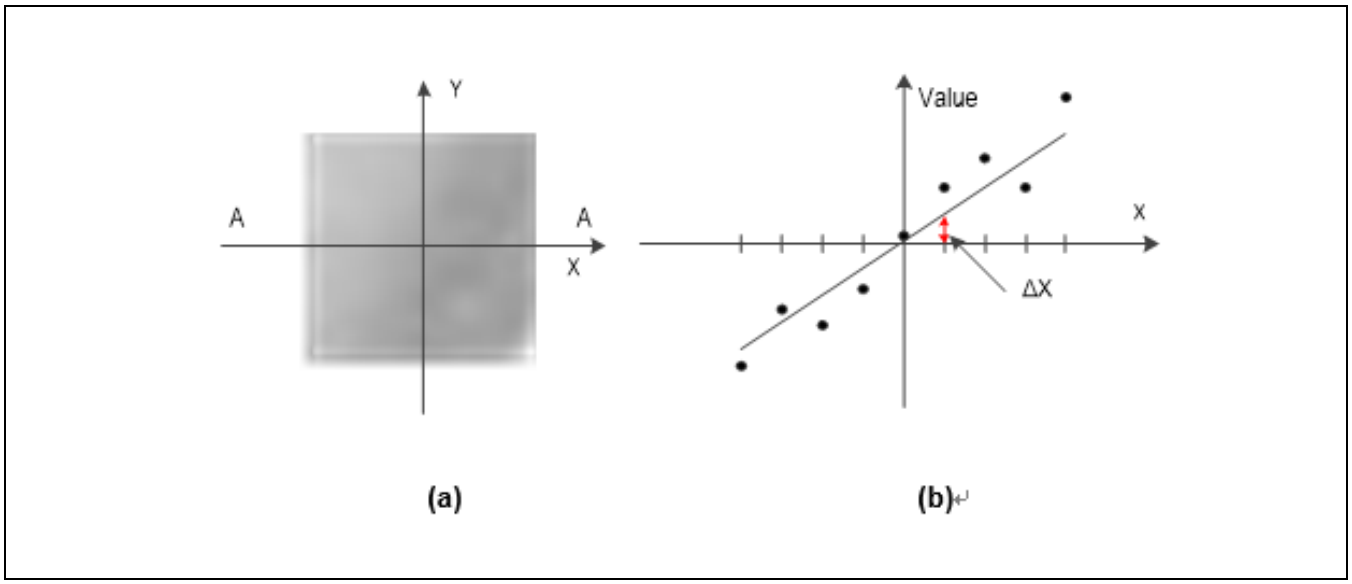


Figure 58 (a) image patch, (b) computed linear curve

The Δx (Δy) is found according to the following equation:

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$$\Delta x = \frac{\sum_{i=1}^{N} (V_i - V_{-i})(i)}{2 * \sum_{i=1}^{N} i^2}$$

Where L Vi - is the average value per column (row)

To correct only real slopes and not correct strong gradients and small slopes. Computation result passes through the following smart limiting function.

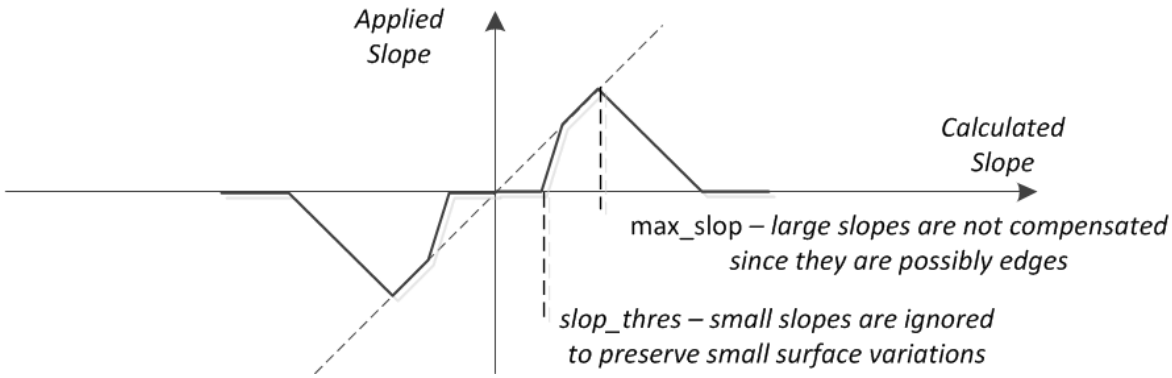


Figure 59 Smart limiting function for slope correction

Complex limit is computed as following figure

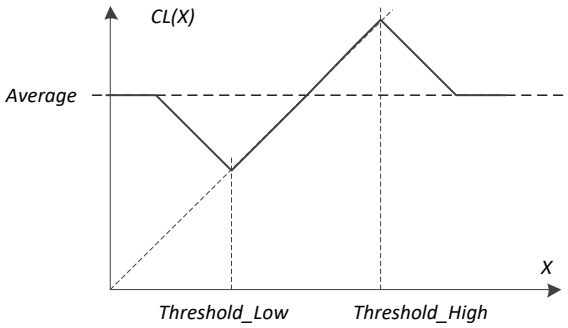


Figure 60 Smart Complex Limiting function for slope correction

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.11.7.6 LFP Green

6.11.7.6.1 Pattern

Pixel averaging is being performed with the following pattern for each pixel in the support.

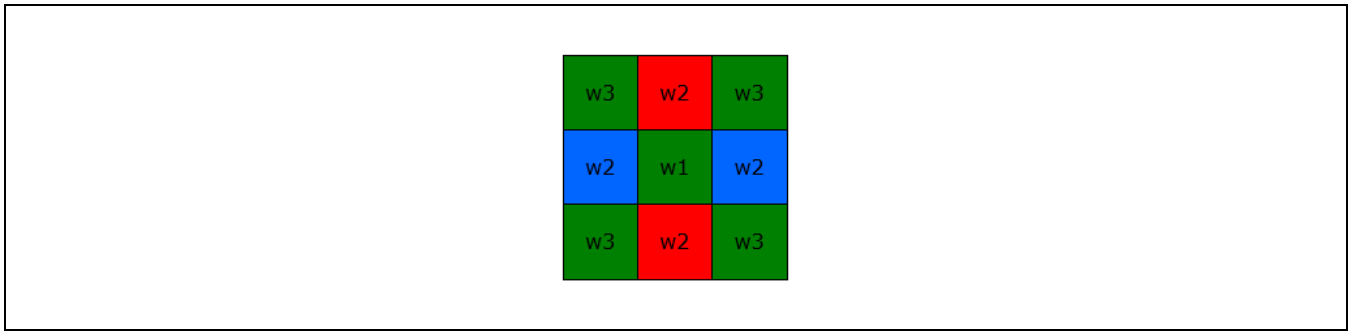


Figure 61 LFP green pattern

6.11.7.6.2 Weights of the pattern

The weights of the pattern are decided by lpf_nb_weight_green [1,4].for the case of green denoising. For the case of UV denoising no LFP is utilized.

The supported weights are defined in the following tables:

Table 56 Weight of Table

lpf_nb_weight_green	w1	w2	w3
1	48	0	4
2	40	0	6
3	32	0	8
4	16	0	12

6.11.7.7 Non local mean

The denoising flow describes in the following figure:

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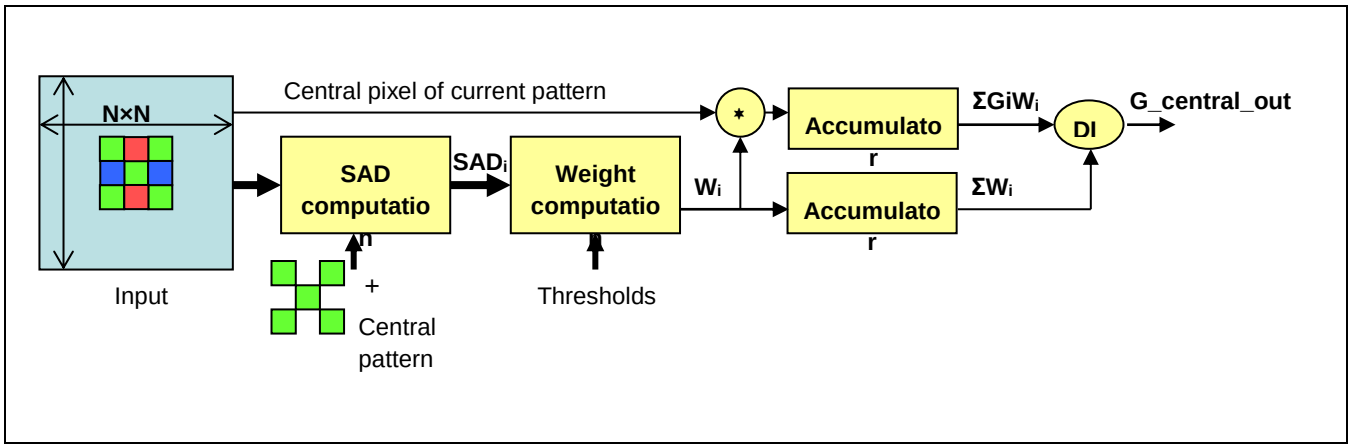


Figure 62 Green denoising scheme

Note: Prior to pattern matching the green averaging can be performed, but note that weighted averaging after pattern matching is always performed on original signal without averaging.

Additional data:
Candidates for averaging with central value are taken from 9*9 support.
The search range for candidate in the support is radial.
Each candidate has a SAD support of 5*5.
The following figures demonstrate candidate search field in the support.

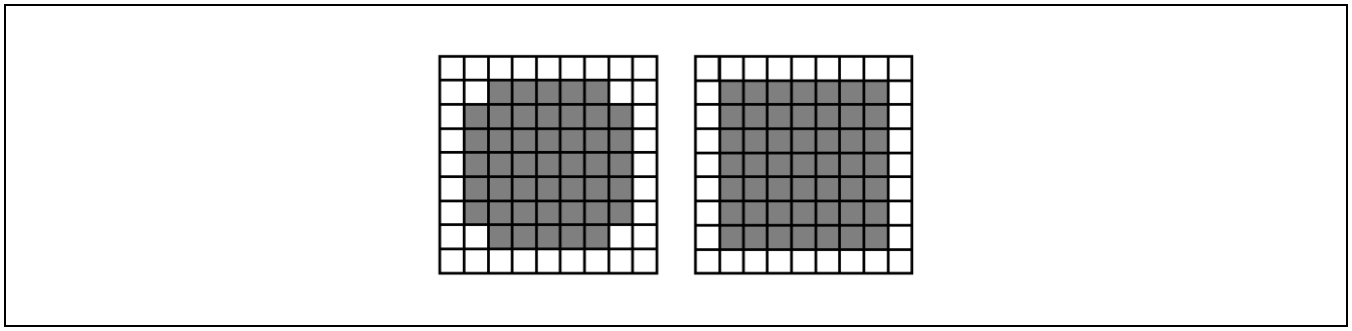


Figure 63 Green denoising search area

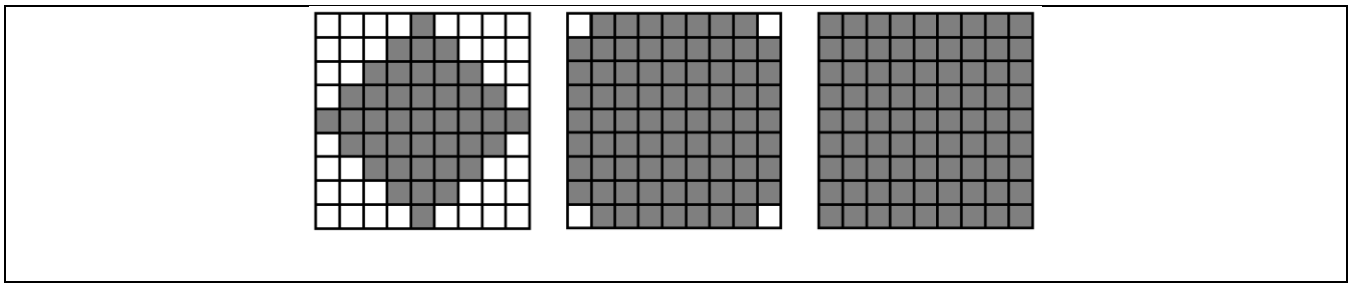


Figure 64 Color denoising search area

NOTE: The following figure shows the principal, actual size may be different from the one shown.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.11.7.7.1 Matching pattern scan procedure

The following figure describes the grade calculation being done on each candidate:

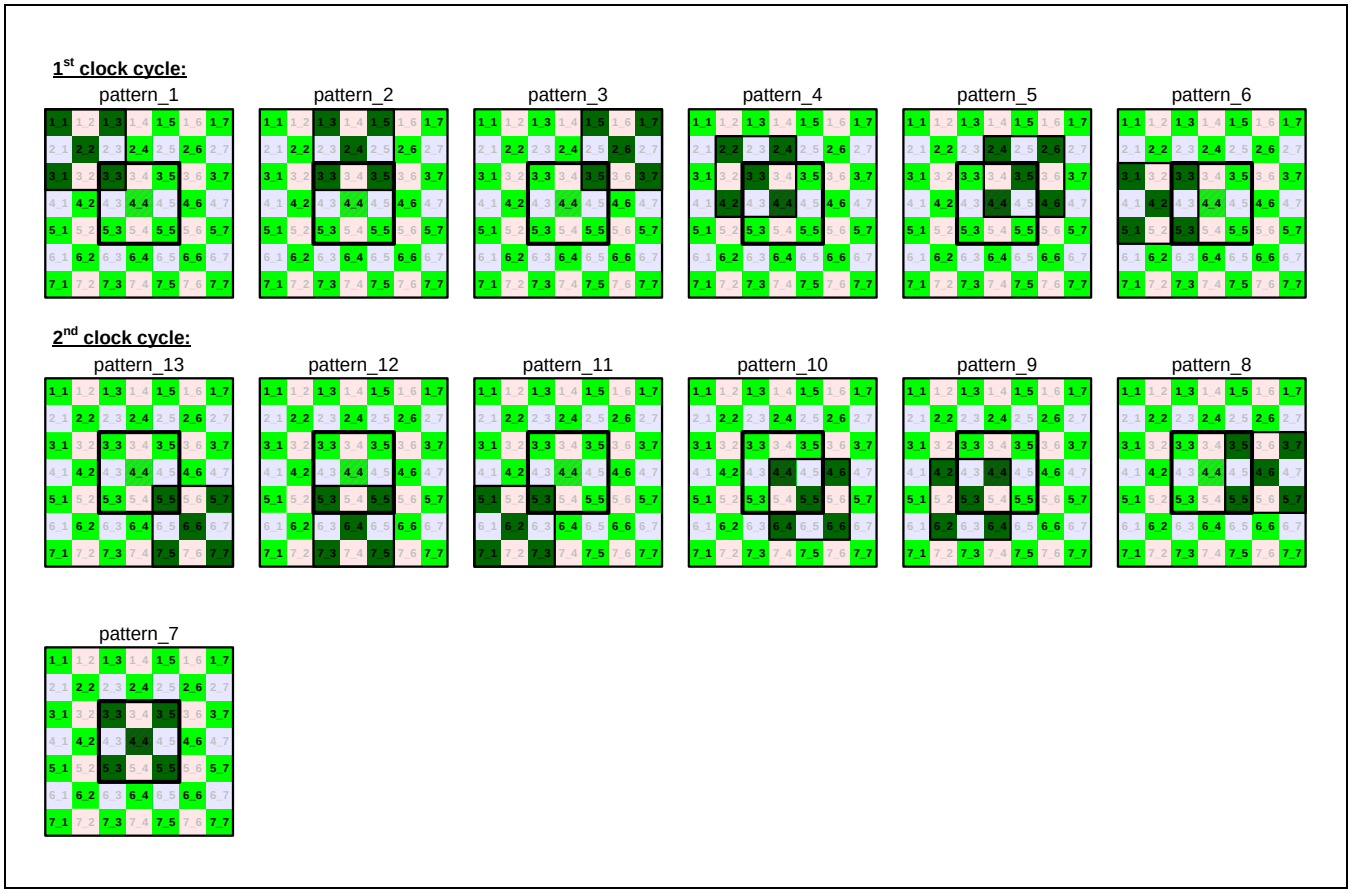


Figure 65 Pattern scan procedure

Each candidate pixel has a 5*5 support. SAD (sum of absolute differences) calculation is performed with the main pixel support. The following figures describe the various SAD patterns available to each color:

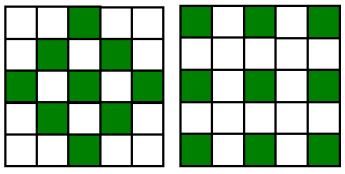


Figure 66 green SAD pattern

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

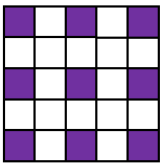


Figure 67 color SAD pattern

SAD grade is calculated as follow:

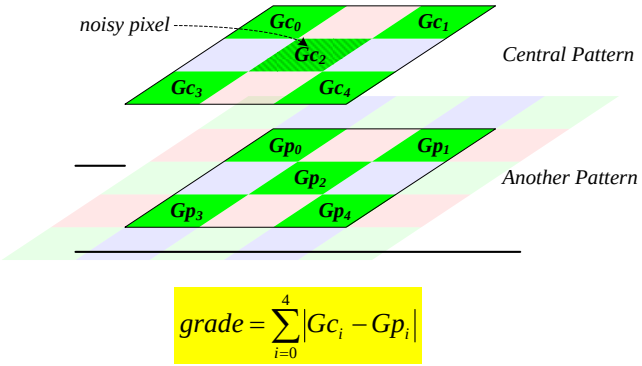
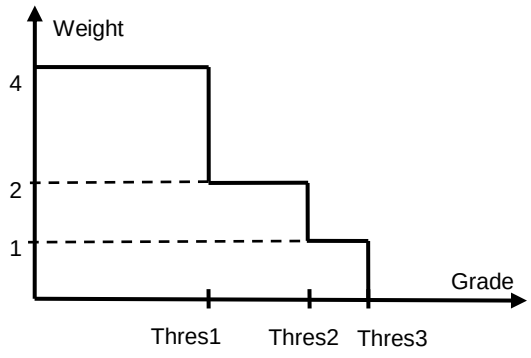


Figure 68 **SAD grade**

SAD accuracy can be set to be less than the blocks bittage in order to reduce power for the subtractions and addition. E.g SAD can be calculated after eliminating 4 LSBs. This is a simulation parameter and not configuration register.

Each grade is compared to the relevant threshold as follows:
Weight computation is performed according to the following function



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Figure 69 Weight computation function

Explanation:
Thres1 = the value calculated with respect to luma and radial distance as described in threshold calculation sub-block.
Thres2 = 1.5*Thres1
Thres3 = 1.75*Thres1

6.11.7.8 Related Registers

Index			Reset value	Bits	Register name	Type	Description
SPD-CDS	SPD-LOF	Merged					
0xa300	0xa372	0x9700	0x0001	[0:0]	bypass	RW1	0 -block is active, 1 -bypass block
0xa306	0xa378	0x9702	0x0001	[0:0]	convert_to_uv_en	RW1	enabling denoie for color pixel after transfering to uv
0xa306	0xa378	0x9702	0x0001	[1:1]	radial_search_range_en	RW1	enable search range
0xa306	0xa378	0x9702	0x0001	[4:4]	green_high_resolution_sad_en	RW1	enable high frequncy pattern in normal mode
0xa306	0xa378	0x9702	0x0001	[6:6]	diamond_candidate_shape_for_uv_en	RW1	allowing in 4 support UV pixels to be compatible with C2 search range shape
0xa306	0xa378	0x9702	0x0001	[7:7]	lpf_green_en	RW1	enable green lpf
0xa306	0xa378	0x9702	0x0001	[8:8]	thresh_calculation_with_mid_thresh_en	RW1	enable 5 node thresh calculation
0xa306	0xa378	0x9702	0x0001	[9:9]	green_donoise_en	RW1	enable denoise on green pixels
0xa306	0xa378	0x9702	0x0001	[10:10]	color_donoise_en	RW1	enable denoise on color pixels
0xa306	0xa378	0x9702	0x0000	[11:11]	slope_estimation_en	RW1	enable slope estimation and removal
0xa306	0xa378	0x9702	0x0000	[12:12]	edge_detection_en	RW1	enabling edge detection. when feature work slope removal and final slope estimation is conditioned by how low max_to_min_diff value is
0xa306	0xa378	0x9702	0x0000	[15:15]	radial_thresholds_en	RW1	enable radial treatment for thresholds
0xa302	0xa374	0x9704	0x0002	[2:0]	green_support	RW1	BNM margin size for green pixels
0xa302	0xa374	0x9704	0x0002	[6:4]	color_support	RW1	BNM margin size for color pixels
0xa302	0xa374	0x9704	0x0003	[2:0]	lpf_nb_weight_green	RW1	weight of the neihbor pixels whe performing Green LPF
0xa310	0xa382		0x0000	[13:0]	g_denoise_thresh_low_short	RW1S	low thresh for green pixel
0xa312	0xa384		0x0000	[13:0]	g_dns_thresh_short_Mid1	RW1S	node 1 thresh for green pixel

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

0xa314	0xa386		0x0000	[13:0]	g_dns_thresh_short_Mid2	RW1S	node 2 thresh for green pixel
0xa316	0xa388		0x0000	[13:0]	g_dns_thresh_short_Mid3	RW1S	node 3 thresh for green pixel
0xa318	0xa38a		0x0000	[13:0]	g_dns_thresh_short_Mid4	RW1S	node 4 thresh for green pixel
0xa31a	0xa38c		0x0000	[13:0]	g_dns_thresh_short_Mid5	RW1S	node 5 thresh for green pixel
0xa31c	0xa38e		0x0000	[13:0]	g_denoise_thresh_high_short	RW1S	high thresh for green pixel
0xa31e	0xa390		0x0000	[13:0]	u_denoise_thresh_low_short	RW1S	low thresh for U pixel
0xa320	0xa392		0x0000	[13:0]	u_dns_thresh_short_Mid1	RW1S	node 1 thresh for U pixel
0xa322	0xa394		0x0000	[13:0]	u_dns_thresh_short_Mid2	RW1S	node 2 thresh for U pixel
0xa324	0xa396		0x0000	[13:0]	u_dns_thresh_short_Mid3	RW1S	node 3 thresh for U pixel
0xa326	0xa398		0x0000	[13:0]	u_dns_thresh_short_Mid4	RW1S	node 4 thresh for U pixel
0xa328	0xa39a		0x0000	[13:0]	u_dns_thresh_short_Mid5	RW1S	node 5 thresh for U pixel
0xa32a	0xa39c		0x0000	[13:0]	u_denoise_thresh_high_short	RW1S	high thresh for U pixel
0xa32c	0xa39e		0x0000	[13:0]	v_denoise_thresh_low_short	RW1S	low thresh for V pixel
0xa32e	0xa3a0		0x0000	[13:0]	v_dns_thresh_short_Mid1	RW1S	node 1 thresh for V pixel
0xa330	0xa3a2		0x0000	[13:0]	v_dns_thresh_short_Mid2	RW1S	node 2 thresh for V pixel
0xa332	0xa3a4		0x0000	[13:0]	v_dns_thresh_short_Mid3	RW1S	node 3 thresh for V pixel
0xa334	0xa3a6		0x0000	[13:0]	v_dns_thresh_short_Mid4	RW1S	node 4 thresh for V pixel
0xa336	0xa3a8		0x0000	[13:0]	v_dns_thresh_short_Mid5	RW1S	node 5 thresh for V pixel
0xa338	0xa3aa		0x0000	[13:0]	v_denoise_thresh_high_short	RW1S	high thresh for V pixel
0xa33a	0xa3ac		0x0000	[13:0]	max_slope_allowed	RW1S	max slope that will be removed at a ratio of 1:1
0xa33c	0xa3ae		0x0000	[13:0]	min_slope_allowed	RW1S	min slope that will be removed
0xa33e	0xa3b0		0x0000	[13:0]	max_to_min_diff_short	RW1S	valid (max difference - min difference). higher values will decrease slope removal
		0x9712	0x0000	[9:0]	g_denoise_thresh_low_short_msb	RW1S	low thresh for green pixel
		0x9714	0x0000	[15:0]	g_denoise_thresh_low_short_lsb	RW1S	
		0x9716	0x0000	[9:0]	g_dns_thresh_short_Mid1_msb	RW1S	node 1 thresh for green pixel
		0x9718	0x0000	[15:0]	g_dns_thresh_short_Mid1_lsb	RW1S	
		0x971a	0x0000	[9:0]	g_dns_thresh_short_Mid2_msb	RW1S	node 2 thresh for green pixel

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		0x971c	0x0000	[15:0]	g_dns_thresh_short_Mid2_lsb	RW1S	
		0x971e	0x0000	[9:0]	g_dns_thresh_short_Mid3_msb	RW1S	node 3 thresh for green pixel
		0x9720	0x0000	[15:0]	g_dns_thresh_short_Mid3_lsb	RW1S	
		0x9722	0x0000	[9:0]	g_dns_thresh_short_Mid4_msb	RW1S	node 4 thresh for green pixel
		0x9724	0x0000	[15:0]	g_dns_thresh_short_Mid4_lsb	RW1S	
		0x9726	0x0000	[9:0]	g_dns_thresh_short_Mid5_msb	RW1S	node 5 thresh for green pixel
		0x9728	0x0000	[15:0]	g_dns_thresh_short_Mid5_lsb	RW1S	
		0x972a	0x0000	[9:0]	g_denoise_thresh_high_short_msb	RW1S	high thresh for green pixel
		0x972c	0x0000	[15:0]	g_denoise_thresh_high_short_lsb	RW1S	
		0x972e	0x0000	[9:0]	u_denoise_thresh_low_short_msb	RW1S	low thresh for U pixel
		0x9730	0x0000	[15:0]	u_denoise_thresh_low_short_lsb	RW1S	
		0x9732	0x0000	[9:0]	u_dns_thresh_short_Mid1_msb	RW1S	node 1 thresh for U pixel
		0x9734	0x0000	[15:0]	u_dns_thresh_short_Mid1_lsb	RW1S	
		0x9736	0x0000	[9:0]	u_dns_thresh_short_Mid2_msb	RW1S	node 2 thresh for U pixel
		0x9738	0x0000	[15:0]	u_dns_thresh_short_Mid2_lsb	RW1S	
		0x973a	0x0000	[9:0]	u_dns_thresh_short_Mid3_msb	RW1S	node 3 thresh for U pixel
		0x973c	0x0000	[15:0]	u_dns_thresh_short_Mid3_lsb	RW1S	
		0x973e	0x0000	[9:0]	u_dns_thresh_short_Mid4_msb	RW1S	node 4 thresh for U pixel
		0x9740	0x0000	[15:0]	u_dns_thresh_short_Mid4_lsb	RW1S	
		0x9742	0x0000	[9:0]	u_dns_thresh_short_Mid5_msb	RW1S	node 5 thresh for U pixel
		0x9744	0x0000	[15:0]	u_dns_thresh_short_Mid5_lsb	RW1S	
		0x9746	0x0000	[9:0]	u_denoise_thresh_high_short_msb	RW1S	high thresh for U pixel
		0x9748	0x0000	[15:0]	u_denoise_thresh_high_short_lsb	RW1S	
		0x974a	0x0000	[9:0]	v_denoise_thresh_low_short_msb	RW1S	low thresh for V pixel
		0x974c	0x0000	[15:0]	v_denoise_thresh_low_short_lsb	RW1S	
		0x974e	0x0000	[9:0]	v_dns_thresh_short_Mid1_msb	RW1S	node 1 thresh for V pixel
		0x9750	0x0000	[15:0]	v_dns_thresh_short_Mid1_lsb	RW1S	
		0x9752	0x0000	[9:0]	v_dns_thresh_short_Mid2_msb	RW1S	node 2 thresh for V pixel

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

		0x9754	0x0000	[15:0]	v_dns_thresh_short_Mid2_lsb	RW1S	
		0x9756	0x0000	[9:0]	v_dns_thresh_short_Mid3_msb	RW1S	node 3 thresh for V pixel
		0x9758	0x0000	[15:0]	v_dns_thresh_short_Mid3_lsb	RW1S	
		0x975a	0x0000	[9:0]	v_dns_thresh_short_Mid4_msb	RW1S	node 4 thresh for V pixel
		0x975c	0x0000	[15:0]	v_dns_thresh_short_Mid4_lsb	RW1S	
		0x975e	0x0000	[9:0]	v_dns_thresh_short_Mid5_msb	RW1S	node 5 thresh for V pixel
		0x9760	0x0000	[15:0]	v_dns_thresh_short_Mid5_lsb	RW1S	
		0x9762	0x0000	[9:0]	v_denoise_thresh_high_short_msb	RW1S	high thresh for V pixel
		0x9764	0x0000	[15:0]	v_denoise_thresh_high_short_lsb	RW1S	
		0x9766	0x0000	[9:0]	max_slope_allowed_msb	RW1S	max slope that will be removed at a ratio of 1:1
		0x9768	0x0000	[15:0]	max_slope_allowed_lsb	RW1S	
		0x976a	0x0000	[9:0]	min_slope_allowed_msb	RW1S	min slope that will be removed
		0x976c	0x0000	[15:0]	min_slope_allowed_lsb	RW1S	
		0x976e	0x0000	[9:0]	max_to_min_diff_short_msb	RW1S	valid (max difference - min difference). higher values will decrease slope removal
		0x9770	0x0000	[15:0]	max_to_min_diff_short_lsb	RW1S	
0xa348	0xa3ba	0x977c	0x0000	[7:0]	radial_thresholds_power	RW1	how strong is the radial effect on the thresholds
0xa34a	0xa3bc	0x977e	0x0000	[15:8]	radial_thresholds_limit	RW1	limiation for the radial effect. camputed radial power cannot be higher than register value

6.11.8 HDR Merger

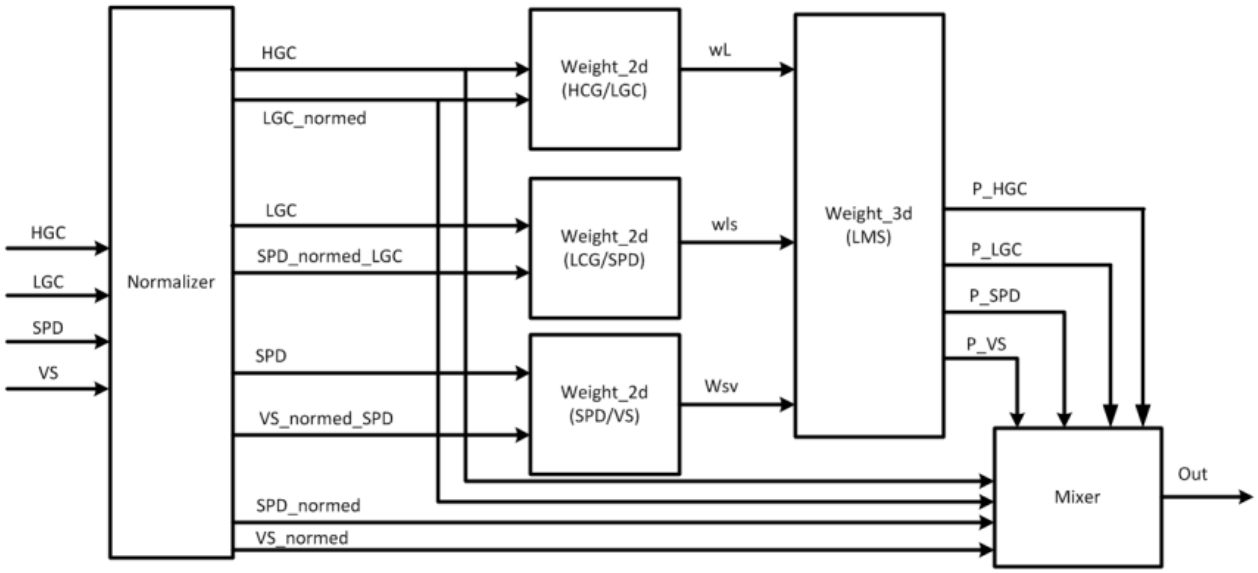
The purpose of the HDR merger is to output a wide dynamic range image from several low dynamic range images captured with different effective exposure times. The effective exposure time is a multiplication of exposure time, conversion gain, photo diode sensitivity, analog gain and digital gain.

The current merger, intended for automotive sensors, uses between 2-4 images (depending on configuration). Each image consists of 12 bits with effective exposure time ratio of ~4096 between the longest/shortest effective exposures to provide an output dynamic range of 140dB in a 24 bits image. In addition, the current merger performs per pixel merging and doesn't share information between pixels.

6.11.8.1 Jet merger

High level diagram:

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6.11.8.1.1 Normalization

S5K3B6SA merger uses long based normalization meaning the highest effective exposure is not normalized at all and each of the other exposures are normalized by the exposure ratio between them and the highest effective exposure:

$$\text{LPD_LCG}_{\text{normed}} = \max\left(0, \left(\text{pixel}_{\text{LPD_LCG}} - \text{pedestal}_{\text{in}}\right) \cdot \frac{\text{LPD_HCG_Exp}}{\text{LPD_LCG_Exp}} + \text{pedestal}_{\text{out}}\right)$$

$$\text{SPD_normed} = \max(0, (\text{pixel}_{\text{SPD_CDS}} - \text{pedestal}_{\text{in}}) \cdot \frac{\text{LPD_HCG_Exp}}{\text{LPD_CDS_Exp}} + \text{pedestal}_{\text{out}})$$

$$\text{VS_normed} = \max(0, (\text{pixel}_{\text{SPD_LOF}} - \text{pedestal}_{\text{in}}) \cdot \frac{\text{LPD_HCG_Exp}}{\text{LPD_LOF_Exp}} + \text{pedestal}_{\text{out}})$$

In addition, for the use of the "weight 2D" as will be explained in the next chapter, each exposure is also normalized by the exposure ratio between itself and the other exposure in the same "Weight 2D":

$$\text{SPD_normed_LCG} = \max(0, (\text{pixel}_{\text{SPD_CDS}} - \text{pedestal}_{\text{in}}) \cdot \frac{\text{LPD_LCG_Exp}}{\text{SPD_CDS_Exp}} + \text{pedestal}_{\text{out}})$$

$$\text{VS_normed_SPD} = \max(0, (\text{pixel}_{\text{SPD_LOF}} - \text{pedestal}_{\text{in}}) \cdot \frac{\text{SPD_CDS_Exp}}{\text{SPD_LOF_Exp}} + \text{pedestal}_{\text{out}})$$

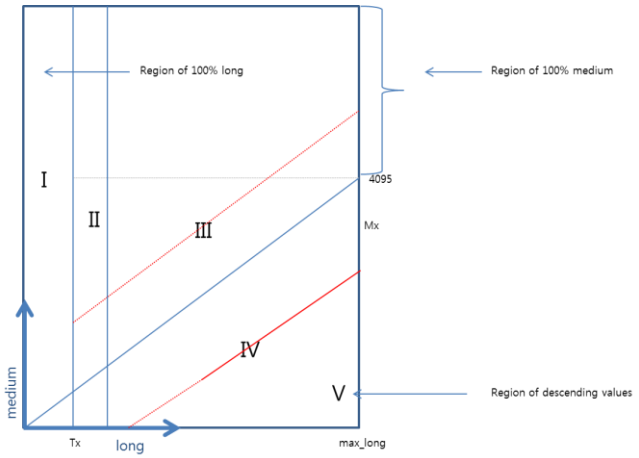
Where for example the LPD-LCG (large photo diode, low conversion gain):

$$\text{LPD_LCG_EXP} = \frac{(LPD_LCG_{\text{exposure_time}} * LPD_LCG_{\text{convertion_gain}} * LPD_LCG_{\text{sensitivity}} * LPD_LCG_{\text{analog_gain}} * LPD_LCG_{\text{digital_gain}})}{LPD_LCG_{\text{ADC_SAT}}}$$

6.11.8.1.2 Weight 2D - Weight calculation

Each Weight 2D map gets 2 normalized streams and outputs weights that will be used in order to create the final merged image. The weight is calculated according to the following diagram:

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Note: in order to simplify the explanations the terms "long" and "medium" will be used as exposure names in this section. Same logic applies to all 3 "weight 2D" sub-blocks.

The weight calculation can create either concave or convex curves as seen in the following figure:

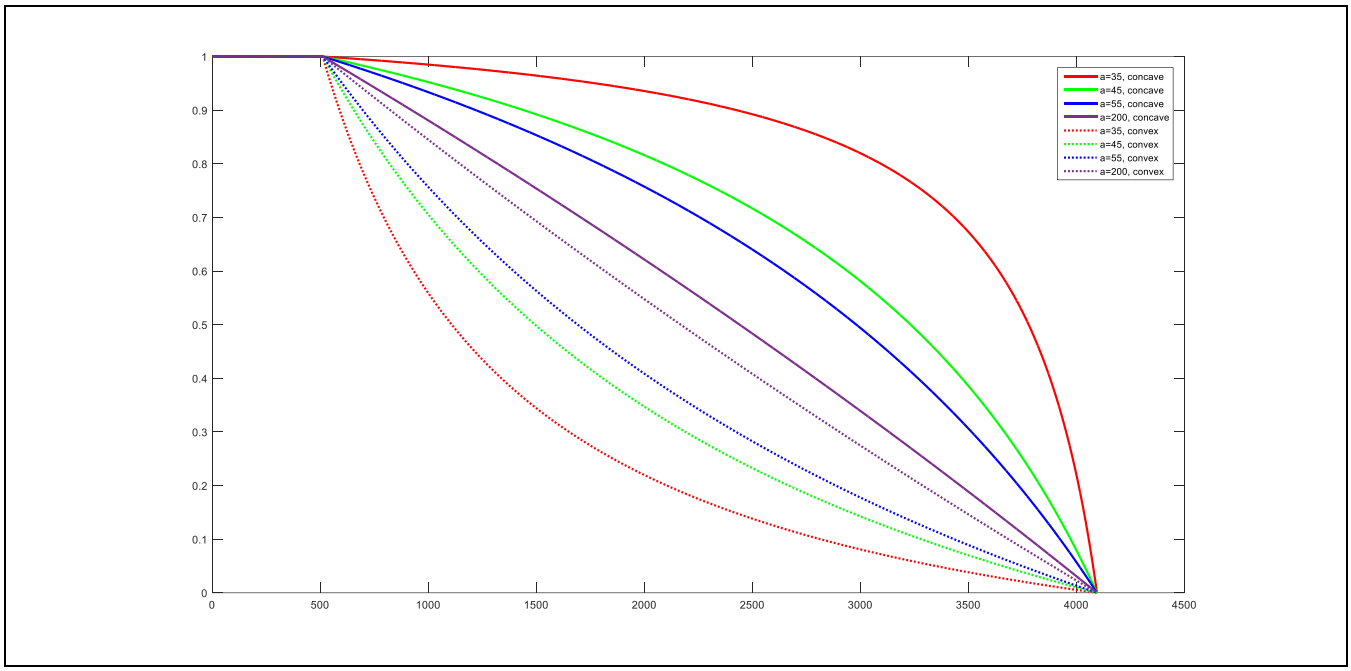


Figure 70 Gamma Curve for Weights Along the Long=Medium Line

6.11.8.1.2.1 Step 1: long = medium line

Along the long=medium line the normalized streams are equal, therefore x=long=medium. The weight on the bottom left point (low code values) is 100% for the long exposure and the weight on the middle-right point (saturation of long exposure) is 100% for the medium exposure. Between these points, the weight gradually moves from the long exposure to the medium exposure. The increase in medium exposure usage can be modified by using the following tuning handles:

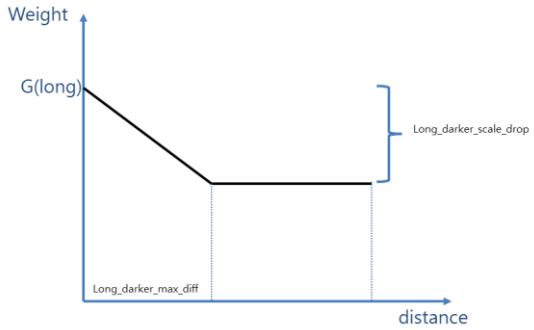
Index	Reset	Bits	Register name	Type	Description

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

	value				
0x08a5a	0x002d	[8:0]	mergerlms_0_jet_curve_param0	RW1S	This parameter controls the curvature of the graph for LPD HCG-LPD LCG
0x08ad2	0x002d	[8:0]	mergerlms_0_jet_curve_param1	RW1S	This parameter controls the curvature of the graph for LPD LCG-SPD CDS
0x08b72	0x002d	[8:0]	mergerlms_0_jet_curve_param2	RW1S	This parameter controls the curvature of the graph for SPD CDS-SPD LOF

6.11.8.1.2.2 Step 2: long < medium

In areas where long < medium (region 3 in above diagram) the weight of the long portion decreases according to the following tuning handles:



The red line in this region, from the weights map, represents that W_L (weight of long exposure) decreases linearly with a defined slope relative to the value calculated on the long=medium line, up to the red line. Beyond that distance, which is defined by the registers mergerlms_0_jet_long_darker_max_diffX, it remains constant. The origin of the graph is the long=medium line.

Where G(long) is value of the weight of the long exposure.

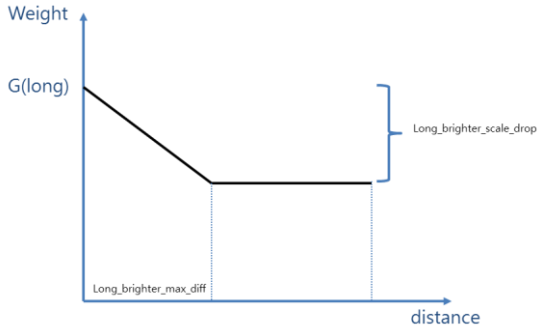
Index	Reset value	Bits	Register name	Type	Description
0x08a9e	0x0800	[11:0]	mergerlms_0_jet_long_darker_max_diff0	0x08a9e	As shown in above image for LPD HCG-LPD LCG
0x08aa0	0x0400	[11:0]	mergerlms_0_jet_long_darker_scale_drop0	0x08aa0	As shown in above image for LPD HCG-LPD LCG
0x08b3e	0x0800	[11:0]	mergerlms_0_jet_long_darker_max_diff1	RW1S	As shown in above image for LPD LCG-SPD CDS
0x08b40	0x0400	[11:0]	mergerlms_0_jet_long_darker_scale_drop1	RW1S	As shown in above image for LPD LCG-SPD CDS
0x08bb6	0x0800	[11:0]	mergerlms_0_jet_long_darker_max_diff2	RW1S	As shown in above image for SPD CDS-SPD LOF

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0x08bb8	0x0400	[11:0]	mergerlms_0_jet_long_darker_scale_drop2	RW1S	As shown in above image for SPD CDS-SPD LOF

6.11.8.1.2.3 Step 3: long > medium

Similarly, in areas where long > medium (region 4 in above diagram) the weight of the long portion decreases according to the following tuning handles:



The red line in this region, from the weights map, represents that W_L (weight of long exposure) decreases linearly with a defined slope relative to the value calculated on the long=medium line, up to the red line. Beyond that distance, it remains constant. The origin of the graph is the long=medium line.

Where G(long) is value of the weight of the long exposure.

Index	Reset value	Bits	Register name	Type	Description
0x08aa2	0x0800	[11:0]	mergerlms_0_jet_long_brighter_max_diff0	RW1S	As shown in above image for LPD HCG-LPD LCG
0x08aa4	0x019a	[11:0]	mergerlms_0_jet_long_brighter_scale_drop0	RW1S	As shown in above image for LPD HCG-LPD LCG
0x08b42	0x0800	[11:0]	mergerlms_0_jet_long_brighter_max_diff1	RW1S	As shown in above image for LPD LCG-SPD CDS
0x08b44	0x019a	[11:0]	mergerlms_0_jet_long_brighter_scale_drop1	RW1S	As shown in above image for LPD LCG-SPD CDS
0x08bba	0x0800	[11:0]	mergerlms_0_jet_long_brighter_max_diff2	RW1S	As shown in above image for SPD CDS-SPD LOF
0x08bbc	0x019a	[11:0]	mergerlms_0_jet_long_brighter_scale_drop2	RW1S	As shown in above image for SPD CDS-SPD LOF

6.11.8.1.2.4 Step 4: Bright area modification

The function bright_func calculates weights near the right bottom corner. Specifically, we want the weight at the point (long, medium) = (4095, 0) to be v_{low}, at the control point (4095,Mx) to be v_{high} and at (4095,4095) we want the weight to be 0.

Near the bottom-right corner (area 5) another function enables setting the weights as needed using the following

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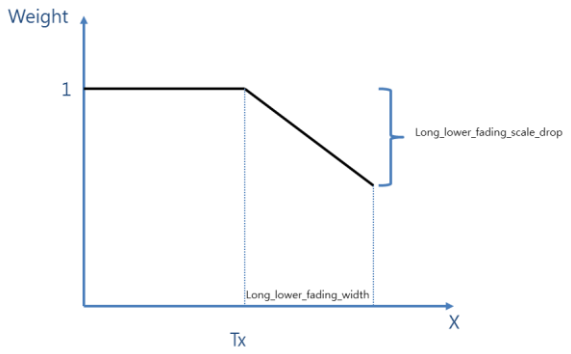
tuning handles:

Index	Reset value	Bits	Register name	Type	Description
0x08aa6	0x0c17	[11:0]	mergerlms_0_jet_mx0	RW1S	Control point position: (4095,Mx) for LPD HCG-LPD LCG
0x08aa8	0x0800	[11:0]	mergerlms_0_jet_v_low0	RW1S	Weight at the right bottom corner (4095,0) for LPD HCG-LPD LCG
0x08aaa	0x04cd	[11:0]	mergerlms_0_jet_v_high0	RW1S	Weight at the control point (4095,Mx) for LPD HCG-LPD LCG
0x08ab4	0x0000	[11:0]	mergerlms_0_jet_bright_func_slope0	RW1S	Rate of long decrease towards (0,0) for LPD HCG-LPD LCG
0x08b46	0x0c17	[11:0]	mergerlms_0_jet_mx1	RW1S	Control point position: (4095,Mx) for LPD LCG-SPD CDS
0x08b48	0x0800	[11:0]	mergerlms_0_jet_v_low1	RW1S	Weight at the right bottom corner (4095,0) for LPD LCG-SPD CDS
0x08b4a	0x04cd	[11:0]	mergerlms_0_jet_v_high1	RW1S	Weight at the control point (4095,Mx) for LPD LCG-SPD CDS
0x08b54	0x0000	[11:0]	mergerlms_0_jet_bright_func_slope1	RW1S	Rate of long decrease towards (0,0) for LPD LCG-SPD CDS
0x08bbe	0x0c17	[11:0]	mergerlms_0_jet_mx2	RW1S	Control point position: (4095,Mx) for SPD CDS-SPD LOF
0x08bc0	0x0800	[11:0]	mergerlms_0_jet_v_low2	RW1S	Weight at the right bottom corner (4095,0) for SPD CDS-SPD LOF
0x08bc2	0x04cd	[11:0]	mergerlms_0_jet_v_high2	RW1S	Weight at the control point (4095,Mx) for SPD CDS-SPD LOF
0x08bcc	0x0000	[11:0]	mergerlms_0_jet_bright_func_slope2	RW1S	Rate of long decrease towards (0,0) for SPD CDS-SPD LOF

6.11.8.1.2.5 Step 5: Dark area modification

At the left part of the diagram (areas 1,2) where long value is weak, another function enables setting the weights as needed using the following tuning handles:

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Where **X** in our general discussion is Long, the origin is at long=0. In region I when func_dark is used (depending on use_low_long_unity_en) W_L is always 1, otherwise W_L is the basic weight calculated at this point. In region II when func_dark is used (depending on use_low_long_unity_en) func_dark calculates a decreasing value for W_L with long_lower_fading_scale_drop and width of this area long_lower_fading_width.

For the transition from area 2 to area 3 we use a blending of results where:

$$\alpha = 1 - \frac{long - Tx}{\text{long_lower_fading_width}}; \qquad total_weight = \alpha \cdot dark_func + (1-\alpha) \cdot basic_weights$$

Index	Reset value	Bits	Register name	Type	Description
0x08aac	0x0200	[11:0]	mergerlms_0_jet_tx0	RW1S	Below this always longer exposure weight is 1 for LPD HCG-LPD LCG
0x08aae	0x03e8	[11:0]	mergerlms_0_jet_long_lower_fading_width0	RW1S	width of weight decreased for LPD HCG-LPD LCG
0x08ab0	0x04cd	[11:0]	mergerlms_0_jet_long_lower_fading_scale_drop0	RW1S	weight value after fading_width for LPD HCG-LPD LCG
0x08b4c	0x0200	[11:0]	mergerlms_0_jet_tx1	RW1S	Below this always longer exposure weight is 1 for LPD LCG-LPD SPD CDS
0x08b4e	0x03e8	[11:0]	mergerlms_0_jet_long_lower_fading_width1	RW1S	width of weight decreased for LPD LCG-SPD CDS
0x08b50	0x04cd	[11:0]	mergerlms_0_jet_long_lower_fading_scale_drop1	RW1S	weight value after fading_width for LPD LCG-SPD CDS
0x08bc4	0x0200	[11:0]	mergerlms_0_jet_tx2	RW1S	Below this always longer exposure weight is 1 for SPD CDS-SPD LOF
0x08bc6	0x03e8	[11:0]	mergerlms_0_jet_long_lower_fading_width2	RW1S	width of weight decreased for SPD CDS-SPD LOF
0x08bc8	0x04cd	[11:0]	mergerlms_0_jet_long_lower_fading_scale_drop2	RW1S	weight value after fading_width for SPD CDS-SPD LOF

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6.11.8.1.2.6 Combining the weights from all functions and calculating the portions

In order to combine the final weight for each exposure we use the following logic:

If max_mode = 0 we take the maximum value of bright_func and basic weight in region IV only. We then use that to apply the correct weights to all regions.

If max_mode = 1 we take the maximum value of bright_func and basic weight over the entire area.

If max_mode = 2 we do not use the value of bright_func and retain the basic weight.

Following this,

If blend_mode=0 we blend the result with the weight obtained from dark_func in region II according to the equations seen above for total_weight.

If blend_mode=1 we take the max weight obtained from dark_func and basic weight.

Index	Reset value	Bits	Register name	Type	Description
0x089b2	0x0000	[1:0]	mergerlms_0_jet_max_mode0	RW1S	Max mode as explained above for LPD HCG-LPD LCG
0x08a38	0x0001	[3:3]	mergerlms_0_jet_blend_mode0	RW1S	Blend mode as explained above for LPD HCG-LPD LCG
0x08a52	0x0000	[1:0]	mergerlms_0_jet_max_mode1	RW1S	Max mode as explained above for LPD LCG-SPD CDS
0x08a38	0x0001	[8:8]	mergerlms_0_jet_blend_mode1	RW1S	Blend mode as explained above for LPD LCG-SPD CDS
0x08aca	0x0000	[1:0]	mergerlms_0_jet_max_mode2	RW1S	Max mode as explained above for SPD CDS-SPD LOF
0x08a38	0x0001	[13:13]	mergerlms_0_jet_blend_mode2	RW1S	Blend mode as explained above for SPD CDS-SPD LOF

Once we have the weights of each exposure we calculate the portions as following:

$$P_{HCG} = W_L$$

$$P_{LCG} = (1 - W_L)\, W_{ls}$$

$$P_{SPD} = (1 - W_L)\,(1 - W_{ls})\quad W_{sv}$$

$$P_{VS} = (1 - W_L)(1 - W_{ls})(1 - W_{sv})$$

Where P_{HCG} P_{LCG} P_{SPD} P_{VS} are the final portions of the LPD_HCG, LPD_LCG, SPD_CDS and SPD_LOF exposures respectively. W_L is the weight calculated for LPD_HCG from (LPD_HCG, LPD_LCG) calculation, W_{ls} is the weight calculated for LPD_LCG from (LPD_LCG, SPD_CDS) calculation, and W_{sv} is the weight calculated for SPD_CDS from (SPD_CDS, SDP_LOF) calculation.

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6.11.8.1.3 Mixer

After all portions are obtained, the final merge is calculated as follows:

$$\mathrm{HDR} = \mathrm{P_{HCG}} * \mathrm{LPD_{HCG}} + \mathrm{P_{LCG}} * \mathrm{LPD_{LCG}} + \mathrm{P_{SPD}} * \mathrm{SPD_{CDS}} + \mathrm{P_{VS}} * \mathrm{SPD}_{LOF}$$

6.11.8.2 GBD - Green based decisions

Sometimes weights calculated for color pixels are different from the weights calculated for the nearby green (or clear) pixels. This is caused by the difference in value of green and color pixels. Normally this isn't a problem but in motion scenes the short and long exposures "see" different images, thereby different weights for green and color may result in selection of pixels from one exposure for the greens and pixels from a different exposure for the color pixels. To overcome this, when use_gbd_en register is checked we calculate weights only for green pixels. Weights for color pixels are then calculated as the mean of their four green neighbors.

Index	Reset value	Bits	Register name	Type	Description
0x08BEA	0x0001	[2:2]	mergerlms_0_use_gbd_en	RW1S	Enable green based correction

6.11.8.3 Apply factor

Apply factor function is used to occupy the dynamic range of an image fully. Trigger the effective merged signal to have higher bits of the available bit-depth.
For example: moving an image that uses only 20 bits out of its available 24 bits to utilize the bit-depth fully.

Index	Reset value	Bits	Register name	Type	Description
0x08BEA	0x0001	[1:1]	mergerlms_0_apply_factor_on_output	RW1S	Apply a factor on Jet output to meet the maximal value
0x08BF2	0x0010	[15:0]	mergerlms_0_factor_on_output	RW1S	factor to multiply Jet output. fixed point 12.4

6.11.9 ASP Statistics

The statistics block generates statistical information on 3 region of interest (ROIs) in the image. Each of the ROIs is defined by the top left pixel location, as well as its height and width.

In each ROI, a histogram for each color channel is calculated according to 256 configurable bins and can be outputted before the image footer.

The block calculates the mean pixel values, for each of the 4 Bayer colors. The block also calculated the mean value of the squares to the pixel values, from which the standard deviation may be calculated.

The values of the bins of the histogram are also able to be updated in the registers.

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6.11.9.1 Configuration

In order to activate the block, the configuration of each ROI needs to be set.

This includes the top left point of the ROI, the height and the width.

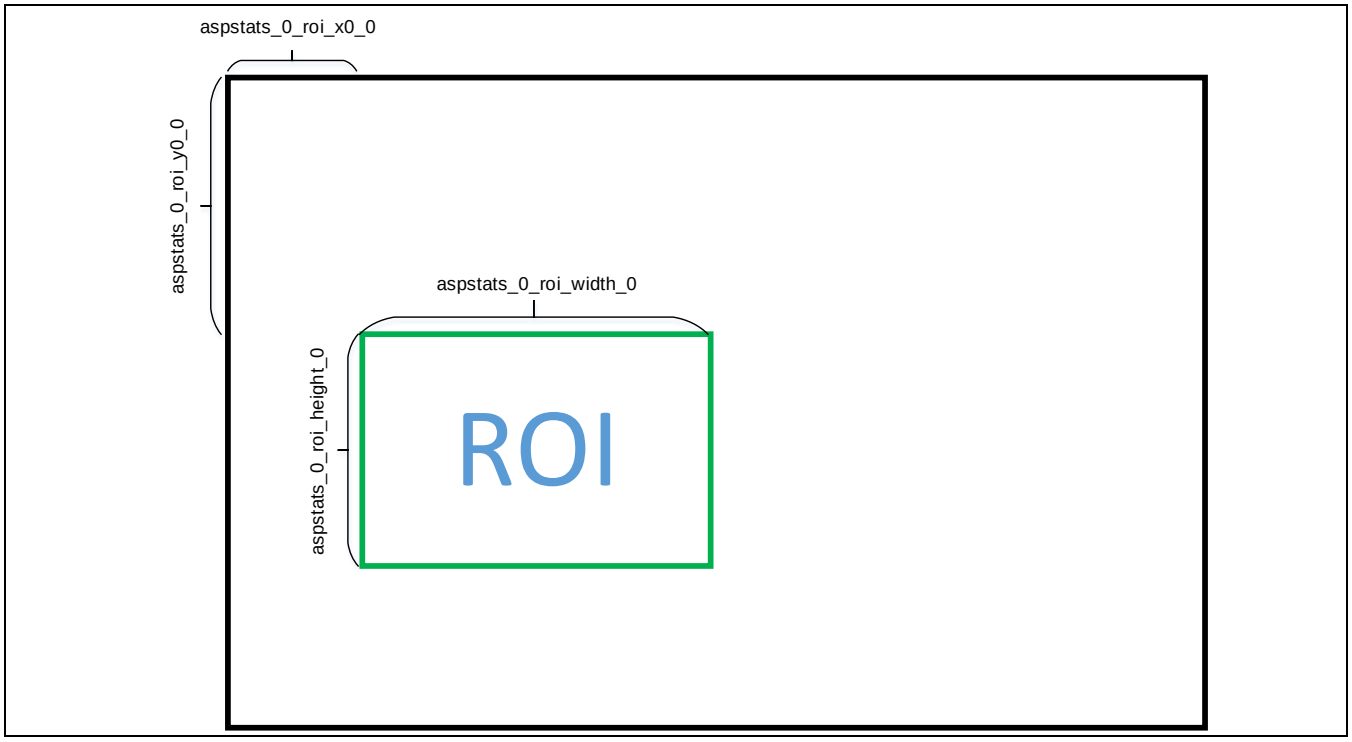


Figure 71 ROI Description for Statistics

The relevant registers are:

Table 57 Registers for Configuring Statistics

Index	Reset value	Bits	Register name	Type	Description
0x9E00	0x0001	[0]	aspstats_0_bypass	RW1S	1'b1 : Bypass enabled. 1'b0 : Bypass Disabled
0xA17C	0x0000	[15:0]	aspstats_0_roi_x0_0	RW1S	ROI[0] Starting Coordinate X
0x A17E	0x0000	[15:0]	aspstats_0_roi_y0_0	RW1S	ROI[0] Starting Coordinate Y
0x A180	0x0f00	[15:0]	aspstats_0_roi_width_0	RW1S	ROI[0] Width
0x A182	0x0870	[15:0]	aspstats_0_roi_height_0	RW1S	ROI[0] Height
0x A184	0x0817	[11:0]	aspstats_0_roi_size_inv_0	RW1S	ROI[0] 1/size, mantissa
0x A186	0x0020	[5:0]	aspstats_0_roi_size_inv_shift_0	RW1S	ROI[0] 1/size, exponent (log2)
0x A188	0x0000	[15:0]	aspstats_0_roi_x0_1	RW1S	ROI[1] Starting Coordinate X
0x A18A	0x0000	[15:0]	aspstats_0_roi_y0_1	RW1S	ROI[1] Starting Coordinate Y
0x A18C	0x0000	[15:0]	aspstats_0_roi_width_1	RW1S	ROI[1] Width

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0x A18E	0x0000	[15:0]	aspstats_0_roi_height_1	RW1S	ROI[1] Height
0x A190	0x0000	[11:0]	aspstats_0_roi_size_inv_1	RW1S	ROI[1] 1/size, mantissa
0x A192	0x0000	[5:0]	aspstats_0_roi_size_inv_shift_1	RW1S	ROI[1] 1/size, exponent (log2)
0x A194	0x0000	[15:0]	aspstats_0_roi_x0_2	RW1S	ROI[2] Starting Coordinate X
0x A196	0x0000	[15:0]	aspstats_0_roi_y0_2	RW1S	ROI[2] Starting Coordinate Y
0x A198	0x0000	[15:0]	aspstats_0_roi_width_2	RW1S	ROI[2] Width
0x A19A	0x0000	[15:0]	aspstats_0_roi_height_2	RW1S	ROI[2] Height
0x A19C	0x0000	[11:0]	aspstats_0_roi_size_inv_2	RW1S	ROI[2] 1/size, mantissa
0x A19E	0x0000	[5:0]	aspstats_0_roi_size_inv_shift_2	RW1S	ROI[2] 1/size, exponent (log2)

The following formula can be used to calculate required “inv” and “inv_shift” values for an ROI, these registers are used for hardware configuration:

$$InvShift = ceil\ [\ \log_2\left(\frac{width * height}{4}\right)] + 11$$

$$Inv = ceil\ \left(\frac{4*2^{InvShift}}{width*height}\right)$$

Note: if ROI size is set to 0, no statistics will be outputted of that ROI.

It is also possible to control the 256 bins used in the histograms.

This is done by setting 255 bin edges which in addition to 2 fixed edges (0 and $2^{24}-1$) defines 256 bins. Each edge is defined by 2 registers, one for the LSBs and one for the MSBs. The final value is calculated as following: $Edge = (MSB_{Val}) * 2^{16} + LSB_{Val}$

The registers controlling these values are:

Table 58 Registers of Histogram Bins

Index	Reset value	Bits	Register name	Type	Description
0x9E06	0x0001	[15:0]	aspstats_0_hist_bins_lsb_0_0	RW1	Histogram bin edges, LSBs
0x9E08	0x0002	[15:0]	aspstats_0_hist_bins_lsb_0_1	RW1	Histogram bin edges, LSBs
…					
0xA002	0x9fac	[15:0]	aspstats_0_hist_bins_lsb_0_254	RW1	Histogram bin edges, LSBs
0xA004	0x0000	[7:0]	aspstats_0_hist_bins_msb_0_0	RW1	Histogram bin edges, MSBs
	0x0000	[15:8]	aspstats_0_hist_bins_msb_0_1	RW1	Histogram bin edges, MSBs
…					
0xA100	0x00d7	[7:0]	aspstats_0_hist_bins_msb_0_252	RW1	Histogram bin edges, MSBs
	0x00e4	[15:8]	aspstats_0_hist_bins_msb_0_253	RW1	Histogram bin edges, MSBs
0xA102	0x00f1	[7:0]	aspstats_0_hist_bins_msb_0_254	RW1	Histogram bin edges, MSBs

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6.11.10 PWL

6.11.10.1 General Description

In order to achieve bit reduction on the output stream, PWL (Piece Wise Linearization) block was developed. A lookup table is needed for mapping between corresponding desired values as shown below:

#	X value	Y value
1	0	0
2	16384	1536
…	…	…
32	16777215	4095

This table should be consisting with range of 32 points (or less) between min and max values according to the number of bits, and in mid points an interpolation is made.

NOTE: It is possible to decide how many points to use through “gamma_1tbl_num_of_points_used” register (see PWL Parameters).

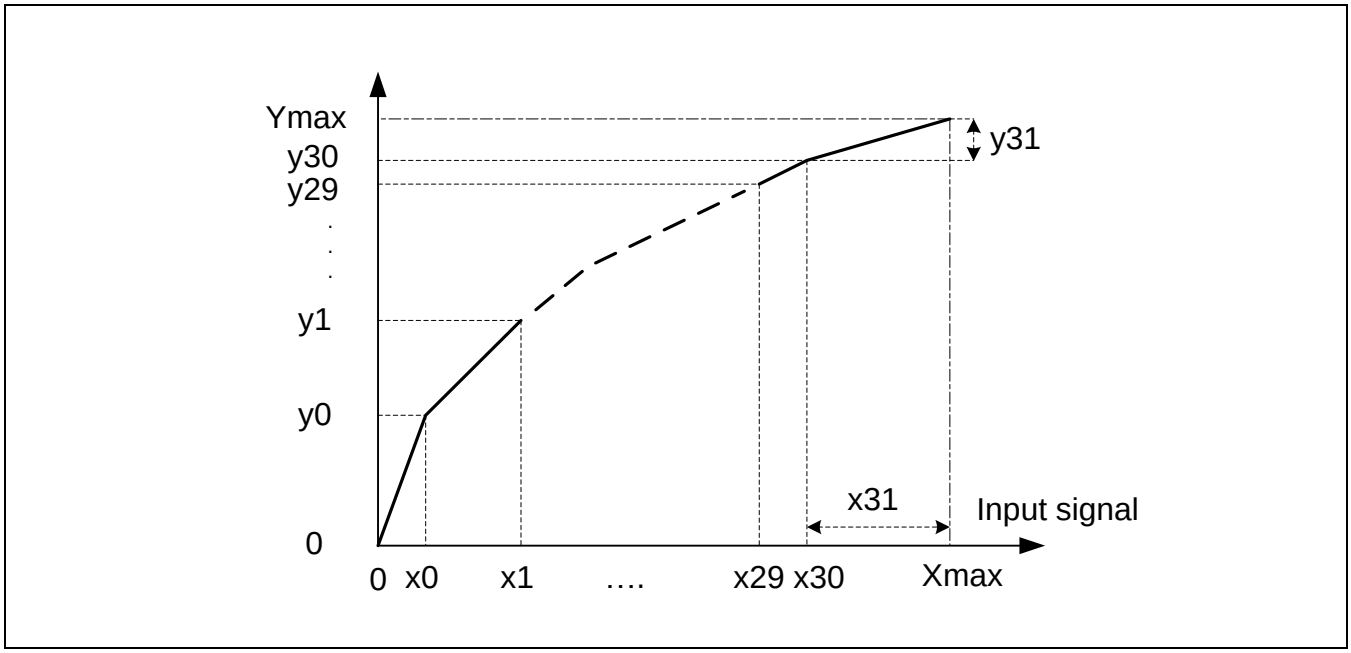


Figure 72 Curve for mapping x & y values

Anyhow, since each register can contain only 16 bits, the data is divided to MSB and LSB separated registers.

Take for an example the converting from the table below:

X Value (24 bits)	Y Value (20 bits)
12,345,678	98,765

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$12{,}345{,}678$ is a decimal number which is displayed binary as follow: $1011\ \ 1100\ \ 0110\ \ 0001\ \ 0100\ \ 1110$.

From this format the data is split to LSB register: $0110\ \ 0001\ \ 0100\ \ 1110$ (first 16 bits),
and to MSB register: $0000\ \ 0000\ \ 1011\ \ 1100$ (left part padded with zeros).

In this situation, each MSB/LSB register meets the limit of 16 bits. Finally, the values are converted to hex units: MSB register will contain: $0x00BC$ and LSB register will contain: $0x614E$. In the same way the y values are handled.

Between two consecutive points it is necessary to calculate the slope required to perform compression to intermediate values not defined in the input data. In order to calculate the slope, one must calculate- $\frac{\Delta y}{\Delta x}$. But to avoid division, multiplication of $\Delta y * \frac{1}{\Delta x}$ is made, while $\frac{1}{\Delta x}$ is calculated as $mult \gg shift$.

The formulas for $shift$ and $mult$ are as follows:

$$shift = accuracy + msb$$

$$mult = \frac{2^{shift} + \Delta x - 1}{\Delta x}$$

While $accuracy$ can reduce quantization error (11 is the default) and msb is the index of the active msb of Δx.

NOTE: Max valid $accuracy$ is 14, $mult$ is limited to value of $2^{16} - 1$ and $shift$ max value is 40.

For example, the following table is given:

X Value	Y Value
100	8
400	10

In this case $\Delta x = 400 - 100 = 300\ \ [dec] \rightarrow\ \ 1\ \ 0010\ \ 1100\ \ [bin]$

So $msb = 9 \rightarrow shift = accuracy + 9 \rightarrow set\ \ accuracy = 11 \rightarrow shift = 11 + 9 = 20\ \ [dec] \rightarrow shift =\ \ 0x0014$

Hence, $mult = \frac{2^{20}+300-1}{300} = 3496\ \ [dec] \rightarrow mult = 0x0DA8$

NOTE: Δx doesn't necessarily have to be number to the power of 2!

NOTE: Since in the ASPv47 chain the MSBs of the 24bit signal are used, Y values should be shifted in order to get the desired output as follows:

$$y_{shifted} = y_{shifted} * 2^{24-output\ \ bittage}$$

6.11.10.2 Related Registers

Table 59 Gamma Parameters

Index	Reset value	Bits	Register name	Type	Description
0x8D00	0x0001	[0:0]	gamma_1tbl_bypass	RW1	1'b1 : Bypass the entire block. 1'b0 :

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					Enable the entire block
0x8D02 - 0x8D40	See next table	[15:0]	gamma_1tbl_x_points_table_lsb_0_0 ~ 31	RW1	X points table default values - LSB
0x8D42 - 0x8D80	See next table	[7:0]	gamma_1tbl_x_points_table_msb_0_0 ~ 31	RW1	X points table default values - MSB
0x8D82 - 0x8DC0	See next table	[15:0]	gamma_1tbl_y_points_table_lsb_0_0 ~ 31	RW1	Y points table default values - LSB
0x8D42 - 0x8D80	See next table	[7:0]	gamma_1tbl_y_points_table_msb_0_0 ~ 31	RW1	Y points table default values - MSB
0x8E02 - 0x8E3E	See next table	[15:0]	gamma_1tbl_mult_points_table_0_0 ~ 30	RW1	LUT intervals multiplier points
0x8E40 - 0x8E7C	See next table	[5:0]	gamma_1tbl_shift_points_table_0_0 ~ 30	RW1	LUT interval shift points
0x8E7E	0x0020	[5:0]	Gamma_1tbl_num_of_points_used	RW1S	Number of X and Y points to use in PWL

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Table 60 Gamma Setting Table X Points Default

Register X MSB	Register X LSB	X_MSB Default	X_LSB Default
gamma_1tbl_x_points_table_msb_0_0	gamma_1tbl_x_points_table_lsb_0_0	0x0000	0x0000
gamma_1tbl_x_points_table_msb_0_1	gamma_1tbl_x_points_table_lsb_0_1	0x0000	0x0002
gamma_1tbl_x_points_table_msb_0_2	gamma_1tbl_x_points_table_lsb_0_2	0x0000	0x0004
gamma_1tbl_x_points_table_msb_0_3	gamma_1tbl_x_points_table_lsb_0_3	0x0000	0x0006
gamma_1tbl_x_points_table_msb_0_4	gamma_1tbl_x_points_table_lsb_0_4	0x0000	0x000a
gamma_1tbl_x_points_table_msb_0_5	gamma_1tbl_x_points_table_lsb_0_5	0x0000	0x0010
gamma_1tbl_x_points_table_msb_0_6	gamma_1tbl_x_points_table_lsb_0_6	0x0000	0x001b
gamma_1tbl_x_points_table_msb_0_7	gamma_1tbl_x_points_table_lsb_0_7	0x0000	0x002e
gamma_1tbl_x_points_table_msb_0_8	gamma_1tbl_x_points_table_lsb_0_8	0x0000	0x004d
gamma_1tbl_x_points_table_msb_0_9	gamma_1tbl_x_points_table_lsb_0_9	0x0000	0x0080
gamma_1tbl_x_points_table_msb_0_10	gamma_1tbl_x_points_table_lsb_0_10	0x0000	0x00d8
gamma_1tbl_x_points_table_msb_0_11	gamma_1tbl_x_points_table_lsb_0_11	0x0000	0x016b
gamma_1tbl_x_points_table_msb_0_12	gamma_1tbl_x_points_table_lsb_0_12	0x0000	0x0261
gamma_1tbl_x_points_table_msb_0_13	gamma_1tbl_x_points_table_lsb_0_13	0x0000	0x0400
gamma_1tbl_x_points_table_msb_0_14	gamma_1tbl_x_points_table_lsb_0_14	0x0000	0x06bb
gamma_1tbl_x_points_table_msb_0_15	gamma_1tbl_x_points_table_lsb_0_15	0x0000	0x0b51
gamma_1tbl_x_points_table_msb_0_16	gamma_1tbl_x_points_table_lsb_0_16	0x0000	0x1307
gamma_1tbl_x_points_table_msb_0_17	gamma_1tbl_x_points_table_lsb_0_17	0x0000	0x2000
gamma_1tbl_x_points_table_msb_0_18	gamma_1tbl_x_points_table_lsb_0_18	0x0000	0x35d2
gamma_1tbl_x_points_table_msb_0_19	gamma_1tbl_x_points_table_lsb_0_19	0x0000	0x5a83
gamma_1tbl_x_points_table_msb_0_20	gamma_1tbl_x_points_table_lsb_0_20	0x0000	0x9838
gamma_1tbl_x_points_table_msb_0_21	gamma_1tbl_x_points_table_lsb_0_21	0x0001	0x0000
gamma_1tbl_x_points_table_msb_0_22	gamma_1tbl_x_points_table_lsb_0_22	0x0001	0xae8a
gamma_1tbl_x_points_table_msb_0_23	gamma_1tbl_x_points_table_lsb_0_23	0x0002	0xd414
gamma_1tbl_x_points_table_msb_0_24	gamma_1tbl_x_points_table_lsb_0_24	0x0004	0xc1c0
gamma_1tbl_x_points_table_msb_0_25	gamma_1tbl_x_points_table_lsb_0_25	0x0008	0x0000
gamma_1tbl_x_points_table_msb_0_26	gamma_1tbl_x_points_table_lsb_0_26	0x000d	0x7450
gamma_1tbl_x_points_table_msb_0_27	gamma_1tbl_x_points_table_lsb_0_27	0x0016	0xa09f
gamma_1tbl_x_points_table_msb_0_28	gamma_1tbl_x_points_table_lsb_0_28	0x0026	0x0dfd
gamma_1tbl_x_points_table_msb_0_29	gamma_1tbl_x_points_table_lsb_0_29	0x0040	0x0000
gamma_1tbl_x_points_table_msb_0_30	gamma_1tbl_x_points_table_lsb_0_30	0x006b	0xa27f
gamma_1tbl_x_points_table_msb_0_31	gamma_1tbl_x_points_table_lsb_0_31	0x00ff	0xffff

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Table 61 Gamma Setting Table Y Points Default

Register Y MSB	Register Y LSB	Y_MSB Default	Y_LSB Default
gamma_1tbl_y_points_table_msb_0_0	gamma_1tbl_y_points_table_lsb_0_0	0x0000	0x0000
gamma_1tbl_y_points_table_msb_0_1	gamma_1tbl_y_points_table_lsb_0_1	0x0001	0x428a
gamma_1tbl_y_points_table_msb_0_2	gamma_1tbl_y_points_table_lsb_0_2	0x0001	0x965f
gamma_1tbl_y_points_table_msb_0_3	gamma_1tbl_y_points_table_lsb_0_3	0x0001	0xd12e
gamma_1tbl_y_points_table_msb_0_4	gamma_1tbl_y_points_table_lsb_0_4	0x0002	0x2789
gamma_1tbl_y_points_table_msb_0_5	gamma_1tbl_y_points_table_lsb_0_5	0x0002	0x8514
gamma_1tbl_y_points_table_msb_0_6	gamma_1tbl_y_points_table_lsb_0_6	0x0002	0xffff
gamma_1tbl_y_points_table_msb_0_7	gamma_1tbl_y_points_table_lsb_0_7	0x0003	0x9542
gamma_1tbl_y_points_table_msb_0_8	gamma_1tbl_y_points_table_lsb_0_8	0x0004	0x411b
gamma_1tbl_y_points_table_msb_0_9	gamma_1tbl_y_points_table_lsb_0_9	0x0005	0x0a28
gamma_1tbl_y_points_table_msb_0_10	gamma_1tbl_y_points_table_lsb_0_10	0x0005	0xffff
gamma_1tbl_y_points_table_msb_0_11	gamma_1tbl_y_points_table_lsb_0_11	0x0007	0x222c
gamma_1tbl_y_points_table_msb_0_12	gamma_1tbl_y_points_table_lsb_0_12	0x0008	0x79ee
gamma_1tbl_y_points_table_msb_0_13	gamma_1tbl_y_points_table_lsb_0_13	0x000a	0x1451
gamma_1tbl_y_points_table_msb_0_14	gamma_1tbl_y_points_table_lsb_0_14	0x000b	0xfd08
gamma_1tbl_y_points_table_msb_0_15	gamma_1tbl_y_points_table_lsb_0_15	0x000e	0x4169
gamma_1tbl_y_points_table_msb_0_16	gamma_1tbl_y_points_table_lsb_0_16	0x0010	0xf390
gamma_1tbl_y_points_table_msb_0_17	gamma_1tbl_y_points_table_lsb_0_17	0x0014	0x28a2
gamma_1tbl_y_points_table_msb_0_18	gamma_1tbl_y_points_table_lsb_0_18	0x0017	0xf92d
gamma_1tbl_y_points_table_msb_0_19	gamma_1tbl_y_points_table_lsb_0_19	0x001c	0x824c
gamma_1tbl_y_points_table_msb_0_20	gamma_1tbl_y_points_table_lsb_0_20	0x0021	0xe720
gamma_1tbl_y_points_table_msb_0_21	gamma_1tbl_y_points_table_lsb_0_21	0x0028	0x5145
gamma_1tbl_y_points_table_msb_0_22	gamma_1tbl_y_points_table_lsb_0_22	0x002f	0xf221
gamma_1tbl_y_points_table_msb_0_23	gamma_1tbl_y_points_table_lsb_0_23	0x0039	0x047d
gamma_1tbl_y_points_table_msb_0_24	gamma_1tbl_y_points_table_lsb_0_24	0x0043	0xce40
gamma_1tbl_y_points_table_msb_0_25	gamma_1tbl_y_points_table_lsb_0_25	0x0050	0xa28b
gamma_1tbl_y_points_table_msb_0_26	gamma_1tbl_y_points_table_lsb_0_26	0x005f	0xe443
gamma_1tbl_y_points_table_msb_0_27	gamma_1tbl_y_points_table_lsb_0_27	0x0072	0x08f8
gamma_1tbl_y_points_table_msb_0_28	gamma_1tbl_y_points_table_lsb_0_28	0x0087	0x9c7d
gamma_1tbl_y_points_table_msb_0_29	gamma_1tbl_y_points_table_lsb_0_29	0x00a1	0x4517
gamma_1tbl_y_points_table_msb_0_30	gamma_1tbl_y_points_table_lsb_0_30	0x00bf	0xc886
gamma_1tbl_y_points_table_msb_0_31	gamma_1tbl_y_points_table_lsb_0_31	0x00ff	0xffff

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6.12 Miscellaneous functions

6.12.1 NVM OTP Memory

The NVM memory module is a non-volatile OTP (one-timing programming) memory module. This module enables the saving of unique data to each chip at the production stage.

OTP memory is used to store the following information:
- Chip ID data – production history data to be stored during die sorting
- Security data - certification, private key and public key
- BPC data
- OTP memory is ordered in pages, 255 pages of 64bytes each.
- Pages 180-255 can be used by users.

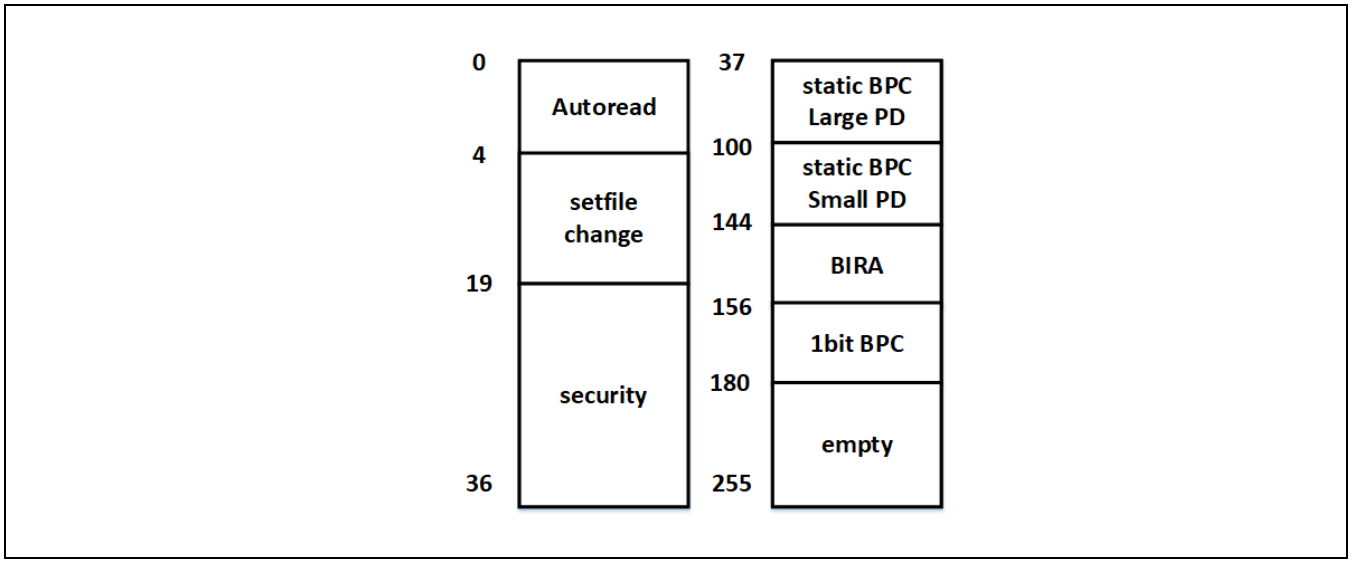


Figure 73 OTP Map Diagram

6.12.1.1 Data read and write interface

6.12.1.1.1 NVM I/F control registers

Table 62 Test Data Color Parameters

Index	Reset value	Bits	Register name	Type	Description
0x5B2E	0x0032	[7:0]	clock_fre_Mhz	RW	system Clock frequency for automatic timing wave
0x5A00	0x0000	[15:0]	data_transfer_if_control	RW	Data transfer if control [0] : 1'b1 - enable 1'b0 - disable [1] : 1'b1 - write enable

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					1'b0 - read enable [2] : 1'b1 - clear error bits Data transfer if status [8] : read if ready [9] : write if ready [10] : data corrupted [11] : improper if usage
0x5A02	0x0000	[7:0]	data_transfer_if_page_select	RW	Pages from 0 - 255
0x5A04 ~0x5A42	0x0000	[15:0]	data_transfer_if_data_0~63	RW	Write or read data 0~63 per each page

6.12.1.1.2 Read sequence

Refer to the order of the figure below. For sequential read, repeat Set PAGE and Read OTP data from buffer without making initial state.

NOTE: To read or write OTP read before streaming on, set the value of 0x1008 register to 0x0000.

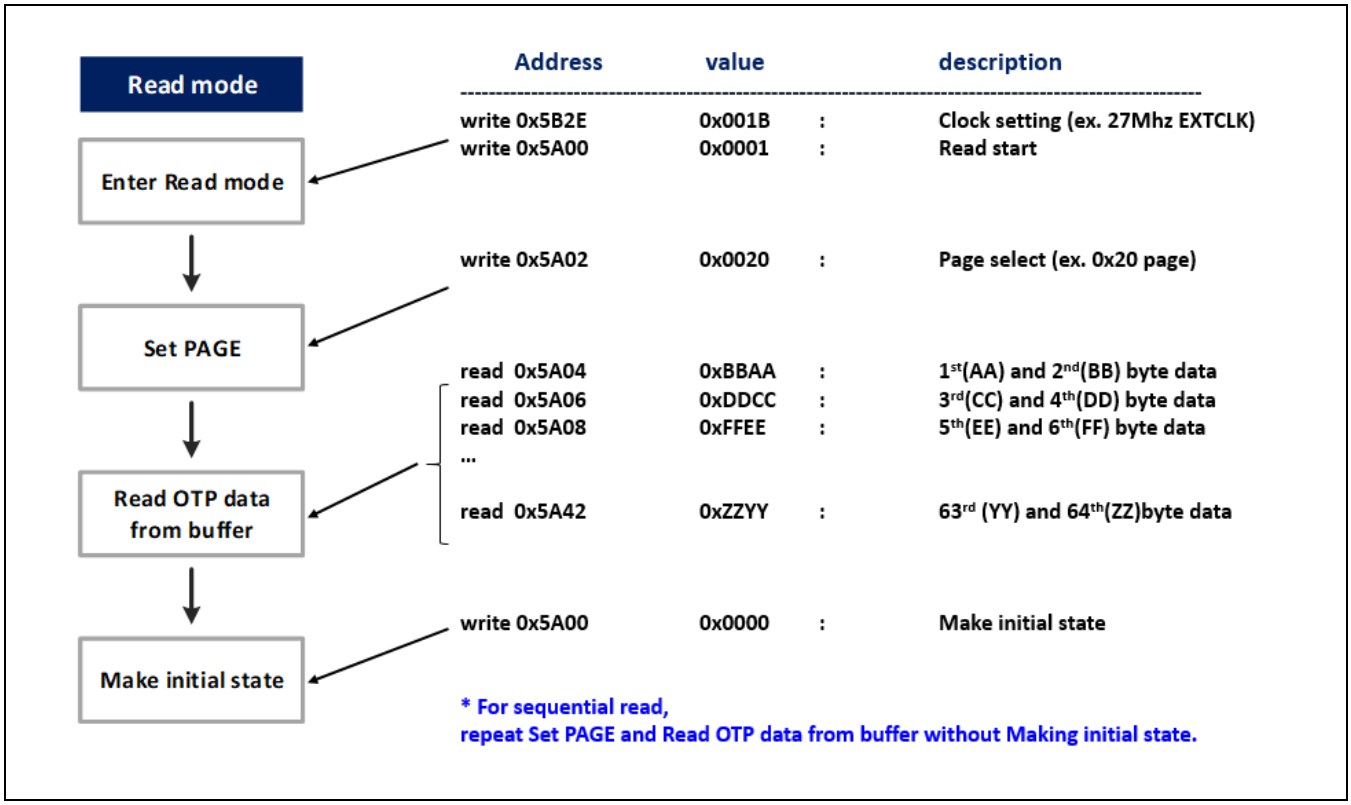


Figure 74 OTP read sequence

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.12.1.1.3 Write sequence

Refer to the order of the figure below. For sequential write, repeat Set PAGE and Write OTP data to buffer without making initial state. Before changing the page, it is necessary to wait 181ms for all data on page 1 to be written.

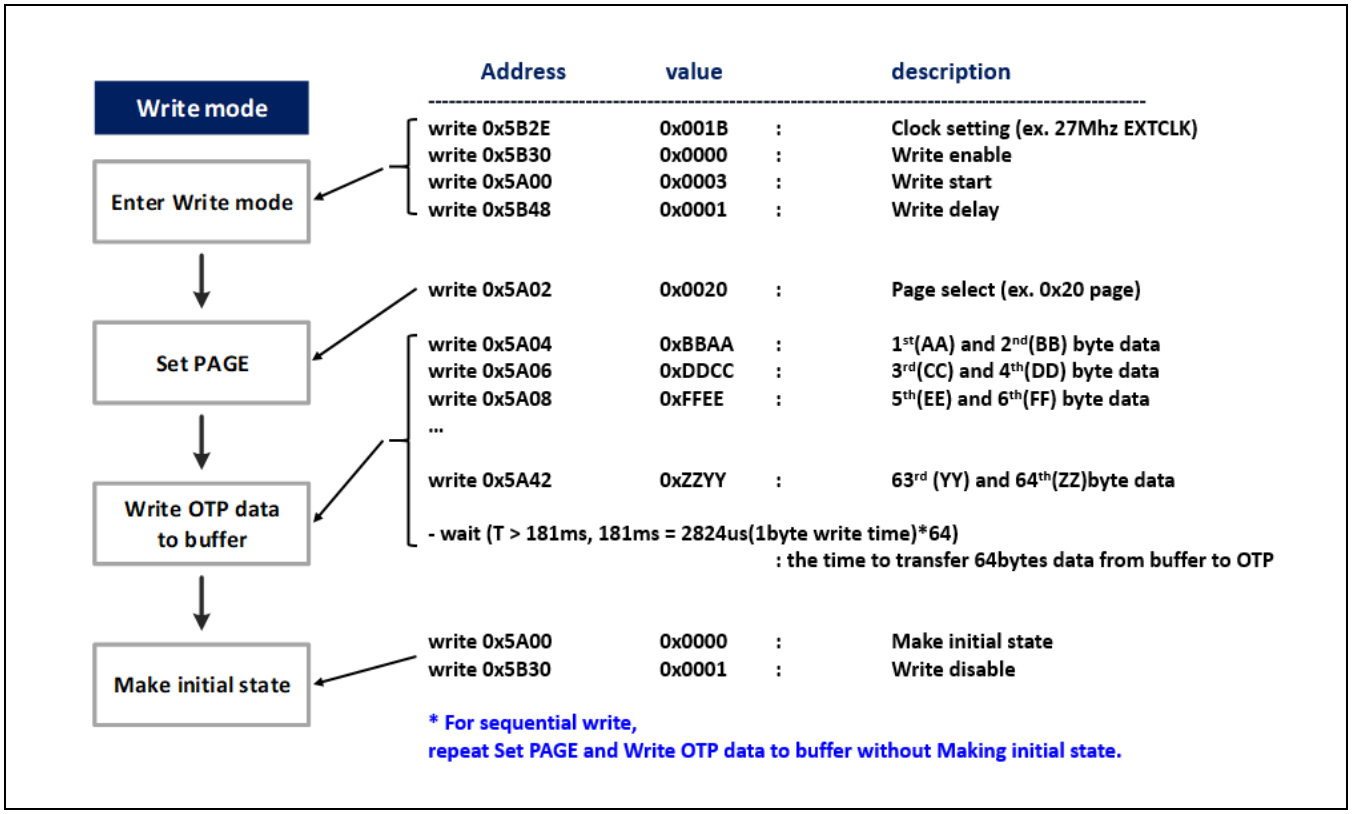


Figure 75 OTP write sequence

6.12.2 TM(Temperature Monitor)

S5K3B6SA has chip-embedded temperature sensor, user can check the junction temperature of sensor by reading the temperature sensor value. S5K3B6SA has 3-point calibrated value tested in Wafer level to meet accuracy of temperature sensor, then the temperature is calculated based on the calibrated value included in NVM memory.

3-point temperature calibrated value makes one trend line like as *Figure 76*.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

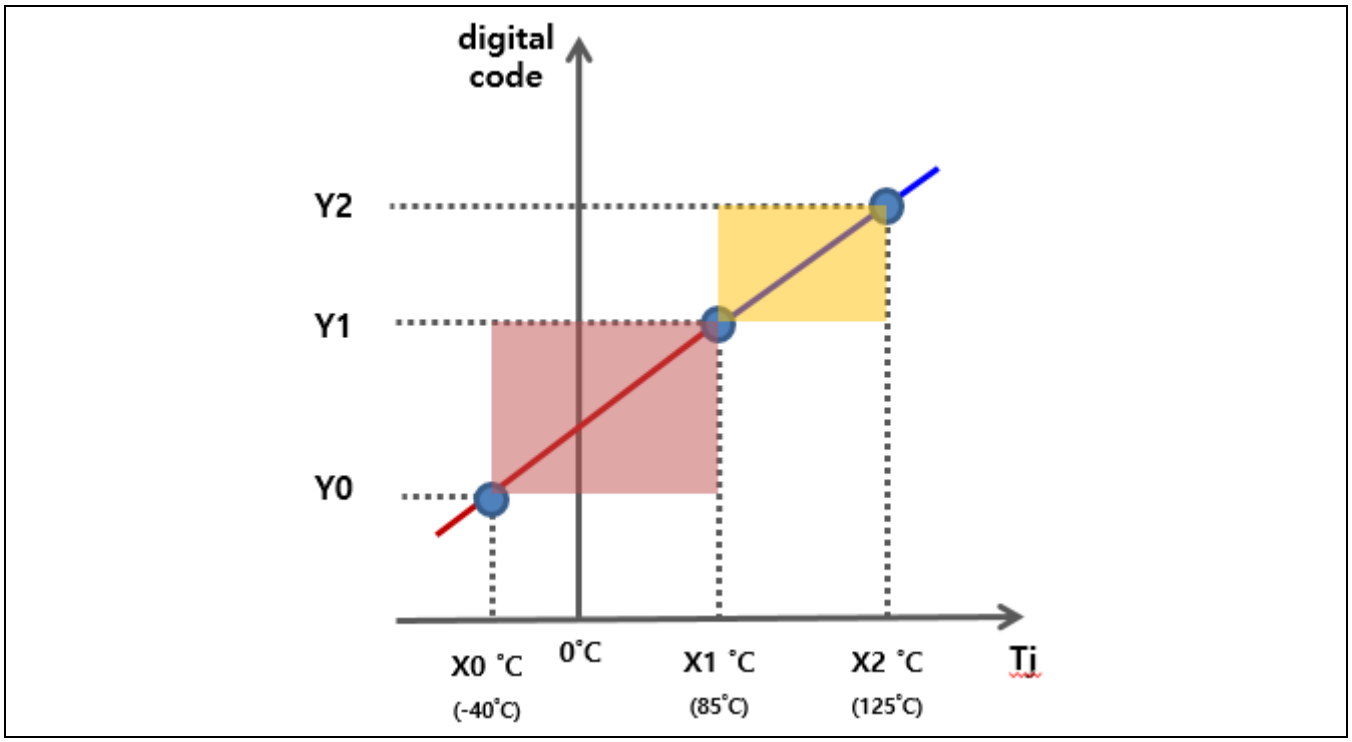


Figure 76 TMC Formula

6.12.2.1 How to read

S5K3B6SA can output both TMC(Temperature Monitoring Circuit) data as ADC temp data (LSB) and calibrated data (℃).

In view of user, reading only this calibrated data is needed. TMC temperature output (14bit) is composed as [13:13] sign bit (1'b1= negative, 1'b0= positive), [12:4] integer bit and [3:0] fraction bit. The value is calculated based on the calibrated value which written in OTP.

TMC calibrated data reflect from -51℃ to 512℃, and represent 13h'2000 when below -51℃, above value from 13h'3000 when above 512℃.

Table 63 Temperature Sensor

Index	Reset value	Bits	Register name	Type	Description
0x6010	0x0A8C	[13:0]	atopif_tmc_stat_data_top	RO	Current TOP TMC data (Physically Bottom TMC)
0x6012	0x0A8C	[13:0]	atopif_tmc_stat_data_top	RO	Current BOT TMC data (Physically TOP TMC)
0x602C	0x2330	[13:0]	atopif_tmc_cal_top	RO	TMC calibration Top [13:13] Sign bit (1'b1= negative, 1'b0= positive)

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

					[12:4] Integer bit (9bit) [3:0] Fraction bit (4bit)
0x602E	0x2330	[13:0]	atopif_tmc_cal_bot	RO	TMC calibration Bottom [13:13] Sign bit (1'b1= negative, 1'b0= positive) [12:4] Integer bit (9bit) [3:0] Fraction bit (4bit)

6.12.2.2 Use case

Example
ex1) atopif_tmc_cal_ = 13'h50D Integer : 'h50 = 'd80, Fraction : 'hD = 13/16 = 0.8125 → temperature : 80.8C ex2) atopif_tmc_cal_ = 13'h2185 Integer : 'h18 = 'd24, Fraction : 'hD = 5/16 = 0.3125 → temperature : -24 + 0.3125 = -23.7C ex2) atopif_tmc_cal_ = 13'h2000 → temperature is under -51C atopif_tmc_cal_ = 13'h3000 → temperature is above 512C

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6.12.3 VM(Voltage Monitor)

S5K3B6SA has chip-embedded temperature sensor, user can check the junction temperature of sensor by reading the temperature sensor value.

6.12.3.1 How to read

S5K3B6SA can output both data as calibrated data. In view of user, reading only this calibrated data is needed. Read the register value of the desired voltage below and convert it into a decimal number. The unit is mV.

Table 64 Temperature Sensor

Index	Reset value	Bits	Register name	Type	Description
0x38b6	0x0000	[12:0]	aig_sfm_data_top_2	RO	TOP VBGR after calibration in real time.
0x38c6	0x0000	[12:0]	aig_sfm_data_top_10	RO	TOP VDDA after calibration in real time.
0x38c8	0x0000	[12:0]	aig_sfm_data_top_11	RO	TOP VDDIO after calibration in real time.
0x38ca	0x0000	[12:0]	aig_sfm_data_top_12	RO	TOP VDDD after calibration in real time.
0x38d6	0x0000	[12:0]	aig_sfm_data_bottom_2	RO	BOTTOM VBGR after calibration in real time.
0x38e6	0x0000	[12:0]	aig_sfm_data_bottom_10	RO	BOTTOM VDDA after calibration in real time.
0x38e8	0x0000	[12:0]	aig_sfm_data_bottom_11	RO	BOTTOM VDDIO after calibration in real time.
0x38ea	0x0000	[12:0]	aig_sfm_data_bottom_12	RO	BOTTOM VDDD after calibration in real time.

6.13 Output interface

6.13.1 Serial Output Interface (MIPI CSI-2)

S5K3B6SA MIPI CSI-2 interface is a four-lane high-speed serial interface to connect the camera sensor to a host processor. S5K3B6SA MIPI core IP is compatible with the MIPI alliance standards for D-PHY and CSI2 v2.1.

The maximum high-speed clock frequency of the MIPI is 4.5 GHz.

MIPI supports following features:

- MIPI DPHY CSI-2: 4-lane (2/3/4-lane modes) with maximum of 4.5 Gbps (bit clock rate)
- RAW8, RAW10, RAW12, RAW14, RAW16, RAW20, RAW24 data types
- MIPI transmitting (Tx) only
- One clock lane
- Up to 15 number of virtual channels for the multiple exposure mode
- Ultra-Low-Power Mode (ULPM)

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6.13.1.1 Lane Control

Table 65 MIPI lane control

Register name	Field Name	Index	Bits	Type	Reset value
outif_mipi_num_of_lanes	outregs_number_of_lanes	0x7090	[1:0]	RW	0x0003
outif_dphy_lane_config	outregs_dphy_continous_clock	0x70A0	[0]	RW	0b1
	outregs_dphy_continous_clock_vblank_disable		[1]	RW	0b0

- outif_mipi_num_of_lanes : Number of lanes //00 : one lane 01: two lanes 10: three lanes 11: 4lanes

Table 66 MIPI lane swap control

Register name	Field Name	Index	Bits	Type	Reset value
outif_general_control	outregs_physical_lane_swap_enable	0x7000	[8]	RW	0b1
outif_physical_lane_swap_control	outregs_physical_lane0_swap_sel	0x7002	[1:0]	RW1	0x0
	outregs_physical_lane1_swap_sel		[5:4]	RW1	0x1
	outregs_physical_lane2_swap_sel		[9:8]	RW1	0x2
	outregs_physical_lane3_swap_sel		[13:12]	RW1	0x3
outif_dphy_swap_pins	outregs_dphy_swap_clk	0x70d2	[0]	RW	0b0
	outregs_dphy_swap_data		[1]	RW	0b0
	outregs_dphy_swap_data0		[4]	RW	0b1
	outregs_dphy_swap_data1		[5]	RW	0b1
	outregs_dphy_swap_data2		[6]	RW	0b1
	outregs_dphy_swap_data3		[7]	RW	0b1

- outif_physical_lane_swap_control : If the lane swap is enabled, each lane can be changed to a different lane. // 0: lane0, 1: lane1, 2: lane2, 3:lane3

- outif_dphy_swap_pins : When dphy_swap_data is enabled, P and N of each lane can be changed. // 0: PN to PN, 1: PN to NP

6.13.2 Output format

There are several output formats, which are pre-defined. These formats are selected the data format register:

Data format

:0x7(RAW14), 0x8(RAW8, RAW16), 0xA(RAW10, RAW20), 0xC(RAW12, RAW24)

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Table 67 Output format register

Register name	Field Name	Index	Bits	Type	Reset value
isp_sthdr_formatter_data_format_1_2	outregs_isp_sthdr_formatter_isp1_data_format	0xf0e0	[7:0]	RW1S	0xC
	outregs_isp_sthdr_formatter_isp2_data_format		[15:8]	RW1S	0xC
isp_sthdr_formatter_data_format_3_4	outregs_isp_sthdr_formatter_isp3_data_format	0xf0e2	[7:0]	RW1S	0x0
	outregs_isp_sthdr_formatter_isp4_data_format		[15:8]	RW1S	0x0
isp_sthdr_formatter_data_format_stat_ddm	outregs_isp_sthdr_formatter_stat_data_format	0xf0e4	[7:0]	RW1S	0x8
	outregs_isp_sthdr_formatter_ddm_data_format		[15:8]	RW1S	0x8
isp_sthdr_formatter_data_format_emb	outregs_isp_sthdr_formatter_emb_header_data_format	0xf0e6	[7:0]	RW1S	0x8
	outregs_isp_sthdr_formatter_emb_footer_data_format		[15:8]	RW1S	0x8
isp_sthdr_formatter_data_format_1_2_s	outregs_isp_sthdr_formatter_isp1_data_format_s	0xf1e0	[7:0]	RW1S	0xC
	outregs_isp_sthdr_formatter_isp2_data_format_s		[15:8]	RW1S	0xC
isp_sthdr_formatter_data_format_3_4_s	outregs_isp_sthdr_formatter_isp3_data_format_s	0xf1e2	[7:0]	RW1S	0x0
	outregs_isp_sthdr_formatter_isp4_data_format_s		[15:8]	RW1S	0x0
isp_sthdr_formatter_data_format_stat_ddm_s	outregs_isp_sthdr_formatter_stat_data_format_s	0xf1e4	[7:0]	RW1S	0x8
	outregs_isp_sthdr_formatter_ddm_data_format_s		[15:8]	RW1S	0x8
isp_sthdr_formatter_data_format_emb_s	outregs_isp_sthdr_formatter_emb_header_data_format_s	0xf1e6	[7:0]	RW1S	0x8
	outregs_isp_sthdr_formatter_emb_footer_data_format_s		[15:8]	RW1S	0x8

6.13.3 Data type

Image data type. RAW24 is packetized as RAW12

:0x2A(RAW8), 0x2B(RAW10), 0x2C(RAW12, RAW24), 0x2D(RAW14), 0x2E(RAW16), 0x2F(RAW20)

Default value of Non-image data type

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: 0x12(Embedded Header), 0x30(Embedded Footer), 0x31(statistics data), 0x32(defect detection map)

Table 68 Data type register

Register name	Field Name	Index	Bits	Type	Reset value
isp_sthdr_formatter_data_type_1_2	outregs_isp_sthdr_formatter_isp1_data_type	0xf020	[5:0]	RW1S	0x2C
	outregs_isp_sthdr_formatter_isp2_data_type		[13:8]	RW1S	0x2C
isp_sthdr_formatter_data_type_3_4	outregs_isp_sthdr_formatter_isp3_data_type	0xf022	[5:0]	RW1S	0x0
	outregs_isp_sthdr_formatter_isp4_data_type		[13:8]	RW1S	0x0
isp_sthdr_formatter_data_type_stat_ddm	outregs_isp_sthdr_formatter_stat_data_type	0xf024	[5:0]	RW1S	0x31
	outregs_isp_sthdr_formatter_ddm_data_type		[13:8]	RW1S	0x32
isp_sthdr_formatter_data_type_emb	outregs_isp_sthdr_formatter_emb_header_data_type	0xf026	[5:0]	RW1S	0x12
	outregs_isp_sthdr_formatter_emb_footer_data_type		[13:8]	RW1S	0x30
isp_sthdr_formatter_data_type_1_2_s	outregs_isp_sthdr_formatter_isp1_data_type_s	0xf120	[5:0]	RW1S	0x2C
	outregs_isp_sthdr_formatter_isp2_data_type_s		[13:8]	RW1S	0x2C
isp_sthdr_formatter_data_type_3_4_s	outregs_isp_sthdr_formatter_isp3_data_type_s	0xf122	[5:0]	RW1S	0x0
	outregs_isp_sthdr_formatter_isp4_data_type_s		[13:8]	RW1S	0x0
isp_sthdr_formatter_data_type_stat_ddm_s	outregs_isp_sthdr_formatter_stat_data_type_s	0xf124	[5:0]	RW1S	0x31
	outregs_isp_sthdr_formatter_ddm_data_type_s		[13:8]	RW1S	0x32
isp_sthdr_formatter_data_type_emb_s	outregs_isp_sthdr_formatter_emb_header_data_type_s	0xf126	[5:0]	RW1S	0x12
	outregs_isp_sthdr_formatter_emb_footer_data_type_s		[13:8]	RW1S	0x30

6.13.4 Virtual Channel control

S5K3B6SA supports max. 15ea of Virtual Channel, which is described in MIPI CSI-2 v2.1. All of below registers

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have shadow registers set for Context Switching, please refer to register map for more information.

Register setting for Number of Virtual Channel

outif_mipi_nof_vc : number of VC – 1

outif_mipi_vc_ch_enable_15_0 : Set N-ea of LSB bits as 0x1 when using N-ea VC.

e.g. N-ea of VC are used. → [N-1]~[0] bits will be all 0x1

Register setting for Virtual Channel Value

outif_mipi_vc_ch_values_N_N+1

: set each MIPI VC values, which match with values of isp_sthdr_formatter_virtual_channel_values_#

isp_sthdr_formatter_virtual_channel_values_# : set each VC value

Table 69 MIPI virtual channel and lane control

Register Name	Field Name	Index	Bits	Type	Reset value
outif_mipi_nof_vc	outregs_mipi_nof_vc	0x7198	[4:0]	RW1S	0x00
outif_mipi_vc_ch_enable_15_0	outregs_mipi_vc_ch_enable_15_0	0x719a	[15:0]	RW1S	0x00
outif_mipi_vc_values_1_2	outregs_mipi_isp1_vc_value	0x7060	[4:0]	RW1S	0x00
	outregs_mipi_isp2_vc_value		[12:8]	RW1S	0x00
outif_mipi_vc_values_3_4	outregs_mipi_isp3_vc_value	0x7062	[4:0]	RW1S	0x00
	outregs_mipi_isp4_vc_value		[12:8]	RW1S	0x00
outif_mipi_vc_values_5_6	outregs_mipi_isp5_vc_value	0x7064	[4:0]	RW1S	0x00
	outregs_mipi_isp6_vc_value		[12:8]	RW1S	0x00
outif_mipi_vc_values_7_8	outregs_mipi_isp7_vc_value	0x7066	[4:0]	RW1S	0x00
	outregs_mipi_isp8_vc_value		[12:8]	RW1S	0x00
outif_mipi_vc_values_9_10	outregs_mipi_isp9_vc_value	0x7068	[4:0]	RW1S	0x00
	outregs_mipi_isp10_vc_value		[12:8]	RW1S	0x00
outif_mipi_vc_values_11_12	outregs_mipi_isp11_vc_value	0x706a	[4:0]	RW1	0x00
	outregs_mipi_isp12_vc_value		[12:8]	RW1	0x00
outif_mipi_vc_values_13_14	outregs_mipi_isp13_vc_value	0x706c	[4:0]	RW1	0x00
	outregs_mipi_isp14_vc_value		[12:8]	RW1	0x00
isp_sthdr_formatter_virtual_channel_values_1_2	outregs_isp_sthdr_formatter_isp1_virtual_channel_value	0xf010	[4:0]	RW1S	0x00
	outregs_isp_sthdr_formatter_isp2_virtual_channel_value		[12:8]	RW1S	0x00
isp_sthdr_formatter_virtual_channel	outregs_isp_sthdr_formatter_isp3	0xf012	[4:0]	RW1S	0x00

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_values_3_4	_virtual_channel_value				
	outregs_isp_sthdr_formatter_isp4 _virtual_channel_value		[12:8]	RW1S	0x00
isp_sthdr_formatter_virtual_channel _values_stat_ddm	outregs_isp_sthdr_formatter_stat _virtual_channel_value	0xf014	[4:0]	RW1S	0x00
	outregs_isp_sthdr_formatter_ddm _virtual_channel_value		[12:8]	RW1S	0x00
isp_sthdr_formatter_virtual_channel _values_emb	outregs_isp_sthdr_formatter_emb _header_virtual_channel_value	0xf016	[4:0]	RW1S	0x00
	outregs_isp_sthdr_formatter_emb _footer_virtual_channel_value		[12:8]	RW1S	0x00
isp_sthdr_formatter_virtual_channel _values_blank	outregs_isp_sthdr_formatter_blan k_virtual_channel_value	0xf018	[4:0]	RW1S	0x00

6.13.5 Output Y size setting

outif_global_y_output_size : total y output size

= outif_num_of_video_lines + outif_num_of_emb_start + (number of statistics lines:non-register) + (number of blank lines:non-register)

outif_num_of_video_lines: Number of Active data lines(total lines of image, additional frame, defect detection map), which excludes embedded data rows, statistics data rows, blank rows.

outif_num_of_emb_start : Number of embedded header lines(which is located before image lines)

Table 70 Output image vertical size setting

Register name	Field Name	Index	Bits	Type	Reset value
outif_global_y_output_size	outregs_global_y_output_size	0x70ba	[15:0]	RW1S	0x0c20
outif_num_of_video_lines	outregs_type_nof_video	0x70f2	[15:0]	RW1S	0x0c20
outif_num_of_emb_start	outregs_type_nof_emb_start	0x70f0	[15:0]	RW1S	0x0000
outif_global_y_output_size_s	outregs_global_y_output_size_s	0x72ba	[15:0]	RW1S	0x0c20
outif_num_of_video_lines_s	outregs_type_nof_video_s	0x72f2	[15:0]	RW1S	0x0c20
outif_num_of_emb_start_s	outregs_type_nof_emb_start_s	0x72f0	[15:0]	RW1S	0x0000

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6.13.6 Output X size setting

X size Setting of Image data

isp_sthdr_formatter_isp1_x_output_size : x output size of LPD HCG data, Merge Compressed data, Merged data, Pixel Reorder data

isp_sthdr_formatter_isp2_x_output_size : x output size of LPD LCG data, additional frame

isp_sthdr_formatter_isp3_x_output_size : x output size of SPD data

isp_sthdr_formatter_isp4_x_output_size : x output size of LPD short exposure data

x output size =

RAW14, RAW16, RAW20, RAW24: readout x size* 2

RAW8, RAW10, RAW12: readout x size

X size Setting of Non-Image data

isp_sthdr_formatter_stat_x_output_size : x output size of statistics rows

isp_sthdr_formatter_emb_header_x_output_size: x output size of Embedded Header

isp_sthdr_formatter_emb_footer_x_output_size : x output size of Embedded Footer

isp_sthdr_formatter_ddm_x_output_size : x output of defect detection map

Table 71 Output image horizontal size setting (size)

Register name	Field Name	Index	Bits	Type	Reset value
isp_sthdr_formatter_isp1_x_output_size	outregs_isp_sthdr_formatter_isp1_x_output_size	0xf030	[15:0]	RW1S	0x0790
isp_sthdr_formatter_isp2_x_output_size	outregs_isp_sthdr_formatter_isp2_x_output_size	0xf032	[15:0]	RW1S	0x0790
isp_sthdr_formatter_isp3_x_output_size	outregs_isp_sthdr_formatter_isp3_x_output_size	0xf034	[15:0]	RW1S	0x0000
isp_sthdr_formatter_isp4_x_output_size	outregs_isp_sthdr_formatter_isp4_x_output_size	0xf036	[15:0]	RW1S	0x0000
isp_sthdr_formatter_stat_x_output_size	outregs_isp_sthdr_formatter_stat_x_output_size	0xf038	[15:0]	RW1S	0x0000
isp_sthdr_formatter_ddm_x_output_size	outregs_isp_sthdr_formatter_ddm_x_output_size	0xf03a	[15:0]	RW1S	0x0000
isp_sthdr_formatter_emb_header_x_output_size	outregs_isp_sthdr_formatter_emb_header_x_output_size	0xf03c	[15:0]	RW1S	0x1030
isp_sthdr_formatter_emb_footer_x_output_size	outregs_isp_sthdr_formatter_emb_footer_x_output_size	0xf03e	[15:0]	RW1S	0x0100

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isp_sthdr_formatter_isp1_x_output_size_s	outregs_isp_sthdr_formatter_isp1_x_output_size_s	0xf130	[15:0]	RW1S	0x0790
isp_sthdr_formatter_isp2_x_output_size_s	outregs_isp_sthdr_formatter_isp2_x_output_size_s	0xf132	[15:0]	RW1S	0x0790
isp_sthdr_formatter_isp3_x_output_size_s	outregs_isp_sthdr_formatter_isp3_x_output_size_s	0xf134	[15:0]	RW1S	0x0000
isp_sthdr_formatter_isp4_x_output_size_s	outregs_isp_sthdr_formatter_isp4_x_output_size_s	0xf136	[15:0]	RW1S	0x0000
isp_sthdr_formatter_stat_x_output_size_s	outregs_isp_sthdr_formatter_stat_x_output_size_s	0xf138	[15:0]	RW1S	0x0000
isp_sthdr_formatter_ddm_x_output_size_s	outregs_isp_sthdr_formatter_ddm_x_output_size_s	0xf13a	[15:0]	RW1S	0x0000
isp_sthdr_formatter_emb_header_x_output_size_s	outregs_isp_sthdr_formatter_emb_header_x_output_size_s	0xf13c	[15:0]	RW1S	0x1030
isp_sthdr_formatter_emb_footer_x_output_size_s	outregs_isp_sthdr_formatter_emb_footer_x_output_size_s	0xf13e	[15:0]	RW1S	0x0100

- isp_sthdr_formatter_#_x_length_size: packet length(in bytes) of each data(#)

(RAW14, RAW16, RAW20, RAW24 data format are packetized as raw7, raw8, raw10,raw12)

= ((isp_sthdr_formatter_#_x_output_size * isp_sthdr_formatter_#_data_format) % 8) ?
(isp_sthdr_formatter_#_x_output_size * isp_sthdr_formatter_#_data_format)/8 + 1 :
(isp_sthdr_formatter_#_x_output_size * isp_sthdr_formatter_#_data_format)

= (x_size * raw7/raw8/raw10/raw12 % 8)? (x_size * raw7/raw8/raw10/raw12) / 8 +1 : (x_size* raw7/raw8/raw10/raw12)/8

Table 72 Output image horizontal size setting (packet length)

Register name	Field Name	Index	Bits	Type	Reset value
isp_sthdr_formatter_isp1_x_length_size	outregs_isp_sthdr_formatter_isp1_x_length_size	0xf050	[15:0]	RW1S	0x0b58
isp_sthdr_formatter_isp2_x_length_size	outregs_isp_sthdr_formatter_isp2_x_length_size	0xf052	[15:0]	RW1S	0x0b58
isp_sthdr_formatter_isp3_x_length_size	outregs_isp_sthdr_formatter_isp3_x_length_size	0xf054	[15:0]	RW1S	0x0000
isp_sthdr_formatter_isp4_x_length_size	outregs_isp_sthdr_formatter_isp4_x_length_size	0xf056	[15:0]	RW1S	0x0000
isp_sthdr_formatter_stat_x_length_size	outregs_isp_sthdr_formatter_stat_x_length_size	0xf058	[15:0]	RW1S	0x0000
isp_sthdr_formatter_ddm_x_length_size	outregs_isp_sthdr_formatter_ddm_x_length_size	0xf05a	[15:0]	RW1S	0x0000

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isp_sthdr_formatter_emb_header_x_length_size	outregs_isp_sthdr_formatter_emb_header_x_length_size	0xf05c	[15:0]	RW1S	0x1030
isp_sthdr_formatter_emb_footer_x_length_size	outregs_isp_sthdr_formatter_emb_footer_x_length_size	0xf05e	[15:0]	RW1S	0x0100
isp_sthdr_formatter_isp1_x_length_size_s	outregs_isp_sthdr_formatter_isp1_x_length_size_s	0xf150	[15:0]	RW1S	0x0b58
isp_sthdr_formatter_isp2_x_length_size_s	outregs_isp_sthdr_formatter_isp2_x_length_size_s	0xf152	[15:0]	RW1S	0x0b58
isp_sthdr_formatter_isp3_x_length_size_s	outregs_isp_sthdr_formatter_isp3_x_length_size_s	0xf154	[15:0]	RW1S	0x0000
isp_sthdr_formatter_isp4_x_length_size_s	outregs_isp_sthdr_formatter_isp4_x_length_size_s	0xf156	[15:0]	RW1S	0x0000
isp_sthdr_formatter_stat_x_length_size_s	outregs_isp_sthdr_formatter_stat_x_length_size_s	0xf158	[15:0]	RW1S	0x0000
isp_sthdr_formatter_ddm_x_length_size_s	outregs_isp_sthdr_formatter_ddm_x_length_size_s	0xf15a	[15:0]	RW1S	0x0000
isp_sthdr_formatter_emb_header_x_length_size_s	outregs_isp_sthdr_formatter_emb_header_x_length_size_s	0xf15c	[15:0]	RW1S	0x1030
isp_sthdr_formatter_emb_footer_x_length_size_s	outregs_isp_sthdr_formatter_emb_footer_x_length_size_s	0xf15e	[15:0]	RW1S	0x0100

- isp_sthdr_formatter_#_num_of_ram_lines
  = (isp_sthdr_formatter_#_x_length_size % 36)? (isp_sthdr_formatter_#_x_length_size) / 36 +1 : (isp_sthdr_formatter_#_x_length_size)/36

Table 73 Output image horizontal size setting (ram lines)

Register name	Field Name	Index	Bits	Type	Reset value
isp_sthdr_formatter_isp1_num_of_ram_lines	outregs_isp_sthdr_formatter_isp1_num_of_ram_lines	0xf080	[15:0]	RW1S	0x0051
isp_sthdr_formatter_isp2_num_of_ram_lines	outregs_isp_sthdr_formatter_isp2_num_of_ram_lines	0xf082	[15:0]	RW1S	0x0051
isp_sthdr_formatter_isp3_num_of_ram_lines	outregs_isp_sthdr_formatter_isp3_num_of_ram_lines	0xf084	[15:0]	RW1S	0x0000
isp_sthdr_formatter_isp4_num_of_ram_lines	outregs_isp_sthdr_formatter_isp4_num_of_ram_lines	0xf086	[15:0]	RW1S	0x0000
isp_sthdr_formatter_stat_num_of_ram_lines	outregs_isp_sthdr_formatter_stat_num_of_ram_lines	0xf088	[15:0]	RW1S	0x0000
isp_sthdr_formatter_ddm_num_of_ram_lines	outregs_isp_sthdr_formatter_ddm_num_of_ram_lines	0xf08a	[15:0]	RW1S	0x0000
isp_sthdr_formatter_emb_header_num_of_ram_lines	outregs_isp_sthdr_formatter_emb_header_num_of_ram_lines	0xf08c	[15:0]	RW1S	0x0074

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isp_sthdr_formatter_emb_footer_num_of_ram_lines	outregs_isp_sthdr_formatter_emb_footer_num_of_ram_lines	0xf08e	[15:0]	RW1S	0x0008
isp_sthdr_formatter_isp1_num_of_ram_lines_s	outregs_isp_sthdr_formatter_isp1_num_of_ram_lines_s	0xf180	[15:0]	RW1S	0x0051
isp_sthdr_formatter_isp2_num_of_ram_lines_s	outregs_isp_sthdr_formatter_isp2_num_of_ram_lines_s	0xf182	[15:0]	RW1S	0x0051
isp_sthdr_formatter_isp3_num_of_ram_lines_s	outregs_isp_sthdr_formatter_isp3_num_of_ram_lines_s	0xf184	[15:0]	RW1S	0x0000
isp_sthdr_formatter_isp4_num_of_ram_lines_s	outregs_isp_sthdr_formatter_isp4_num_of_ram_lines_s	0xf186	[15:0]	RW1S	0x0000
isp_sthdr_formatter_stat_num_of_ram_lines_s	outregs_isp_sthdr_formatter_stat_num_of_ram_lines_s	0xf188	[15:0]	RW1S	0x0000
isp_sthdr_formatter_ddm_num_of_ram_lines_s	outregs_isp_sthdr_formatter_ddm_num_of_ram_lines_s	0xf18a	[15:0]	RW1S	0x0000
isp_sthdr_formatter_emb_header_num_of_ram_lines_s	outregs_isp_sthdr_formatter_emb_header_num_of_ram_lines_s	0xf18c	[15:0]	RW1S	0x0074
isp_sthdr_formatter_emb_footer_num_of_ram_lines_s	outregs_isp_sthdr_formatter_emb_footer_num_of_ram_lines_s	0xf18e	[15:0]	RW1S	0x0008

- isp_sthdr_formatter_offset_# : offset value of each data(#)
  = ((isp_sthdr_formatter_#_x_length_size % 36) == 0) ? 'd36 : isp_sthdr_formatter_#_x_length_size % 36

Table 74 Output image horizontal size setting (offset)

Register name	Field Name	Index	Bits	Type	Reset value
isp_sthdr_formatter_offset_1_2	outregs_isp_sthdr_formatter_isp1_offset	0xf070	[5:0]	RW1S	0x0018
	outregs_isp_sthdr_formatter_isp2_offset		[13:8]	RW1S	0x0018
isp_sthdr_formatter_offset_3_4	outregs_isp_sthdr_formatter_isp3_offset	0xf072	[5:0]	RW1S	0x0000
	outregs_isp_sthdr_formatter_isp4_offset		[13:8]	RW1S	0x0000
isp_sthdr_formatter_offset_stat_ddm	outregs_isp_sthdr_formatter_stat_offset	0xf074	[5:0]	RW1S	0x0000
	outregs_isp_sthdr_formatter_ddm_offset		[13:8]	RW1S	0x0000
isp_sthdr_formatter_offset_emb	outregs_isp_sthdr_formatter_emb_header_offset	0xf076	[5:0]	RW1S	0x0004
	outregs_isp_sthdr_formatter_emb_footer_offset		[13:8]	RW1S	0x0004

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isp_sthdr_formatter_offset_1_2_s	outregs_isp_sthdr_formatter_isp1_offset_s	0xf170	[5:0]	RW1S	0x0018
	outregs_isp_sthdr_formatter_isp2_offset_s		[13:8]	RW1S	0x0018
isp_sthdr_formatter_offset_3_4_s	outregs_isp_sthdr_formatter_isp3_offset_s	0xf172	[5:0]	RW1S	0x0000
	outregs_isp_sthdr_formatter_isp4_offset_s		[13:8]	RW1S	0x0000
isp_sthdr_formatter_offset_stat_ddm_s	outregs_isp_sthdr_formatter_stat_offset_s	0xf174	[5:0]	RW1S	0x0000
	outregs_isp_sthdr_formatter_ddm_offset_s		[13:8]	RW1S	0x0000
isp_sthdr_formatter_offset_emb_s	outregs_isp_sthdr_formatter_emb_header_offset_s	0xf176	[5:0]	RW1S	0x0004
	outregs_isp_sthdr_formatter_emb_footer_offset_s		[13:8]	RW1S	0x0004

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7 Embedded Data

Sensor information can come out as embedded data output such as the following figure.

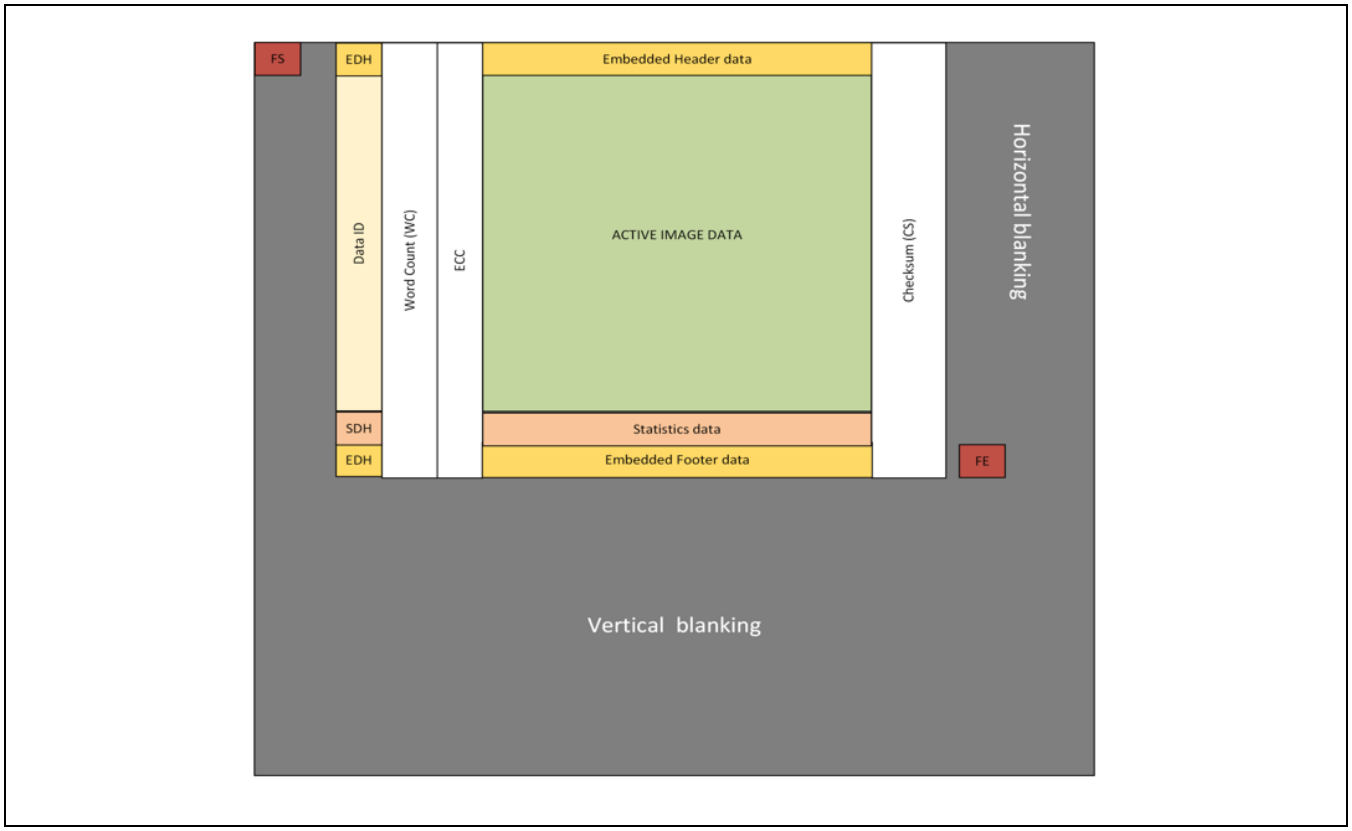


Figure 77 Frame structure with Embedded data

Following table shows S5K3B6SA's basic embedded information. (This value can be changed by register setting)

Table 75 Embedded data information

Embedded header	Embedded footer
EDH(Embedded Data type Header) : 0x12	EDF(Embedded Data type Footer) : 0x30
Embedded Header line : 2-line	Embedded Footer line : 1-line
Embedded Header Size : 5360 pixels(byte)	Embedded Footer Size : 512 pixels(byte)

NOTE: First embedded-header line is filled with dummy value, second embedded-header line is valid as described excel file at default set.

NOTE: If user wants to get detailed embedded lists, contact to FAE.

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S5K3B6SA follows the MIPI CSI-2 rule, so basically Embedded data transmission will be RAW8 Data transmission as follows. Also, Embedded data can transmit the same as image packing format optionally. In case of RAW12, embedded data LSB 4-bit padded to filler data (default 0x1).

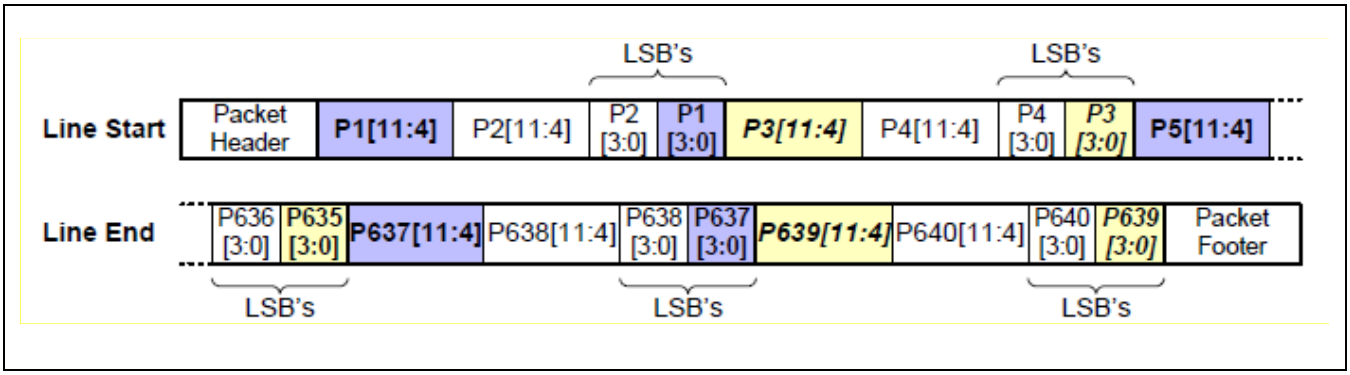


Figure 78 Byte packing of RAW12 data (Reference from MIPI CSI-2 v1.2)

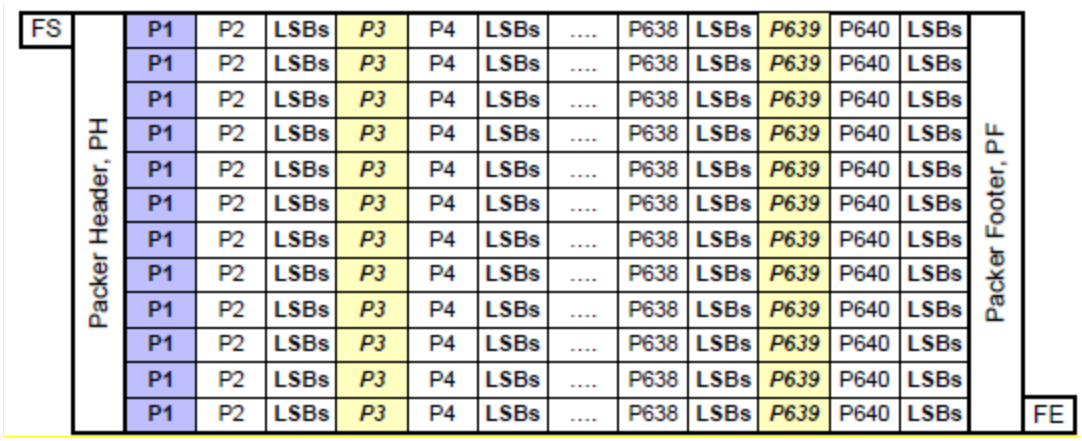


Figure 79 RAW12 Frame Format (Reference from MIPI CSI-2 v1.2)

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7.1 Frame Format

7.1.1 Default mode

- Embedded header : 5360bytes * 2lines (default : 1 dummy line, 1 data line)
- Embedded footer : 512 bytes

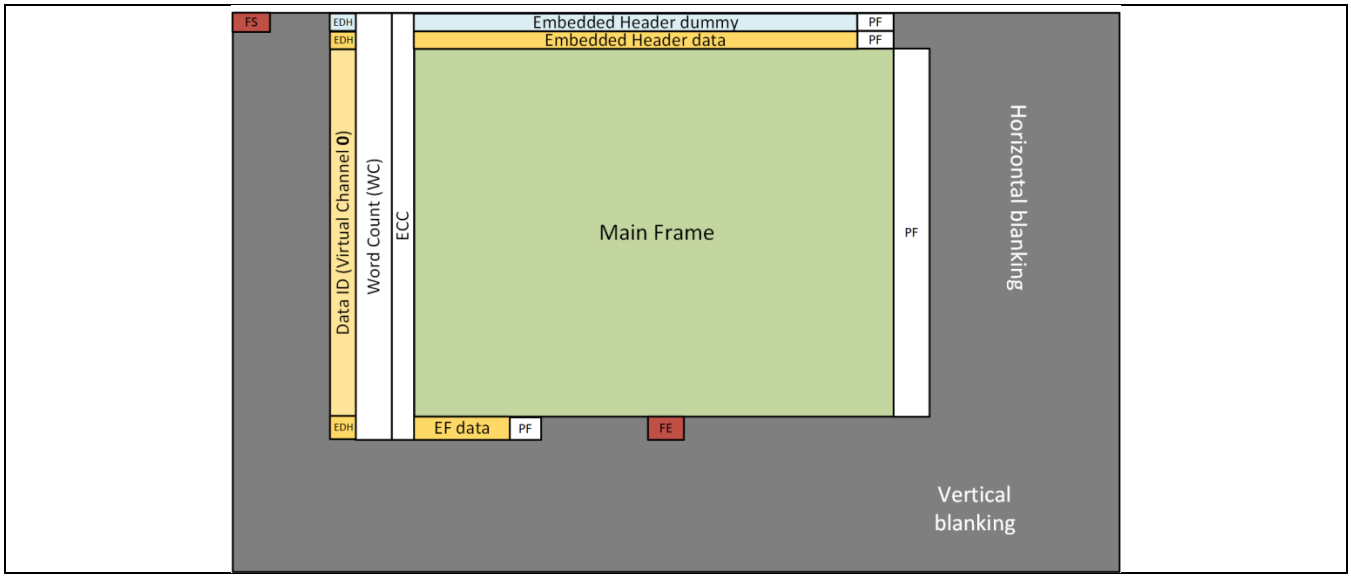


Figure 80 Single output frame format (default size)

7.1.2 Embedded header split mode

- Embedded header : 5360bytes * 2lines (2 data lines: No dummy line)
- Embedded footer : 512 bytes

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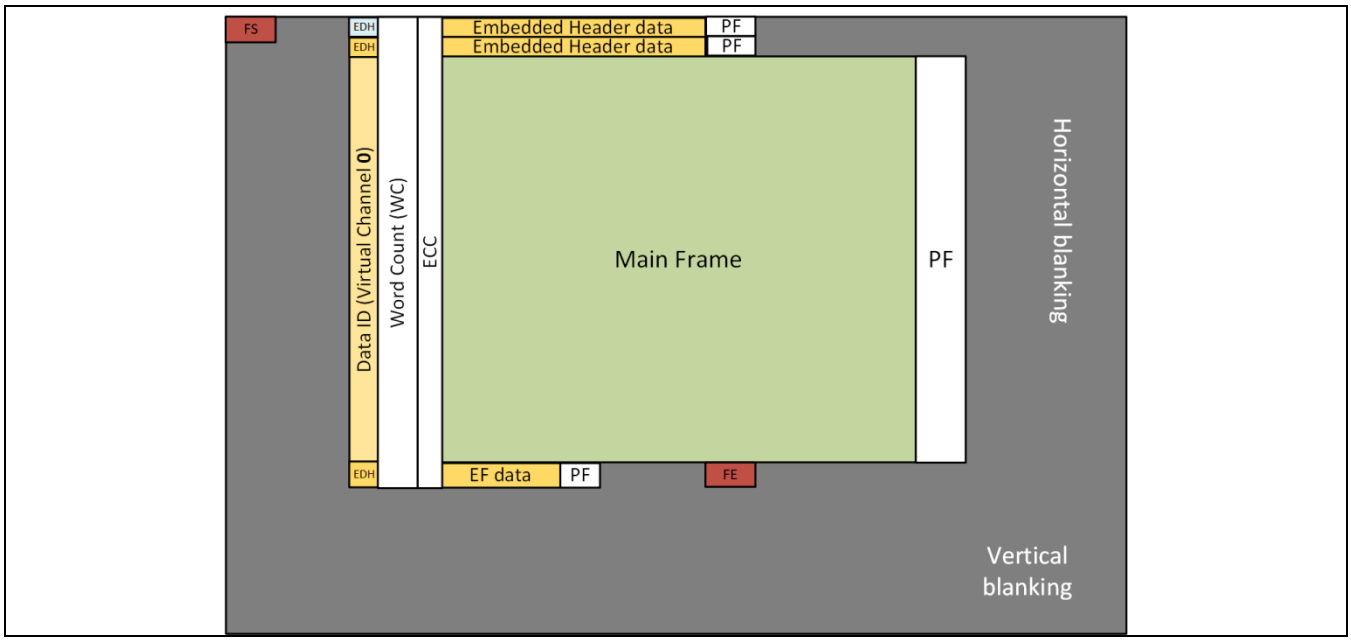


Figure 81 Single output frame format (embedded header split mode)

7.1.3 Embedded data & Active data aligned mode

- Embedded header : Nbytes * 3lines (3data lines. After 5360bytes, datas are padded w/ dummy value('h07))
- Embedded footer : Nbytes

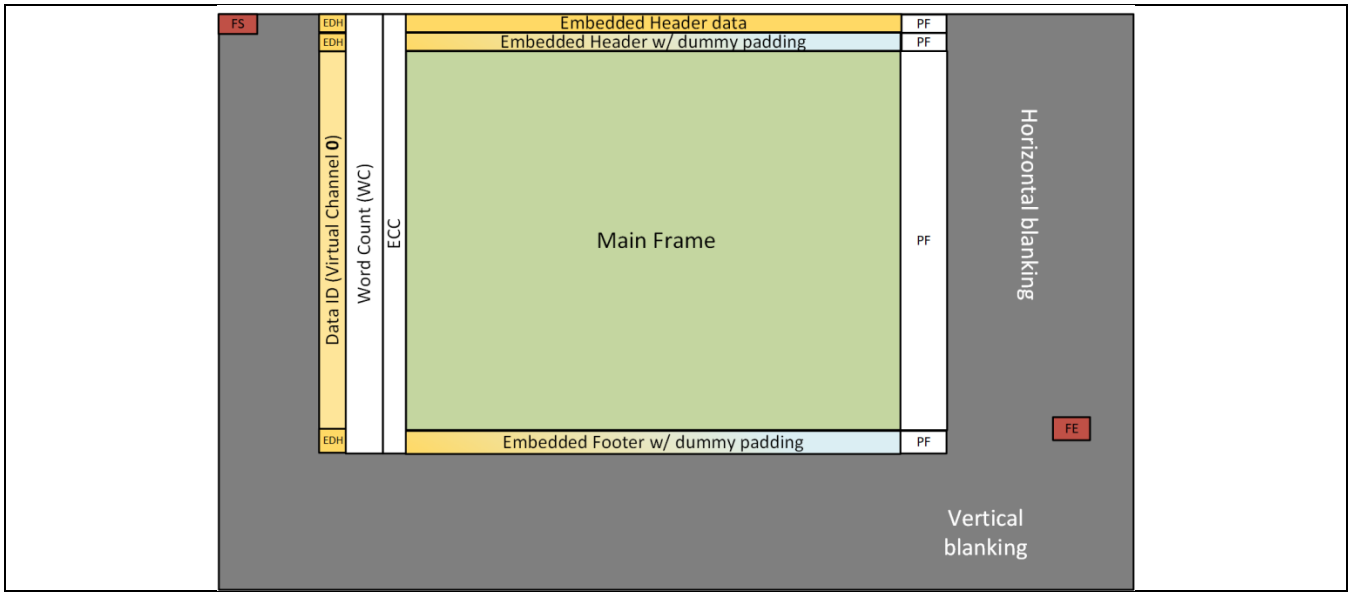


Figure 82 Single output frame format (embedded-active align mode)

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.